

MEASUREMENTS & TECHNIQUES

1.3 Errors and uncertainties

Candidates should be able to:

- 1 understand and explain the effects of systematic errors (including zero errors) and random errors in measurements
- 2 understand the distinction between precision and accuracy
- 3 assess the uncertainty in a derived quantity by simple addition of absolute or percentage uncertainties

1 9702/1/O/N/02/Q5

A student carries out a series of determinations of the acceleration of free fall g . The table shows the results.

What can be said about this experiment?

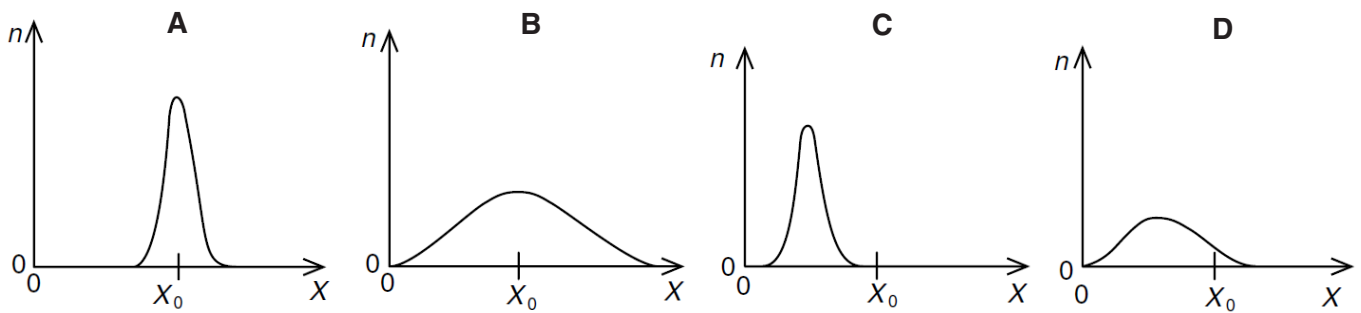
- A** It is accurate and precise.
- B** It is accurate but not precise.
- C** It is not accurate and not precise.
- D** It is not accurate but is precise.

g/ms^{-2}
4.91
4.89
4.88
4.90
4.93
4.92

2 9702/1/O/N/02/Q6

A quantity X is measured many times. A graph is plotted showing the number n of times a particular value of X is obtained. X has a true value X_0 .

Which graph could be obtained if the measurement of X has a large systematic error but a small random error?

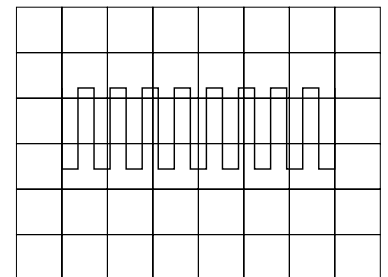


3 9702/1/O/N/02/Q7*

The diagram shows a square-wave trace on the screen of a cathode-ray oscilloscope. A grid of 1 cm squares covers the screen. The time-base setting is 10 ms cm^{-1} .

What is the approximate frequency of the square-wave?

- A** 70 Hz **B** 140 Hz
- C** 280 Hz **D** 1400 Hz



4 9702/01/O/N/03/Q4*

A thermometer can be read to an accuracy of $\pm 0.5^\circ\text{C}$. This thermometer is used to measure a temperature rise from 40°C to 100°C .

What is the percentage uncertainty in the measurement of the temperature rise?

- A** 0.5 % **B** 0.8 % **C** 1.3 % **D** 1.7 %

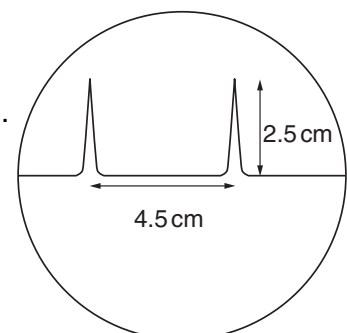
5 9702/01/O/N/03/Q5

The time-base on a cathode-ray oscilloscope is set at 6 ms/cm .

A trace consisting of two pulses is recorded as shown in the diagram.

What is the time interval between the two pulses?

- A** 0.42 ms **B** 0.75 ms
- C** 1.33 ms **D** 27 ms



6 9702/01/O/N/03/Q6

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

What is the diameter of the wire?

- A** 1.90 mm **B** 2.45 mm
C 2.59 mm **D** 2.73 mm

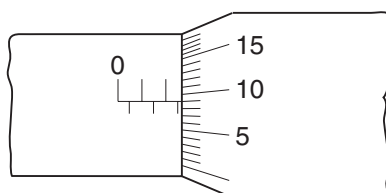


diagram 1

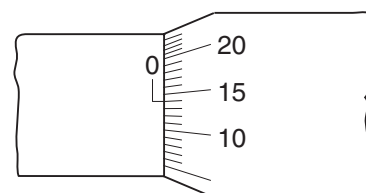
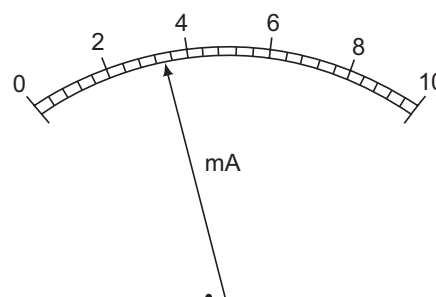


diagram 2

7 9702/01/M/J/04/Q4

What is the reading shown on this milliammeter?

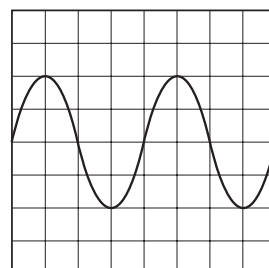
- A** 2.35 mA **B** 2.7 mA
C 3.4 mA **D** 3.7 mA



8 9702/13/M/J/16/Q5

The following trace is seen on the screen of a cathode-ray oscilloscope.

The setting of the time base is then changed from 10 ms cm^{-1} to 20 ms cm^{-1} and the Y-sensitivity is unaltered.



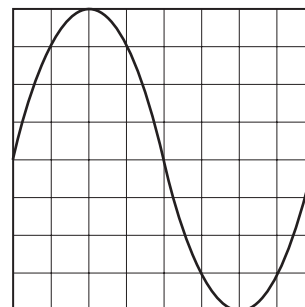
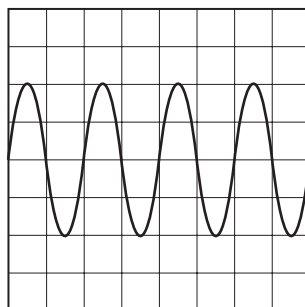
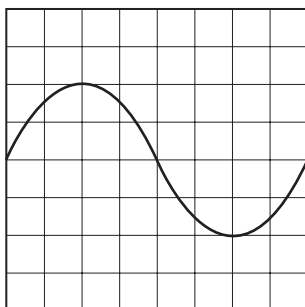
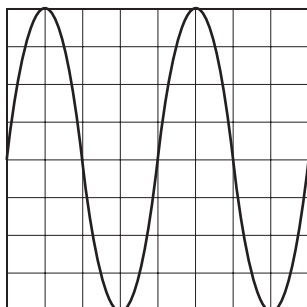
Which trace is now seen on the screen?

A

B

C

D



9 9702/13/M/J/16/Q5

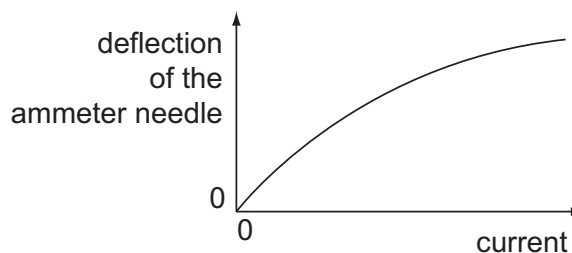
In a simple electrical circuit, the current in a resistor is measured as $(2.50 \pm 0.05) \text{ mA}$. The resistor is marked as having a value of $4.7 \Omega \pm 2\%$.

If these values were used to calculate the power dissipated in the resistor, what would be the percentage uncertainty in the value obtained?

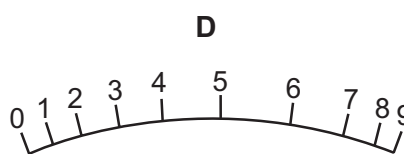
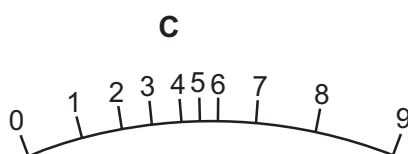
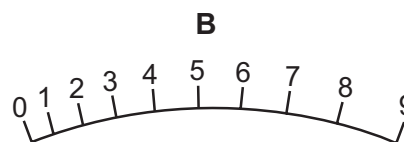
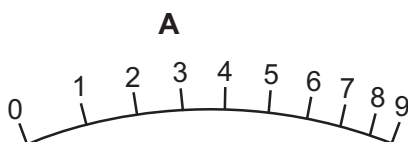
- A** 2% **B** 4% **C** 6% **D** 8%

10 9702/01/O/N/04/Q4

The deflection of the needle of an ammeter varies with the current passing through the ammeter as shown in the graph.



Which diagram could represent the appearance of the scale of this meter?

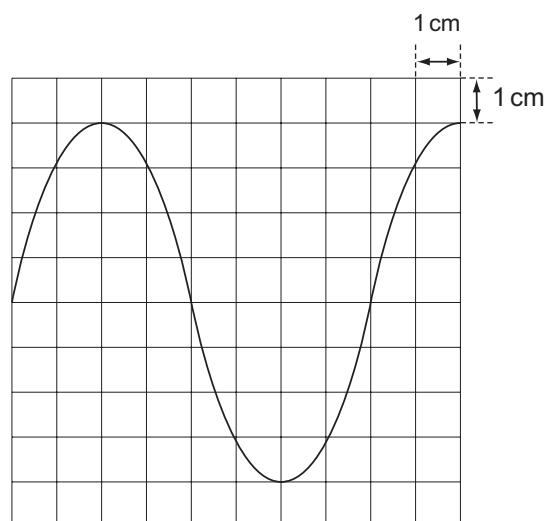


11 9702/01/O/N/04/Q5

When a 12 V 50 Hz supply is connected to the Y-terminals of an oscilloscope, the trace in the diagram is obtained.

What is the setting of the time-base control?

- A** 2.0 ms cm⁻¹ **B** 2.5 ms cm⁻¹
C 5 ms cm⁻¹ **D** 20 ms cm⁻¹



12 9702/01/O/N/04/Q6

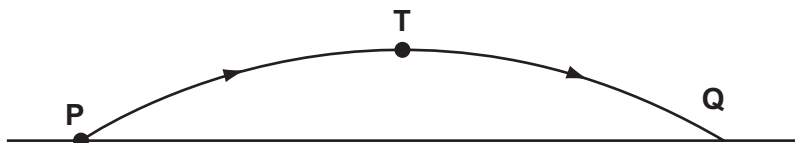
Four students each made a series of measurements of the acceleration of free fall g . The table shows the results obtained.

Which student obtained a set of results that could be described as precise but not accurate?

student	results, $g / \text{m s}^{-2}$			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	8.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

13 9702/01/O/N/04/Q7

In the absence of air resistance, a stone is thrown from **P** and follows a parabolic path in which the highest point reached is **T**. The stone reaches point **Q** just before landing.



The vertical component of acceleration of the stone is

- A** zero at **T**. **B** greatest at **T**. **C** greatest at **Q**. **D** the same at **Q** as at **T**.

14 9702/01/M/J/05/Q4*

In an experiment, a radio-controlled car takes 2.50 ± 0.05 s to travel 40.0 ± 0.1 m.

What is the car's average speed and the uncertainty in this value?

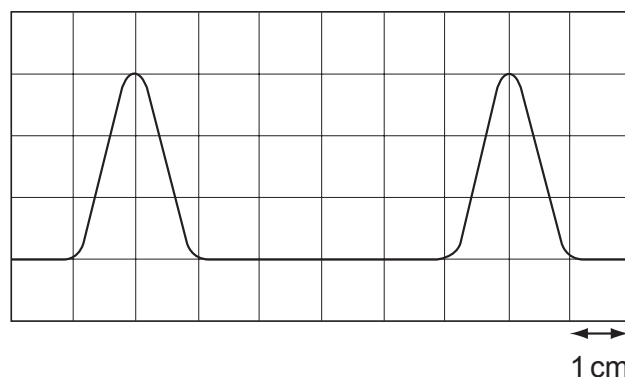
- A** $16 \pm 1 \text{ m s}^{-1}$ **B** $16.0 \pm 0.2 \text{ m s}^{-1}$ **C** $16.0 \pm 0.4 \text{ m s}^{-1}$ **D** $16.00 \pm 0.36 \text{ m s}^{-1}$

15 9702/01/M/J/05/Q5

The diagram shows two pulses on the screen of a cathode ray oscilloscope. A grid of 1 cm squares covers the screen. The time base setting is $1 \mu\text{s cm}^{-1}$.

How long does each pulse last?

- A** $2 \mu\text{s}$ **B** $3 \mu\text{s}$
C $4 \mu\text{s}$ **D** $6 \mu\text{s}$



16 9702/01/M/J/05/Q6

Which feature of a graph allows acceleration to be determined?

- A** the area under a displacement-time graph **C** the slope of a displacement-time graph
B the area under a velocity-time graph **D** the slope of a velocity-time graph

17 9702/01/M/J/05/Q7

A boy throws a ball vertically upwards. It rises to a maximum height, where it is momentarily at rest, and falls back to his hands.

Which of the following gives the acceleration of the ball at various stages in its motion? Take vertically upwards as positive. Neglect air resistance.

	rising	at maximum height	falling
A	-9.81 m s^{-2}	0	$+9.81 \text{ m s}^{-2}$
B	-9.81 m s^{-2}	-9.81 m s^{-2}	-9.81 m s^{-2}
C	$+9.81 \text{ m s}^{-2}$	$+9.81 \text{ m s}^{-2}$	$+9.81 \text{ m s}^{-2}$
D	$+9.81 \text{ m s}^{-2}$	0	-9.81 m s^{-2}

18 9702/01/O/N/05/Q4

A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm 892, 891, 892, 891, 891, 892

Are the readings accurate and precise to within 1 mm?

	results are accurate to within 1 mm	results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

19 9702/01/O/N/05/Q5

The density of the material of a rectangular block is determined by measuring the mass and linear dimensions of the block. The table shows the results obtained, together with their uncertainties.

mass = (25.0 ± 0.1) g

breadth = (2.00 ± 0.01) cm

length = (5.00 ± 0.01) cm

height = (1.00 ± 0.01) cm

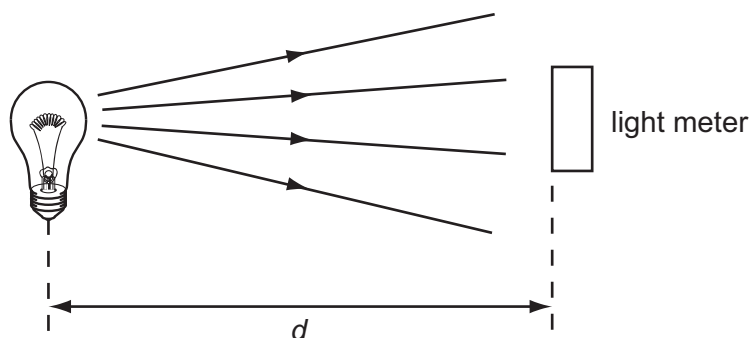
The density is calculated to be 2.50 g cm^{-3} .

What is the uncertainty in this result?

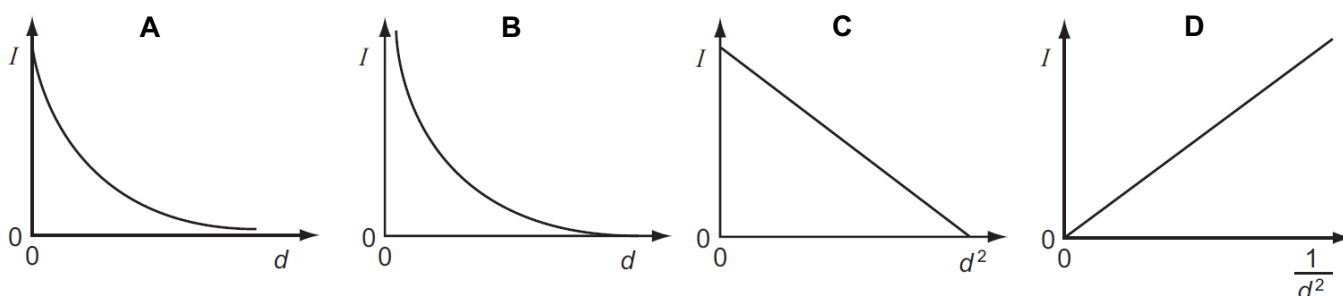
A $\pm 0.01 \text{ g cm}^{-3}$ **B** $\pm 0.02 \text{ g cm}^{-3}$ **C** $\pm 0.05 \text{ g cm}^{-3}$ **D** $\pm 0.13 \text{ g cm}^{-3}$

20 9702/01/M/J/06/Q4

A light meter measures the intensity I of the light falling on it. Theory suggests that this varies as the inverse square of the distance d .



Which graph of the results supports this theory?



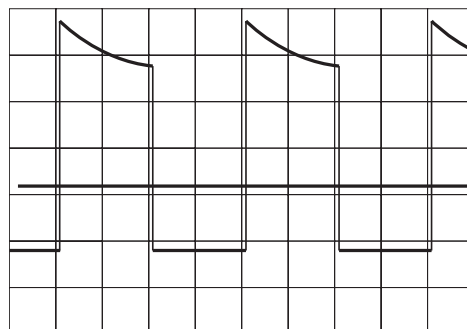
26 9702/01/M/J/07/Q4

An oscilloscope display consists of two separate traces, a waveform and a long horizontal line. The horizontal line may be taken as the zero level.

The grid on the screen is calibrated in cm squares, the timebase setting is 2.5 ms cm^{-1} , and the Y-sensitivity is 5 mV cm^{-1} .

What are the period and the peak positive voltage of the waveform in the diagram?

	period / ms	peak positive voltage / mV
A	5	17
B	5	25
C	10	17
D	10	25

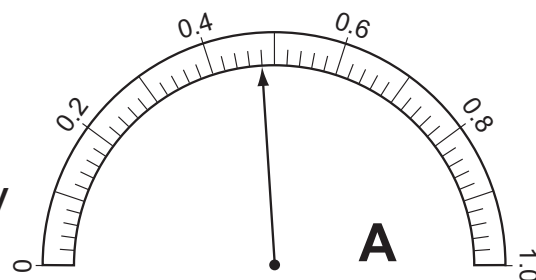


27 9702/01/M/J/07/Q5

The resistance of an electrical component is measured. The following meter readings are obtained.

What is the resistance?

- A 2.5Ω B 2.7Ω
C 2500Ω D 2700Ω



28 9702/01/O/N/07/Q4

A series of measurements of the acceleration of free fall g is shown in the table. Which set of results is precise but not accurate?

	g/ms^{-2}				
A	9.81	9.79	9.84	9.83	9.79
B	9.81	10.12	9.89	8.94	9.42
C	9.45	9.21	8.99	8.76	8.51
D	8.45	8.46	8.50	8.41	8.47

29 9702/01/O/N/07/Q5

A mass m has acceleration a . It moves through a distance s in time t . The power used in accelerating the mass is equal to the product of force and velocity. The percentage uncertainties are

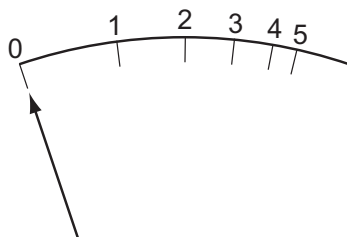
- 0.1 % in m ,
1 % in a ,
1.5 % in s ,
0.5 % in t .

What is the percentage uncertainty in the average power?

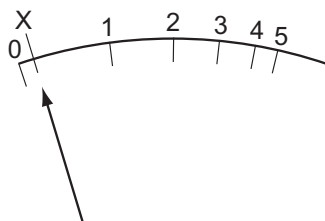
- A 2.1 % B 2.6 % C 3.1 % D 4.1 %

30 9702/01/O/N/07/Q6

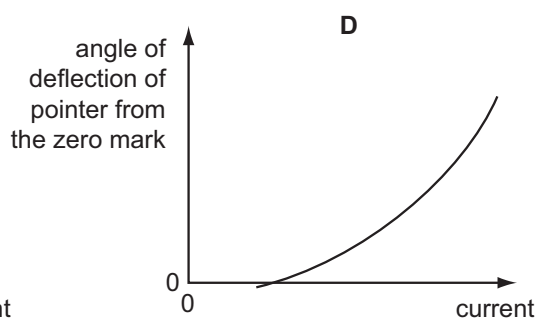
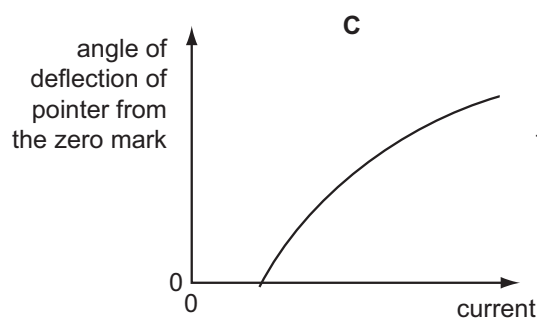
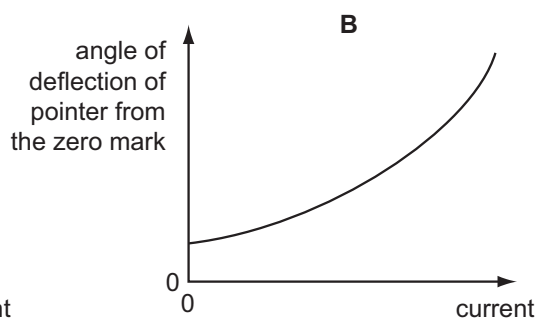
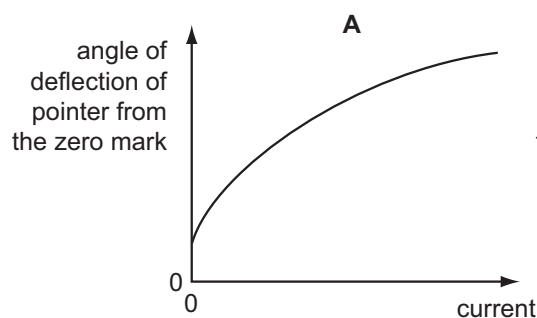
The diagram shows the graduations of a correctly calibrated ammeter. When the current is zero, the pointer is at 0.



The ammeter is accidentally readjusted so that when the current is zero, the pointer is at X.



Which calibration graph best represents the response of the readjusted ammeter?



31 9702/01/M/J/08/Q4

The resistance R of a resistor is determined by measuring the potential difference V across it and the current I in it. The value of R is then calculated using the equation

$$R = \frac{V}{I}.$$

The values measured are $V = 1.00 \pm 0.05 \text{ V}$ and $I = 0.50 \pm 0.01 \text{ A}$.

What is the percentage uncertainty in the value of R ?

- A** 2.5% **B** 3.0% **C** 7.0% **D** 10.0%

32 9702/01/M/J/08/Q5

Four students each made a series of measurements of the acceleration of free fall g . The table shows the results obtained.

Which set of results could be described as precise but **not** accurate?

	g/ms^{-2}			
A	9.81	9.79	9.84	9.83
B	9.81	10.12	9.89	8.94
C	9.45	9.21	8.99	8.76
D	8.45	8.46	8.50	8.41

33 9702/01/O/N/08/Q4

A student uses a digital ammeter to measure a current. The reading of the ammeter is found to fluctuate between 1.98 A and 2.02 A.

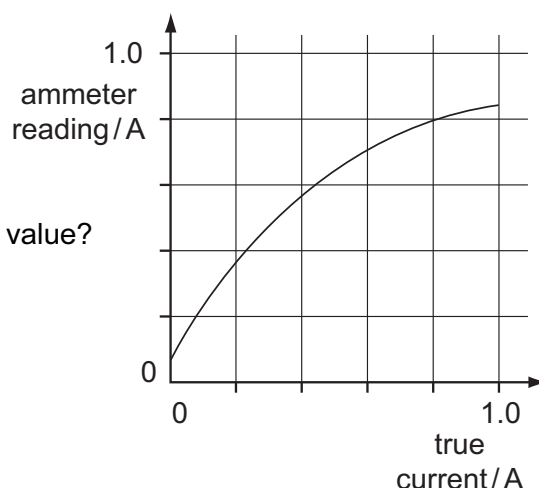
The manufacturer of the ammeter states that any reading has a systematic uncertainty of $\pm 1\%$.

Which value of current should be quoted by the student?

- A** $(2.00 \pm 0.01)\text{A}$ **B** $(2.00 \pm 0.02)\text{A}$ **C** $(2.00 \pm 0.03)\text{A}$ **D** $(2.00 \pm 0.04)\text{A}$

34 9702/11/M/J/16/Q5

A calibration graph is produced for a faulty ammeter.

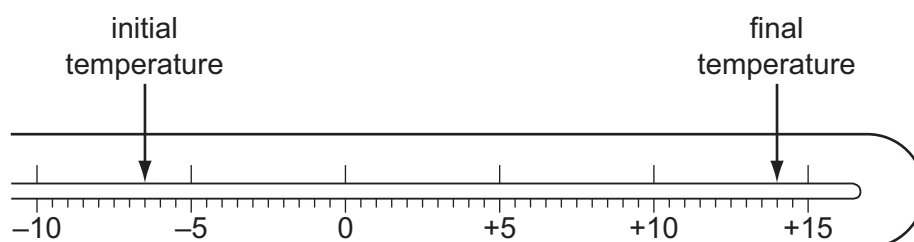


Which ammeter reading will be nearest to the correct value?

- A** 0.2 A **B** 0.4 A
C 0.6 A **D** 0.8 A

35 9702/13/M/J/13/Q6

The diagram shows the stem of a Celsius thermometer marked to show initial and final temperature values.



What is the temperature change expressed to an appropriate number of significant figures?

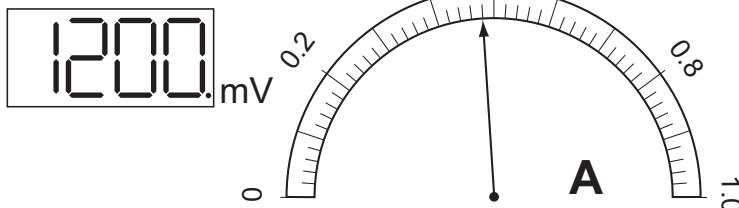
- A** 14°C **B** 20.5°C **C** 21°C **D** 22.0°C

36 9702/01/M/J/09/Q4

The diagrams show digital voltmeter and analogue ammeter readings from a circuit in which electrical heating is occurring.

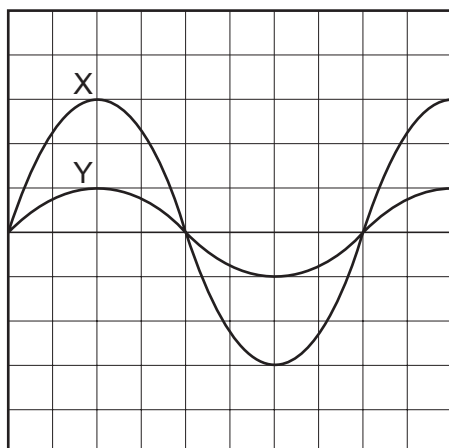
What is the electrical power of the heater?

- A 0.53 W B 0.58 W
C 530 W D 580 W



37 9702/12/O/N/09/Q3

The diagram shows an oscilloscope screen displaying two signals.



Signal X has a frequency of 50 Hz and peak voltage of 12 V.

What is the period and peak voltage of signal Y?

	period / ms	peak voltage / V
A	20	4
B	20	12
C	50	4
D	50	12

38 9702/12/M/J/10/Q3

A student finds the density of a liquid by measuring its mass and its volume. The following is a summary of his measurements.

mass of empty beaker = (20 ± 1) g

mass of beaker + liquid = (70 ± 1) g

volume of liquid = (10.0 ± 0.6) cm³

He correctly calculates the density of the liquid as 5.0 g cm⁻³.

What is the uncertainty in this value?

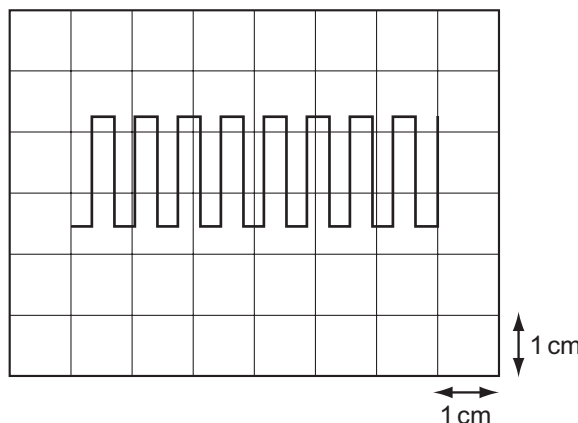
- A 0.3 g cm⁻³ B 0.5 g cm⁻³ C 0.6 g cm⁻³ D 2.6 g cm⁻³

39 9702/12/M/J/10/Q5

The diagram shows a square-wave trace on the screen of a cathode-ray oscilloscope. A grid of 1 cm squares covers the screen. The time-base setting is 10 ms cm⁻¹.

What is the approximate frequency of the square wave?

- A 70 Hz B 140 Hz
C 280 Hz D 1400 Hz



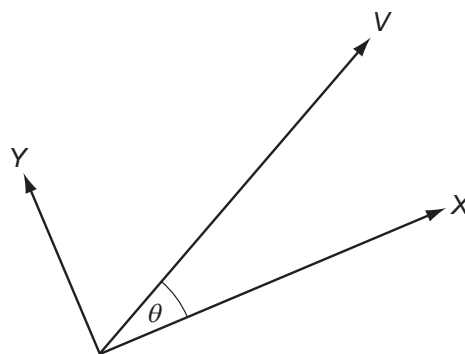
40 9702/12/M/J/10/Q6

A vector quantity V is resolved into two perpendicular components X and Y . The angle between V and component X is θ .

The angle between component X and the vector V is increased from 0° to 90° .

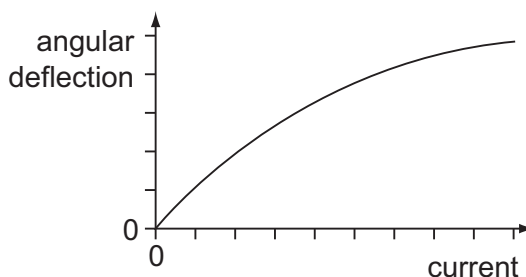
How do the magnitudes of X and Y change as the angle θ is increased in this way?

	X	Y
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

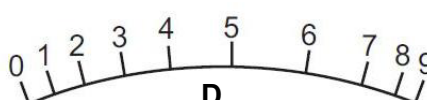
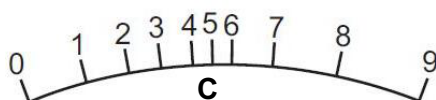
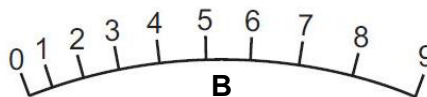


41 9702/11/O/N/10/Q4

The angular deflection of the needle of an ammeter varies with the current passing through the ammeter as shown in the graph.

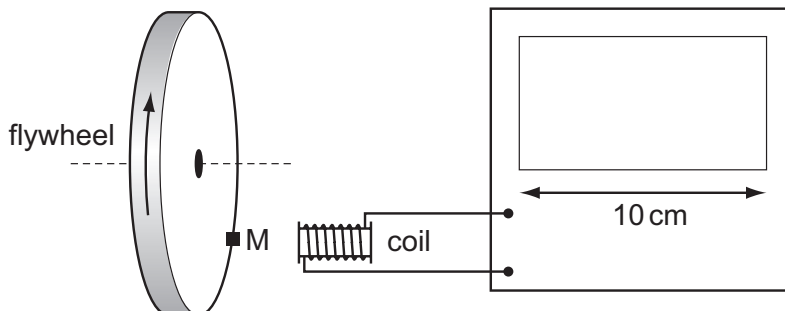


Which diagram could represent the appearance of the scale on this meter?



42 9702/11/O/N/10/Q5

The diagram shows a cathode-ray oscilloscope (c.r.o.) being used to measure the rate of rotation of a flywheel.



The flywheel has a small magnet M mounted on it. Each time the magnet passes the coil, a voltage pulse is generated, which is passed to the c.r.o. The display of the c.r.o. is 10 cm wide. The flywheel is rotating at a rate of about 3000 revolutions per minute.

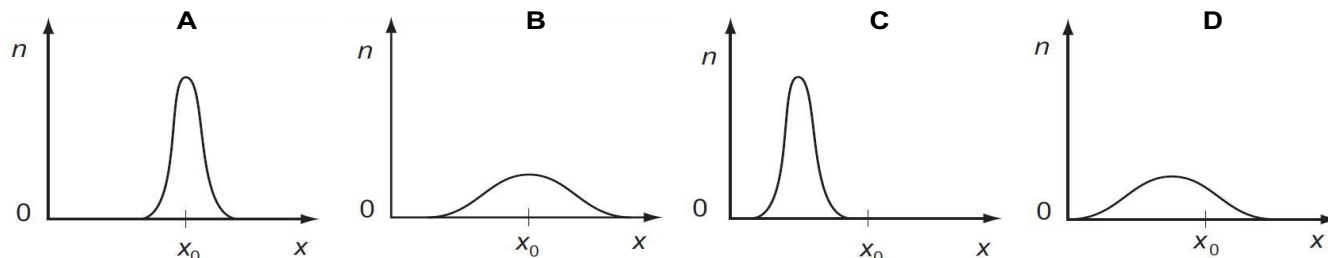
Which time-base setting will display clearly separate pulses on the screen?

- A 1 s cm^{-1} B 10 ms cm^{-1} C $100 \mu\text{s cm}^{-1}$ D $1 \mu\text{s cm}^{-1}$

43 9702/11/O/N/10/Q6

A fixed quantity x_0 is measured many times in an experiment that has experimental uncertainty. A graph is plotted to show the number n of times that a particular value x is obtained.

Which graph could be obtained if the measurement of x_0 has a large systematic error but a small random error?



44 9702/12/O/N/10/Q4

A metre rule is used to measure the length of a piece of wire. It is found to be 70 cm long to the nearest millimetre.

How should this result be recorded in a table of results?

- A 0.7 m B 0.70 m C 0.700 m D 0.7000 m

45 9702/12/O/N/10/Q5

A quantity x is to be determined from the equation

$$x = P - Q.$$

P is measured as 1.27 ± 0.02 m.

Q is measured as 0.83 ± 0.01 m.

What is the percentage uncertainty in x to one significant figure?

- A 0.4 % B 2 % C 3 % D 7 %

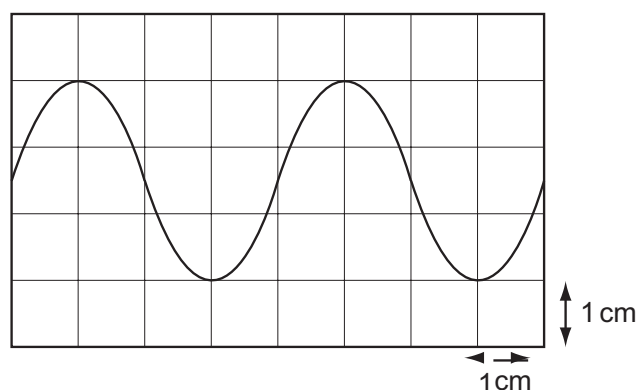
46 9702/11/M/J/11/Q4

The diagram shows a trace of a wave on a cathode-ray oscilloscope.

The vertical and horizontal gridlines have a spacing of 1.0 cm. The voltage scaling is 4 V cm^{-1} and the time scaling is 5 ms cm^{-1} .

What are the amplitude and period of the wave?

	amplitude / V	period / ms
A	1.5	4
B	5.0	10
C	6.0	20
D	12.0	20



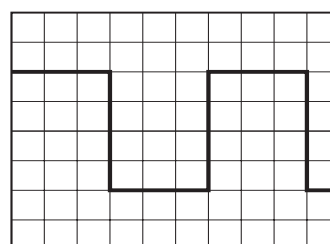
47 9702/13/O/N/12/Q5

A cathode-ray oscilloscope displays a square wave, as shown in the diagram.

The time-base setting is $0.20 \text{ ms per division}$.

What is the frequency of the square wave?

- A 8.3 Hz B 830 Hz
C 1300 Hz D 1700 Hz



48 9702/11/M/J/11/Q5

The diagram shows an experiment to measure the speed of a small ball falling at constant speed through a clear liquid in a glass tube.

There are two marks on the tube. The top mark is positioned at 115 ± 1 mm on the adjacent rule and the lower mark at 385 ± 1 mm. The ball passes the top mark at 1.50 ± 0.02 s and passes the lower mark at 3.50 ± 0.02 s.

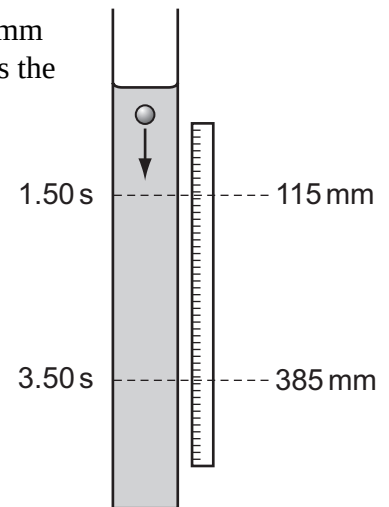
The constant speed of the ball is calculated by

$$\frac{385 - 115}{3.50 - 1.50} = \frac{270}{2.00} = 135 \text{ mm s}^{-1}.$$

Which expression calculates the fractional uncertainty in the value of this speed?

A $\frac{2}{270} + \frac{0.04}{2.00}$ **B** $\frac{2}{270} - \frac{0.04}{2.00}$

C $\frac{1}{270} \times \frac{0.02}{2.00}$ **D** $\frac{1}{270} \div \frac{0.02}{2.00}$



49 9702/12/M/J/11/Q4

The uncertainty in the value of the momentum of a trolley passing between two points X and Y varies with the choice of measuring devices.

Measurements for the same trolley made by different instruments were recorded.

- 1 distance between X and Y using a metre rule with cm divisions = 0.55 m
- 2 distance between X and Y using a metre rule with mm divisions = 0.547 m
- 3 timings using a wristwatch measuring to the nearest 0.5 s at X = 0.0 s and at Y = 4.5 s
- 4 timings using light gates measuring to the nearest 0.1 s at X = 0.0 s and at Y = 4.3 s
- 5 mass of trolley using a balance measuring to the nearest g = 6.4×10^{-2} kg
- 6 mass of trolley using a balance measuring to the nearest 10 g = 6×10^{-2} kg

Which measurements, one for each quantity measured, lead to the least uncertainty in the value of the momentum of the trolley?

- A** 1, 3 and 6 **B** 1, 4 and 6 **C** 2, 3 and 6 **D** 2, 4 and 5

50 9702/12/M/J/11/Q6

A bullet is fired horizontally with speed v from a rifle. For a short time t after leaving the rifle, the only force affecting its motion is gravity. The acceleration of free fall is g .

Which expression gives the value of $\frac{\text{the horizontal distance travelled in time } t}{\text{the vertical distance travelled in time } t}$?

A $\frac{vt}{g}$ **B** $\frac{v}{gt}$ **C** $\frac{2vt}{g}$ **D** $\frac{2v}{gt}$

51 9702/12/O/N/11/Q4

A micrometer is used to measure the diameters of two cylinders.

diameter of first cylinder = 12.78 ± 0.02 mm

diameter of second cylinder = 16.24 ± 0.03 mm

The difference in the diameters is calculated.

What is the uncertainty in this difference?

- A** ± 0.01 mm **B** ± 0.02 mm **C** ± 0.03 mm **D** ± 0.05 mm

52 9702/13/O/N/15/Q6

A cylindrical tube rolling down a slope of inclination θ moves a distance L in time T . The equation relating these quantities is

$$L \left(3 + \frac{a^2}{P} \right) = QT^2 \sin \theta$$

Where a is the internal radius of the tube and P and Q are constants.

Which line gives the correct units for P and Q ?

	P	Q
A	m^2	$\text{m}^2 \text{s}^{-2}$
B	m^2	m s^{-2}
C	m^2	$\text{m}^3 \text{s}^{-2}$
D	m^3	m s^{-2}

53 9702/12/O/N/11/Q5

The speedometer in a car consists of a pointer which rotates. The pointer is situated several millimetres from a calibrated scale.

What could cause a random error in the driver's measurement of the car's speed?

- A The car's speed is affected by the wind direction.
- B The driver's eye is not always in the same position in relation to the pointer.
- C The speedometer does not read zero when the car is at rest.
- D The speedometer reads 10 % higher than the car's actual speed.

54 9702/12/M/J/12/Q2

For which quantity is the magnitude a reasonable estimate?

- A frequency of a radio wave 500 pHz
- B mass of an atom 500 μg
- C the Young modulus of a metal 500 kPa
- D wavelength of green light 500 nm

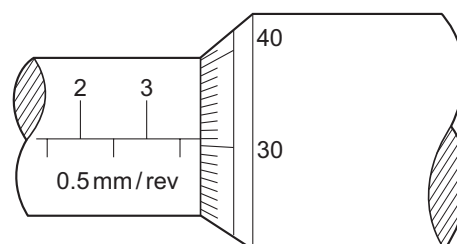
55 9702/12/M/J/12/Q4*

The diameter of a cylindrical metal rod is measured using a micrometer screw gauge.

The diagram below shows an enlargement of the scale on the micrometer screw gauge when taking the measurement.

What is the cross-sectional area of the rod?

- A 3.81 mm^2
- B 11.4 mm^2
- C 22.8 mm^2
- D 45.6 mm^2

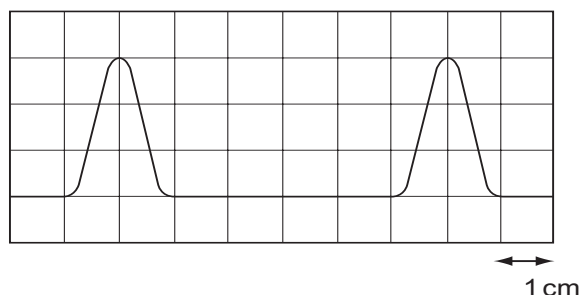


56 9702/12/M/J/12/Q6

The diagram shows two complete pulses on the screen of a cathode-ray oscilloscope. A grid of 1 cm squares covers the screen. The time-base setting is $1 \mu\text{s cm}^{-1}$.

How long does each pulse last?

- A 2 μs
- B 3 μs
- C 4 μs
- D 6 μs



57 9702/12/M/J/12/Q5

A mass is dropped from rest, and falls through a distance of 2.0 m in a vacuum. An observer records the time taken for the mass to fall through this distance using a manually operated stopwatch and repeats the measurements a further two times. The average result of these measured times, displayed in the table below, was used to determine a value for the acceleration of free fall. This was calculated to be 9.8 m s^{-2} .

	first measurement	second measurement	third measurement	average
time/s	0.6	0.73	0.59	0.64

Which statement best relates to the experiment?

- A The measurements are precise and accurate with no evidence of random errors.
- B The measurements are not accurate and not always recorded to the degree of precision of the measuring device but the calculated experimental result is accurate.
- C The measurements are not always recorded to the degree of precision of the measuring device but are accurate. Systematic errors may be present.
- D The range of results shows that there were random errors made but the calculated value is correct so the experiment was successful.

58 9702/13/M/J/12/Q4*

In an experiment, a radio-controlled car takes $2.50 \pm 0.05 \text{ s}$ to travel $40.0 \pm 0.1 \text{ m}$.

What is the car's average speed and the uncertainty in this value?

- A $16 \pm 1 \text{ m s}^{-1}$ B $16.0 \pm 0.2 \text{ m s}^{-1}$ C $16.0 \pm 0.4 \text{ m s}^{-1}$ D $16.00 \pm 0.36 \text{ m s}^{-1}$

A student is given a reel of wire of diameter less than 0.2 mm and is asked to find the density of the metal.

Which pair of instruments would be most suitable for finding the **volume** of the wire?

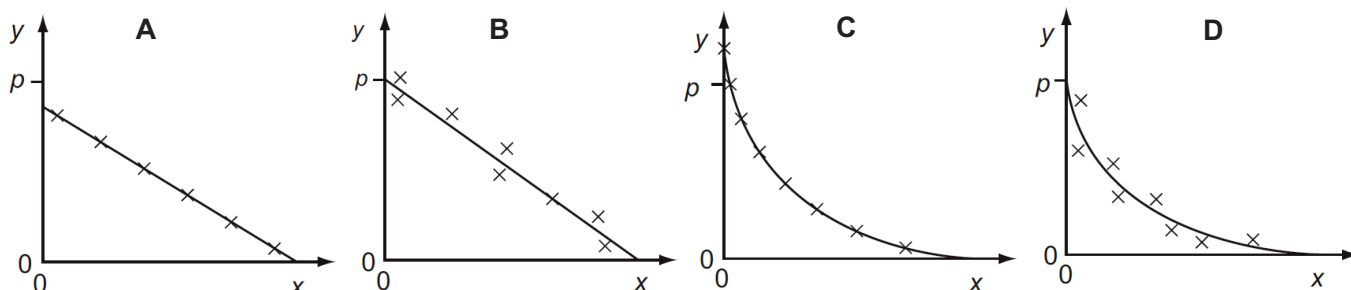
- A balance and micrometer
- B metre rule and micrometer
- C metre rule and vernier calipers
- D micrometer and vernier calipers

59 9702/12/O/N/12/Q6

Variables x and y are related by the equation $y = p - qx$ where p and q are constants.

Values of x and y are measured experimentally. The results contain a systematic error.

Which graph best represents these results?



60 9702/12/O/N/12/Q8

A science museum designs an experiment to show the fall of a feather in a vertical glass vacuum tube.

The time of fall from rest is to be close to 0.5 s.

What length of tube is required?

- A 1.3 m B 2.5 m C 5.0 m D 10.0 m

61 9702/12/O/N/12/Q7

The speed of a car is calculated from measurements of the distance travelled and the time taken. The distance is measured as 200 m, with an uncertainty of ± 2 m. The time is measured as 10.0 s, with an uncertainty of ± 0.2 s. What is the percentage uncertainty in the calculated speed?

- A** $\pm 0.5\%$ **B** $\pm 1\%$ **C** $\pm 2\%$ **D** $\pm 3\%$

62 9702/13/O/N/12/Q6*

What will reduce the systematic errors when taking a measurement?

- A** adjusting the needle on a voltmeter so that it reads zero when there is no potential difference across it
B measuring the diameter of a wire at different points and taking the average
C reducing the parallax effects by using a marker and a mirror when measuring the amplitude of oscillation of a pendulum
D timing 20 oscillations, rather than a single oscillation, when finding the period of a pendulum

63 9702/13/O/N/12/Q7

In an experiment to determine the acceleration of free fall g , the time t taken for a ball to fall through distance s was measured. The uncertainty in the measurement of s is estimated to be 2%. The uncertainty in the measurement of t is estimated to be 3%.

The value of g is determined using the equation $g = \frac{2s}{t^2}$.

What is the uncertainty in the calculated value of g ?

- A** 1% **B** 5% **C** 8% **D** 11%

64 9702/11/M/J/13/Q5

In an experiment to determine the acceleration of free fall g , the period of oscillation T and length l of a simple pendulum were measured. The uncertainty in the measurement of l is estimated to be 4%, and the uncertainty in the measurement of T is estimated to be 1%.

The value of g is determined using the formula $g = \frac{4\pi^2 l}{T^2}$.

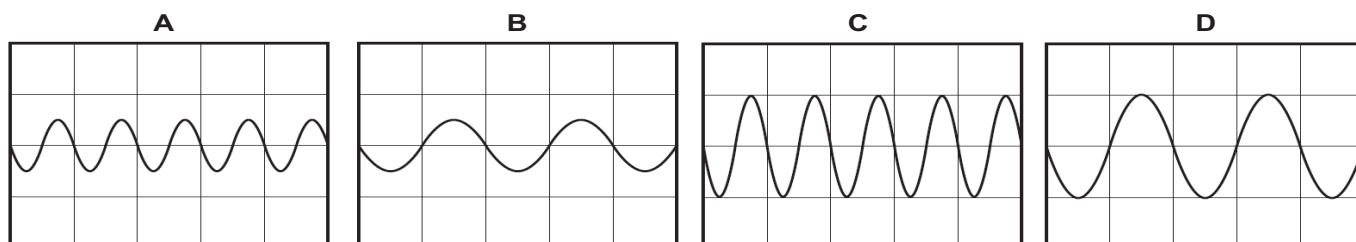
What is the uncertainty in the calculated value for g ?

- A** 2% **B** 3% **C** 5% **D** 6%

65 9702/11/M/J/13/Q6

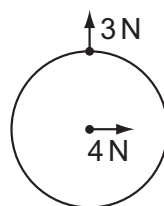
The Y-input terminals of a cathode-ray oscilloscope (c.r.o.) are connected to a supply of amplitude 5.0 V and frequency 50 Hz. The time-base is set at 10 ms per division and the Y-gain at 5.0 V per division.

Which trace is obtained?

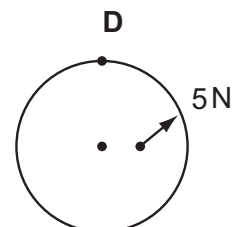
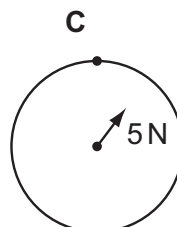
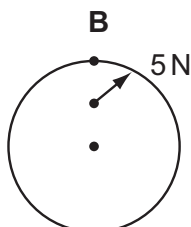
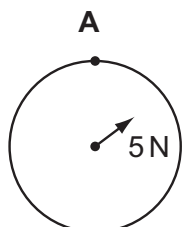


66 9702/12/M/J/13/Q3

Two forces act on a circular disc as shown.



Which diagram shows the line of action of the resultant force?



67 9702/12/M/J/13/Q4

A student carried out an experiment in which an electric current was known to decrease with time. The readings he found, from first to last, were 3.62 mA, 2.81 mA, 1.13 mA, 1.76 mA and 0.90 mA.

Which statement could **not** explain the anomalous 1.13 mA reading?

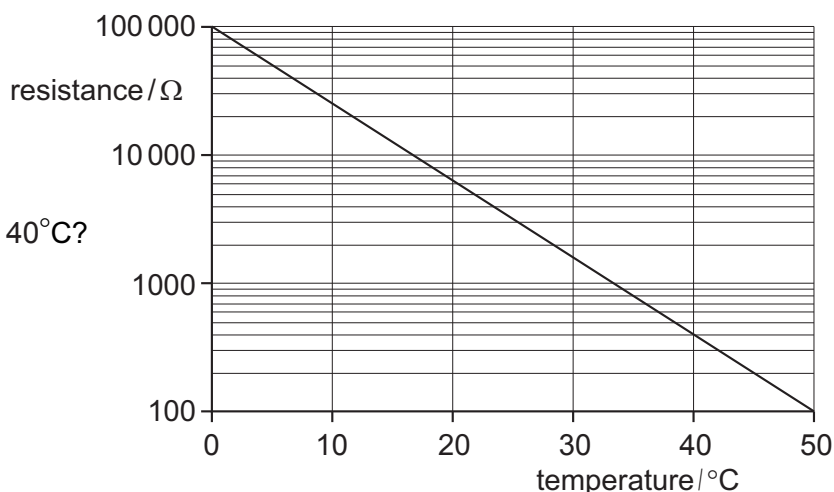
- A** He has reversed the third and fourth readings in the results table.
- B** He read the ammeter incorrectly; the reading should have been 2.13 mA.
- C** He took the current reading at the wrong time.
- D** There was a systematic error in the readings from the ammeter.

68 9702/12/M/J/13/Q5

The diagram shows a calibration curve for a thermistor, drawn with an unusual scale on the vertical axis.

What is the thermistor resistance corresponding to a temperature of 40°C?

- A** 130 Ω **B** 150 Ω
- C** 400 Ω **D** 940 Ω



69 9702/13/M/J/13/Q3

Which statement is **incorrect** by a factor of 100 or more?

- A** Atmospheric pressure is about 1×10^5 Pa.
- B** Light takes 5×10^2 s to reach us from the Sun.
- C** The frequency of ultra-violet light is 3×10^{12} Hz.
- D** The life-span of a man is about 2×10^9 s.

70 9702/13/M/J/13/Q5

A student takes measurements of the current in a resistor of constant resistance and the potential difference (p.d.) across it. The readings are then used to plot a graph of current against p.d.

There is a systematic error in the current readings.

How could this be identified from the graph?

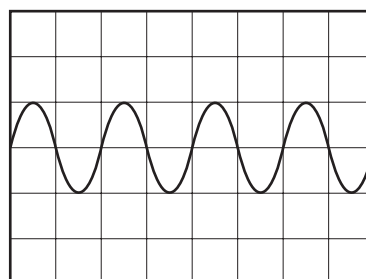
- A At least one anomalous data point can be identified.
- B The data points are scattered about the straight line of best fit.
- C The graph is a curve, not a straight line.
- D The straight line graph does not pass through the origin.

71 9702/11/M/J/15/Q4

A whale produces sound waves of frequency 5 Hz. The waves are detected by a microphone and displayed on an oscilloscope.

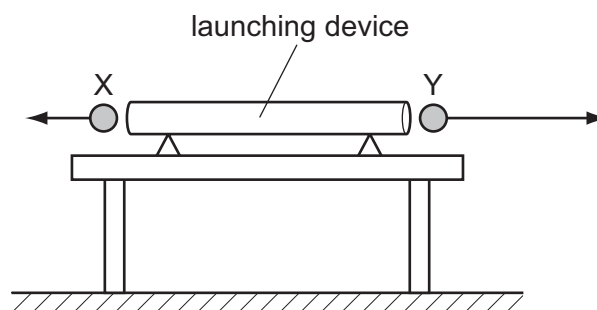
What is the time-base setting on the oscilloscope?

- A 0.1 ms div^{-1}
- B 1 ms div^{-1}
- C 10 ms div^{-1}
- D 100 ms div^{-1}



72 9702/13/M/J/13/Q7

A double-ended launching device fires two identical steel balls X and Y at exactly the same time. The diagram shows the initial velocities of the balls. They are both launched horizontally, but Y has greater speed.



Which statement explains what an observer would see?

- A Both X and Y reach the ground simultaneously, because air resistance will cause both to have the same final speed.
- B Both X and Y reach the ground simultaneously, because gravitational acceleration is the same for both.
- C X reaches the ground before Y, because X lands nearer to the launcher.
- D Y reaches the ground before X, because Y has greater initial speed.

73 9702/11/M/J/14/Q4

An experiment is carried out to measure the resistance of a wire.

The current in the wire is $(1.0 \pm 0.2) \text{ A}$ and the potential difference across the wire is $(8.0 \pm 0.4) \text{ V}$.

What is the resistance of the wire and its uncertainty?

- A $(8.0 \pm 0.2) \Omega$
- B $(8.0 \pm 0.6) \Omega$
- C $(8 \pm 1) \Omega$
- D $(8 \pm 2) \Omega$

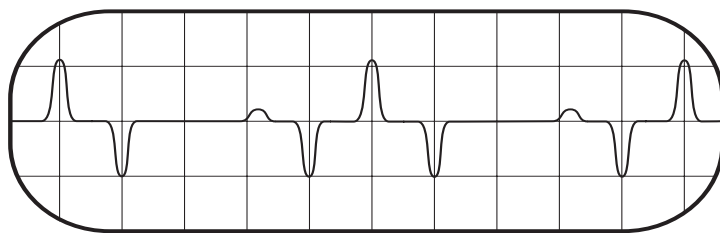
74 **9702/12/O/N/13/Q4**

A signal that repeats periodically is displayed on the screen of a cathode-ray oscilloscope.

The screen has 1 cm squares and the time base is set at 2.00 ms cm^{-1} .

What is the frequency of this periodic signal?

- A 50 Hz B 100 Hz
C 125 Hz D 200 Hz



75 **9702/12/O/N/13/Q5**

A micrometer screw gauge is used to measure the diameter of a small uniform steel sphere. The micrometer reading is $5.00 \text{ mm} \pm 0.01 \text{ mm}$.

What will be the percentage uncertainty in a calculation of the volume of the sphere, using these values?

- A 0.2% B 0.4% C 0.6% D 1.2%

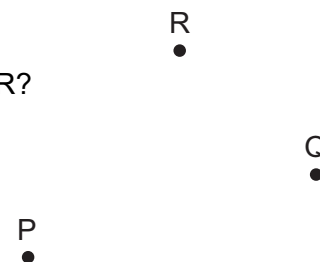
76 **9702/12/O/N/13/Q6**

One object moves directly from P to R.

In a shorter time, a second object moves from P to Q to R.

Which statement about the two objects is correct for the journey from P to R?

- A They have the same average speed.
B They have the same average velocity.
C They have the same displacement.
D They travel the same distance.



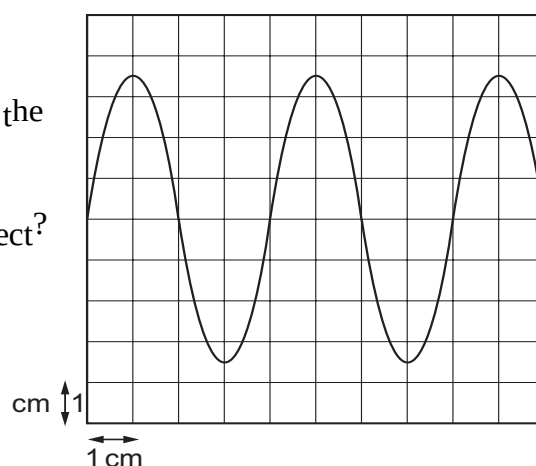
77 **9702/12/M/J/14/Q3***

A cathode-ray oscilloscope (c.r.o.) is connected to an alternating voltage. The following trace is produced on the screen.

The oscilloscope time-base setting is 0.5 ms cm^{-1} and the Y-plate sensitivity is 2 V cm^{-1} .

Which statement about the alternating voltage is correct?

- A The amplitude is 3.5 cm.
B The frequency is 0.5 kHz.
C The period is 1 ms.
D The wavelength is 4 cm.



78 **9702/13/O/N/13/Q6**

A student wishes to determine the density ρ of lead. She measures the mass and diameter of a small sphere of lead:

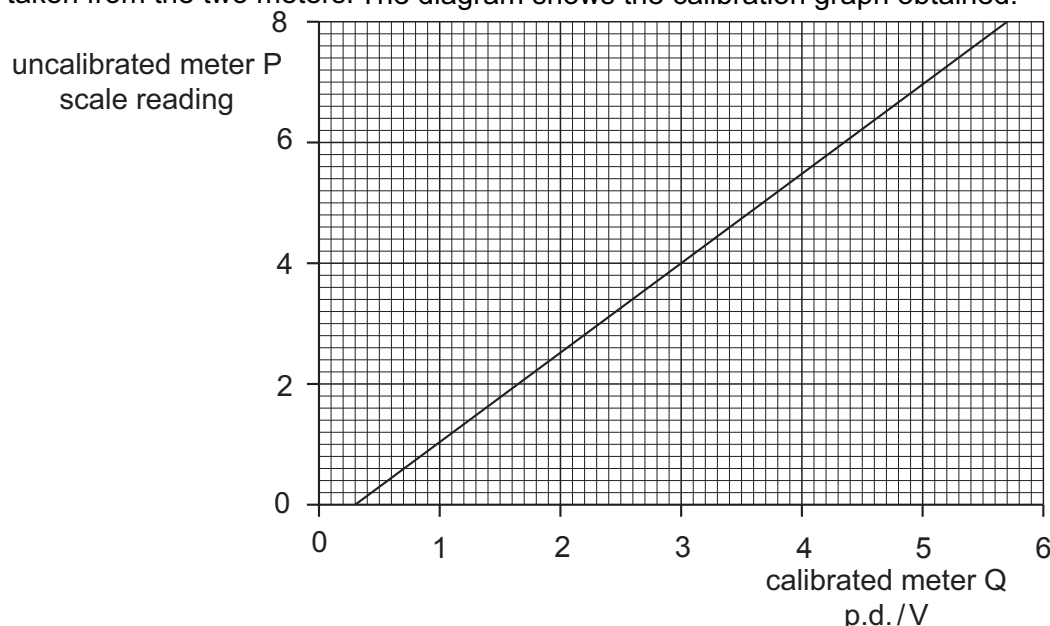
$$\text{mass} = (0.506 \pm 0.005) \text{ g} \qquad \text{diameter} = (2.20 \pm 0.02) \text{ mm}.$$

What is the best estimate of the percentage uncertainty in her value of ρ ?

- A 1.9% B 2.0% C 2.8% D 3.7%

79 9702/13/O/N/13/Q5

An uncalibrated analogue voltmeter P is connected in parallel with another voltmeter Q which is known to be accurately calibrated. For a range of values of potential difference (p.d.), readings are taken from the two meters. The diagram shows the calibration graph obtained.



The graph shows that meter P has a zero error. This meter is now adjusted to remove this zero error. When the meter is recalibrated, the gradient of the calibration graph is found to be unchanged.

What is the new scale reading on meter P, when it is used to measure a p.d. of 5.0 V?

The graph shows that meter P has a zero error. This meter is now adjusted to remove this zero error. When the meter is recalibrated, the gradient of the calibration graph is found to be unchanged.

What is the new scale reading on meter P when it is used to measure a p.d. of 5.0 V?

- A** 6.6 **B** 6.7 **C** 7.2 **D** 7.4

80 9702/11/M/J/14/Q5

The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation below.

$$E = \frac{4Fl}{\pi d^2 x}$$

The wire is extended by a known force and the following measurements are made.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	measurement	symbol	value
A	length of wire before force applied	l	2.043 ± 0.002 m
B	diameter of wire	d	0.54 ± 0.02 mm
C	force applied	F	19.62 ± 0.01 N
D	extension of wire with force applied	x	5.2 ± 0.2 mm

81 9702/12/M/J/14/Q4

A quantity y is to be determined from the equation shown. $y = \frac{px}{q^2}$

The percentage uncertainties in p , x and q are shown.

What is the percentage uncertainty in y ?

- A** 0.5 % **B** 1 %
C 16 % **D** 192 %

	percentage uncertainty
p	6%
x	2%
q	4%

82 9702/12/M/J/14/Q5

A thermometer can be read to an accuracy of $\pm 0.5^\circ\text{C}$. This thermometer is used to measure a temperature rise from 40°C to 100°C .

What is the percentage uncertainty in the measurement of the temperature rise?

- A** 0.5 % **B** 0.8 % **C** 1.3 % **D** 1.7 %

83 9702/13/M/J/14/Q4*

The resistance of a lamp is calculated from the value of the potential difference (p.d.) across it and the value of the current passing through it.

Which statement correctly describes how to combine the uncertainties in the p.d. and in the current?

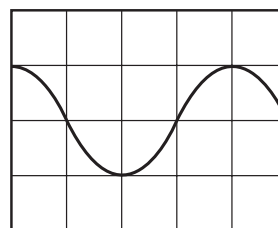
- A** Add together the actual uncertainty in the p.d. and the actual uncertainty in the current.
B Add together the percentage uncertainty in the p.d. and the percentage uncertainty in the current.
C Subtract the actual uncertainty in the current from the actual uncertainty in the p.d.
D Subtract the percentage uncertainty in the current from the percentage uncertainty in the p.d.

84 9702/13/M/J/14/Q5*

The display on a cathode-ray oscilloscope shows the signal produced by an electronic circuit. The time-base is set at 5.0 ns per division and the Y-gain at 10 V per division.

What is the frequency of the signal?

- A** $2.0 \times 10^{-8}\text{ Hz}$ **B** $2.5 \times 10^{-2}\text{ Hz}$
C $5.0 \times 10^7\text{ Hz}$ **D** $3.1 \times 10^8\text{ Hz}$



85 9702/13/M/J/14/Q4

The resistance of a lamp is calculated from the value of the potential difference (p.d.) across it and the value of the current passing through it.

Which statement correctly describes how to combine the uncertainties in the p.d. and in the current?

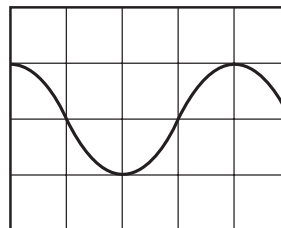
- A** Add together the actual uncertainty in the p.d. and the actual uncertainty in the current.
B Add together the percentage uncertainty in the p.d. and the percentage uncertainty in the current.
C Subtract the actual uncertainty in the current from the actual uncertainty in the p.d.
D Subtract the percentage uncertainty in the current from the percentage uncertainty in the p.d.

86 9702/13/M/J/14/Q5

The display on a cathode-ray oscilloscope shows the signal produced by an electronic circuit. The time-base is set at 5.0 ns per division and the Y-gain at 10 V per division.

What is the frequency of the signal?

- A 2.0×10^{-8} Hz C 5.0×10^7 Hz
B 2.5×10^{-2} Hz D 3.1×10^8 Hz



87 9702/13/M/J/14/Q6

A digital caliper is used to measure the 28.50 mm width of a plastic ruler. The digital caliper reads to the nearest 0.01 mm.

What is the correct way to record this reading?

- A 0.02850 ± 0.01 m C $(2.850 \pm 0.001) \times 10^{-2}$ m
B 0.0285 ± 0.001 m D $(2.850 \pm 0.001) \times 10^{-3}$ m

88 9702/13/M/J/14/Q7

An experiment is performed to measure the acceleration of free fall g . A body falls between two fixed points. The four measurements shown below are taken.

Which measurement is **not** required for the calculation of g ?

- A the distance fallen by the body C the mass of the body
B the initial velocity of the body D the time taken for the body to fall

89 9702/13/M/J/14/Q8

In a cathode-ray tube, an electron is accelerated uniformly in a straight line from a speed of $4 \times 10^3 \text{ ms}^{-1}$ to $2 \times 10^7 \text{ ms}^{-1}$ over a distance of 10 mm.

What is the acceleration of the electron?

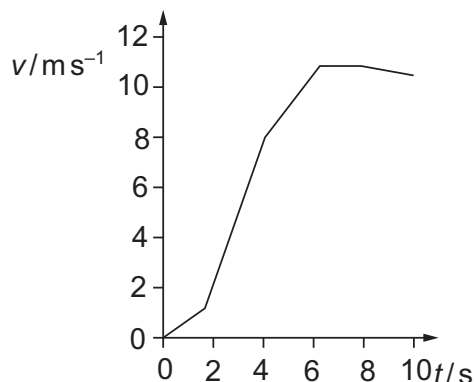
- A $2 \times 10^3 \text{ ms}^{-2}$ B $2 \times 10^6 \text{ ms}^{-2}$ C $2 \times 10^{13} \text{ ms}^{-2}$ D $2 \times 10^{16} \text{ ms}^{-2}$

90 9702/13/M/J/14/Q9

The graph shows how the speed v of a sprinter changes with time t during a 100 m race.

What is the best estimate of the maximum acceleration of the sprinter?

- A 0.5 ms^{-2} B 1 ms^{-2}
C 3 ms^{-2} D 10 ms^{-2}



91 9702/11/O/N/14/Q4

A steel wire is stretched in an experiment to determine the Young modulus for steel.
The uncertainties in the measurements are given below.

What is the percentage uncertainty in the Young modulus?

- A 1.3% B 1.8%
C 4.7% D 6.2%

measurement	uncertainty
load on wire	$\pm 2\%$
length of wire	$\pm 0.2\%$
diameter of wire	$\pm 1.5\%$
extension	$\pm 1\%$

92 9702/12/O/N/14/Q5

The acceleration of free fall on the Moon is one-sixth of that on Earth.
On Earth it takes time t for a stone to fall from rest a distance of 2 m.
What is the time taken for a stone to fall from rest a distance of 2 m on the Moon?

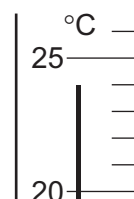
- A $6t$ B $\frac{t}{6}$ C $t\sqrt{6}$ D $\frac{t}{\sqrt{6}}$

93 9702/13/O/N/14/Q4

The diagram shows part of a thermometer.

What is the correct reading on the thermometer and the uncertainty in this reading?

	reading / °C	uncertainty in reading / °C
A	24	± 1
B	24	± 0.5
C	24	± 0.2
D	24.0	± 0.5



94 9702/13/O/N/14/Q5

The resistance R of a resistor is to be determined. The current I in the resistor and the potential difference V across it are measured.

The results, with their uncertainties, are

$$I = (2.0 \pm 0.2) \text{ A} \qquad V = (15.0 \pm 0.5) \text{ V.}$$

The value of R is calculated to be 7.5Ω .

What is the uncertainty in this value for R ?

- A $\pm 0.3 \Omega$ B $\pm 0.5 \Omega$ C $\pm 0.7 \Omega$ D $\pm 1 \Omega$

95 9702/13/O/N/14/Q6

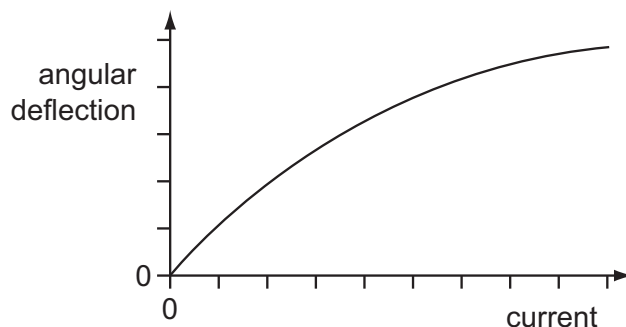
In an experiment to determine the acceleration of free fall g , a ball-bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball-bearing falls. After falling a distance h , the ball-bearing strikes a switch to stop the clock which measures the time t of the fall.

Which expression can be used to calculate the value of g ?

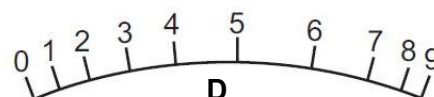
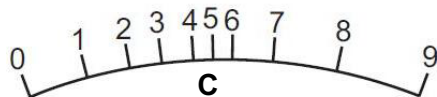
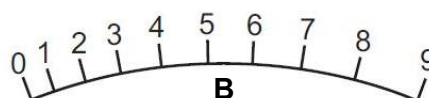
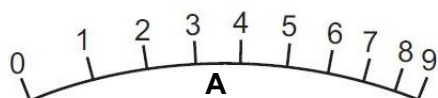
- A $\frac{ht^2}{2}$ B $\frac{th^2}{2}$ C $\sqrt{\frac{2t}{h^2}}$ D $\frac{2h}{t^2}$

96 9702/11/M/J/15/Q5

The angular deflection of the needle of an ammeter varies with the current in the ammeter as shown in the graph.



Which diagram could represent the appearance of the scale on this meter?



97 9702/11/M/J/15/Q6

The strain energy W of a spring is determined from its spring constant k and extension x . The spring obeys Hooke's law and the value of W is calculated using the equation shown. $W = \frac{1}{2} kx^2$

The spring constant is $100 \pm 2 \text{ N m}^{-1}$ and the extension is $0.050 \pm 0.002 \text{ m}$.

What is the percentage uncertainty in the calculated value of W ?

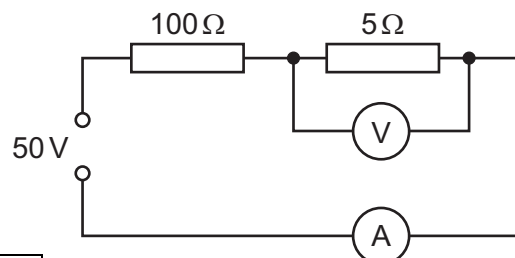
- A 6% B 10% C 16% D 32%

98 9702/12/M/J/15/Q5

A power supply of electromotive force (e.m.f.) 50 V and negligible internal resistance is connected in series with resistors of resistance 100Ω and 5Ω , as shown.

A voltmeter measures the potential difference (p.d.) across the 5Ω resistor and an ammeter measures the current in the circuit.

What are suitable ranges for the ammeter and for the voltmeter?



	ammeter range / A	voltmeter range / V
A	0–0.1	0–1
B	0–0.1	0–3
C	0–1.0	0–1
D	0–1.0	0–3

99 9702/12/M/J/15/Q6

A single sheet of aluminium foil is folded twice to produce a stack of four sheets. The total thickness of the stack of sheets is measured to be (0.80 ± 0.02) mm. This measurement is made using a digital caliper with a zero error of (-0.20 ± 0.02) mm.

What is the percentage uncertainty in the calculated thickness of a single sheet?

- A 1.0% B 2.0% C 4.0% D 6.7%

100 9702/12/M/J/15/Q7

In an experiment to determine the acceleration of free fall g , a ball bearing is held by an electromagnet. When the current to the electromagnet is switched off, a clock starts and the ball bearing falls. After falling a distance h , the ball bearing strikes a switch to stop the clock which measures the time t of the fall.

If systematic errors cause t and h to be measured incorrectly, which error **must** cause g to appear greater than 9.81 ms^{-2} ?

- A h measured as being **smaller** than it actually is and t is measured correctly
 B h measured as being **smaller** than it actually is and t measured as being **larger** than it actually is
 C h measured as being **larger** than it actually is and t measured as being **larger** than it actually is
 D h is measured correctly and t measured as being **smaller** than it actually is

101 9702/13/M/J/15/Q4

A student is given a reel of wire of diameter less than 0.2 mm and is asked to find the density of the metal.

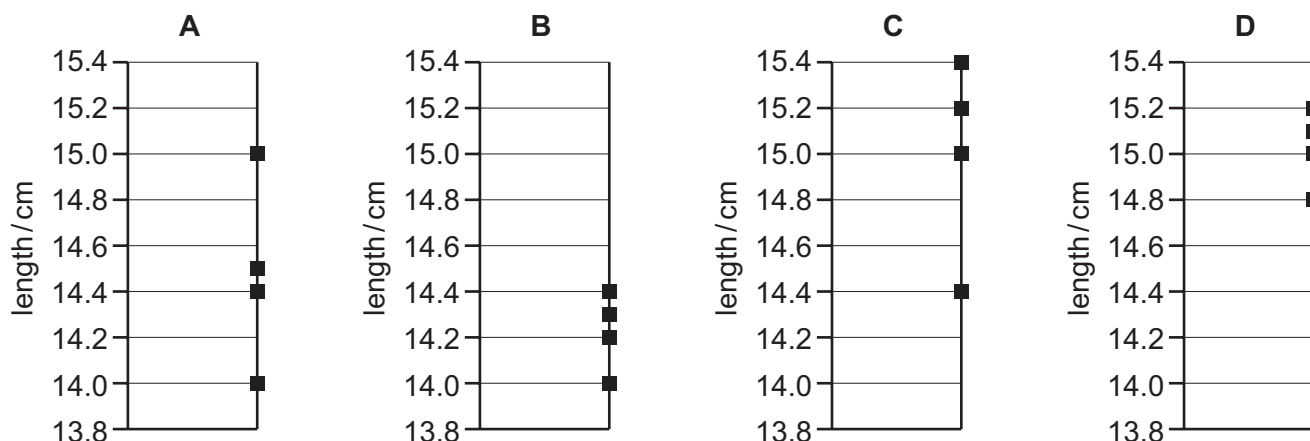
Which pair of instruments would be most suitable for finding the **volume** of the wire?

- A balance and micrometer C metre rule and vernier calipers
 B metre rule and micrometer D micrometer and vernier calipers

102 9702/13/M/J/15/Q5

Four different students use a ruler to measure the length of a 15.0 cm pencil. Their measurements are recorded on four different charts.

Which chart shows measurements that are precise but **not** accurate?



103 9702/11/M/J/17/Q3

The speed v of a liquid leaving a tube depends on the change in pressure ΔP and the density ρ of the liquid. The speed is given by the equation

$$v = k \left(\frac{\Delta P}{\rho} \right)^n$$

where k is a constant that has no units.

What is the value of n ?

- A** $\frac{1}{2}$ **B** 1 **C** $\frac{3}{2}$ **D** 2

104 9702/11/M/J/17/Q9

A student attempts to find the density ρ of aluminium by taking measurements of a rectangular sheet.

$$\text{mass } m = 51.6 \pm 0.1 \text{ g}$$

$$\text{length } l = 100.0 \pm 0.1 \text{ cm}$$

$$\text{width } w = 10.0 \pm 0.1 \text{ cm}$$

$$\text{thickness } t = 0.20 \pm 0.01 \text{ mm}$$

He uses the equation $\rho = \frac{m}{wlt}$ to calculate the density.

What is the calculated value of density with its uncertainty?

- A** $0.26 \pm 0.01 \text{ g cm}^{-3}$ **C** $2.6 \pm 0.1 \text{ g cm}^{-3}$
B $0.26 \pm 0.02 \text{ g cm}^{-3}$ **D** $2.6 \pm 0.2 \text{ g cm}^{-3}$

105 9702/11/M/J/17/Q1

A student creates a table to show reasonable estimates of some physical quantities.

Which row is **not** a reasonable estimate?

	quantity	value
A	current in a fan heater	12 A
B	mass of an adult person	70 kg
C	speed of an Olympic sprint runner	10 m s^{-1}
D	water pressure at the bottom of a garden pond	10^6 Pa

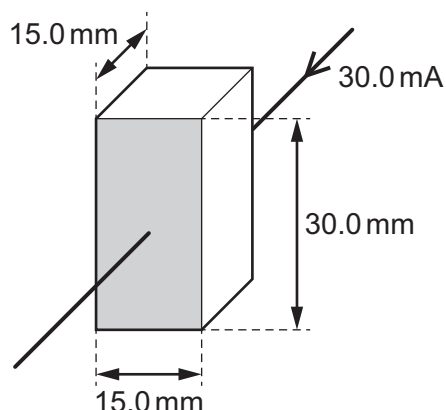
106 9702/12/M/J/17/Q1

What is the approximate average speed of a winning female Olympic athlete running a 100 m race?

- A** 6 m s^{-1} **B** 9 m s^{-1} **C** 12 m s^{-1} **D** 15 m s^{-1}

107 9702/12/M/J/17/Q4

The current in a block of semiconductor is 30.0 mA when there is a potential difference (p.d.) of 10.0 V across it. The dimensions of the block and the direction of the current in it are as shown.



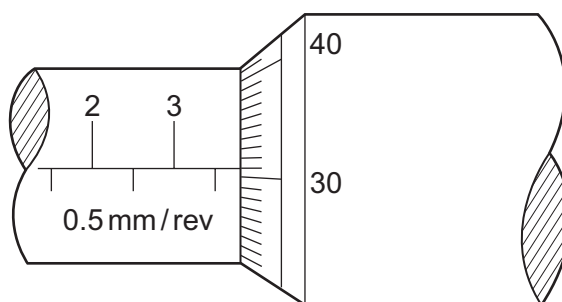
The electrical meters used are accurate to ± 0.1 mA and ± 0.1 V. The dimensions of the block are accurate to ± 0.2 mm.

What is the resistivity of the semiconductor?

- A** $10.0 \pm 0.2 \Omega \text{ m}$ **C** $10.0 \pm 0.5 \Omega \text{ m}$
B $10.0 \pm 0.3 \Omega \text{ m}$ **D** $10.0 \pm 0.8 \Omega \text{ m}$

108 9702/12/M/J/17/Q5

The diameter of a cylindrical metal rod is measured using a micrometer screw gauge. The diagram below shows an enlargement of the scale on the micrometer screw gauge when taking the measurement.



What is the cross-sectional area of the rod?

- A** 3.81 mm^2 **B** 11.4 mm^2 **C** 22.8 mm^2 **D** 45.6 mm^2

109 9702/11/O/N/17/Q4

What is a typical value of the wavelength of a microwave travelling in a vacuum?

- A** 3 000 000 pm **B** 30 nm **C** 30 000 μm **D** 3000 mm

110 9702/11/O/N/17/Q5

A double-slit interference experiment is used to determine the wavelength of light from a monochromatic source.

The following measurements are used.

$$\text{slit separation } a = 0.50 \pm 0.02 \text{ mm}$$

$$\text{fringe separation } x = 1.7 \pm 0.1 \text{ mm}$$

$$\text{distance between slits and screen } D = 2.000 \pm 0.002 \text{ m}$$

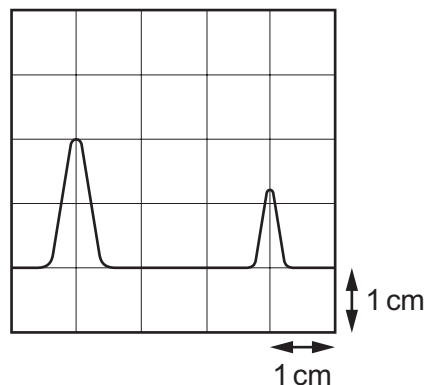
What is the percentage uncertainty in the calculated wavelength?

- A** 0.1% **B** 1% **C** 6% **D** 10%

111 9702/12/O/N/17/Q5

A transmitter emits a pulse of electromagnetic waves towards a reflector. The pulse is reflected and returns to the transmitter.

A detector is located at the transmitter. The emitted pulse and the reflected pulse are displayed on a cathode-ray oscilloscope (c.r.o.) as shown.



The pulse takes $6.3 \mu\text{s}$ to travel from the transmitter to the reflector.

What is the time-base setting of the c.r.o.?

- A** $2.1 \mu\text{s cm}^{-1}$ **B** $3.2 \mu\text{s cm}^{-1}$ **C** $4.2 \mu\text{s cm}^{-1}$ **D** $6.3 \mu\text{s cm}^{-1}$

112 9702/12/O/N/17/Q6

A hot-air balloon is moving vertically upwards with a constant speed of 3.00 m s^{-1} . A sandbag is dropped from the balloon. It takes 5.00 s for the sandbag to fall to the ground.

What was the height of the balloon when the sandbag was released?

- A** 29 m **B** 108 m **C** 123 m **D** 138 m

113 9702/13/O/N/17/Q4

A school has a piece of aluminium that it uses for radioactivity experiments. Its thickness is marked as 3.2 mm. A student decides to check this value. He has vernier calipers which give measurements to 0.1 mm and a micrometer which gives measurements to 0.01 mm.

Which statement **must** be correct?

- A** The micrometer gives a more accurate measurement.
- B** The micrometer gives a more precise measurement.
- C** The vernier calipers give a more accurate measurement.
- D** The vernier calipers give a more precise measurement.

114 9702/13/O/N/17/Q5

Four possible sources of error in a series of measurements are listed.

- 1 an analogue meter whose scale is read from different angles
- 2 a meter which always measures 5% too high
- 3 a meter with a needle that is not frictionless, so the needle sometimes sticks slightly
- 4 a meter with a zero error

Which errors are random and which are systematic?

	random error	systematic error
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	3 and 4	1 and 2

115 9702/13/O/N/17/Q24

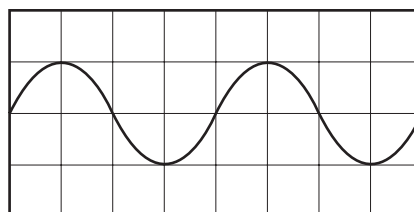
An electrical signal is displayed on a cathode-ray oscilloscope (c.r.o.).

The time-base setting is 50 ms div^{-1} .

The Y-gain setting is 2 V div^{-1} .

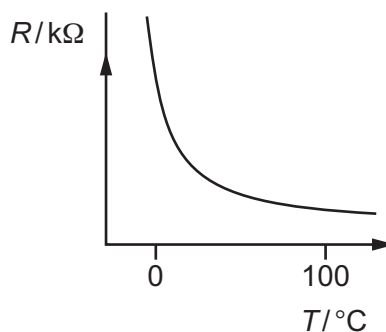
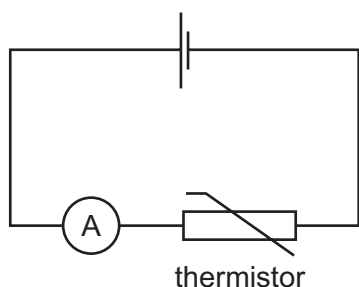
What is the amplitude of the signal?

- A** 2 V
- B** 4 V
- C** 5 V
- D** 10 V

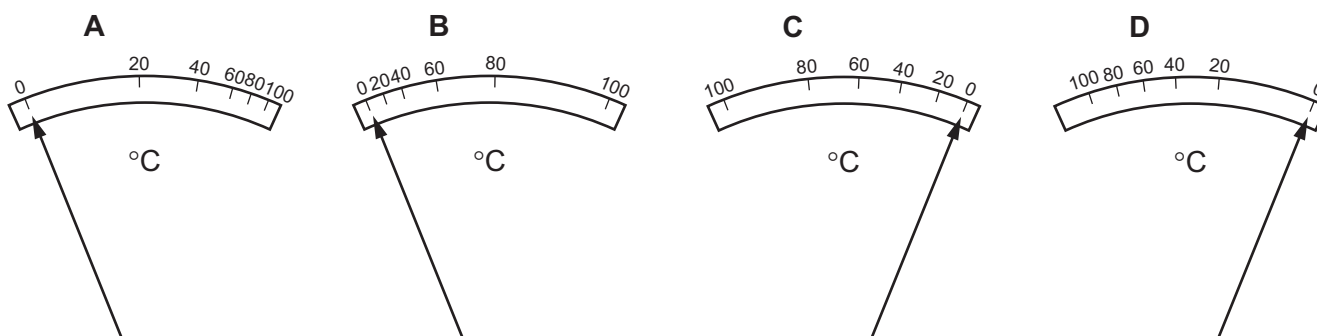


116 9702/11/M/J/18/Q4

In the circuit shown, an analogue ammeter is to be recalibrated as a thermometer. The ammeter is connected in series with a thermistor. The thermistor is a component with a resistance that varies with temperature. The graph shows how the resistance R of the thermistor changes with temperature T .



Which diagram could represent the temperature scale on the ammeter?



117 9702/11/M/J/18/Q5

The sides of a cube are measured with calipers.

The measured length of each side is (30.0 ± 0.1) mm.

The measurements are used to calculate the volume of the cube.

What is the percentage uncertainty in the calculated value of the volume?

- A** 0.01% **B** 0.3% **C** 1% **D** 3%

118 9702/12/M/J/18/Q3

A student measures the current through a resistor and the potential difference (p.d.) across it. There is a 4% uncertainty in the current reading and a 1% uncertainty in the p.d. reading. The student calculates the resistance of the resistor.

What is the percentage uncertainty in the calculated resistance?

- A** 0.25% **B** 3% **C** 4% **D** 5%

119 9702/12/M/J/18/Q4

A student applies a potential difference V of (4.0 ± 0.1) V across a resistor of resistance R of $(10.0 \pm 0.3) \Omega$ for a time t of (50 ± 1) s.

The student calculates the energy E dissipated using the equation below.

$$E = \frac{V^2 t}{R} = \frac{4.0^2 \times 50}{10.0} = 80 \text{ J}$$

What is the absolute uncertainty in the calculated energy value?

- A** 1.5 J **B** 3 J **C** 6 J **D** 8 J

120 9702/13/M/J/18/Q4

What will reduce the systematic errors when taking a measurement?

- A** adjusting the needle on a voltmeter so that it reads zero when there is no potential difference across it
- B** measuring the diameter of a wire at different points and taking the average
- C** reducing the parallax effects by using a marker and a mirror when measuring the amplitude of oscillation of a pendulum
- D** timing 20 oscillations, rather than a single oscillation, when finding the period of a pendulum

121 9702/13/M/J/18/Q5

In an experiment to determine the Young modulus E of the material of a wire, the measurements taken are shown.

mass hung on end of wire	$m = 2.300 \pm 0.002 \text{ kg}$
original length of wire	$l = 2.864 \pm 0.005 \text{ m}$
diameter of wire	$d = 0.82 \pm 0.01 \text{ mm}$
extension of wire	$e = 7.6 \pm 0.2 \text{ mm}$

The Young modulus is calculated using $E = \frac{4mgl}{\pi d^2 e}$

where g is the acceleration of free fall.

The calculated value of E is $1.61 \times 10^{10} \text{ N m}^{-2}$.

How should the calculated value of E and its uncertainty be expressed?

- A** $(1.61 \pm 0.04) \times 10^{10} \text{ N m}^{-2}$
- B** $(1.61 \pm 0.05) \times 10^{10} \text{ N m}^{-2}$
- C** $(1.61 \pm 0.07) \times 10^{10} \text{ N m}^{-2}$
- D** $(1.61 \pm 0.09) \times 10^{10} \text{ N m}^{-2}$

122 9702/12/F/M/18/Q4

The density of paper is 800 kg m^{-3} . A typical sheet of paper has a width of 210 mm and a length of 300 mm.

The thickness of a pack of 500 sheets of paper is 50 mm.

What is the mass of a single sheet of paper?

- A** 0.5 g **B** 5 g **C** 50 g **D** 500 g

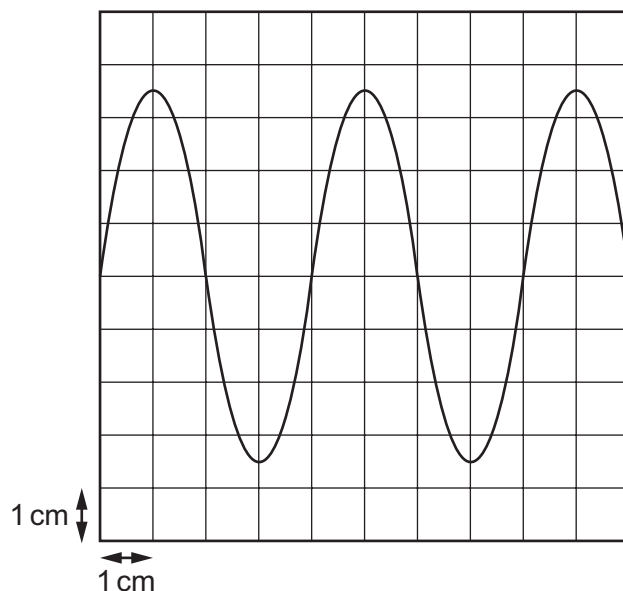
123 9702/12/F/M/18/Q5

A person calculates the potential difference across a wire by using the measurements shown. Which measured quantity has the greatest contribution to the percentage uncertainty in the calculated potential difference?

	quantity	value	uncertainty
A	current / A	5.0	± 0.5
B	diameter of wire / mm	0.8	± 0.1
C	length of wire / m	150	± 5
D	resistivity of metal in wire / Ωm	1.6×10^{-8}	$\pm 0.2 \times 10^{-8}$

124 9702/12/F/M/18/Q6

A cathode-ray oscilloscope (c.r.o.) is connected to an alternating voltage. The following trace is produced on the screen.



The oscilloscope time-base setting is 0.5 ms cm^{-1} and the Y-plate sensitivity is 2 V cm^{-1} .

Which statement about the alternating voltage is correct?

- | | |
|------------------------------------|----------------------------------|
| A The amplitude is 3.5 cm. | C The period is 1 ms. |
| B The frequency is 0.5 kHz. | D The wavelength is 4 cm. |

125 9702/11/O/N/18/Q4

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

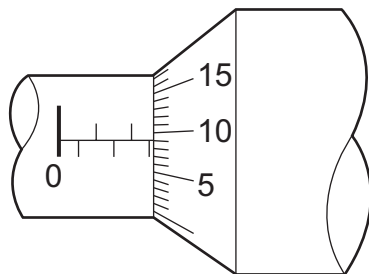


diagram 1

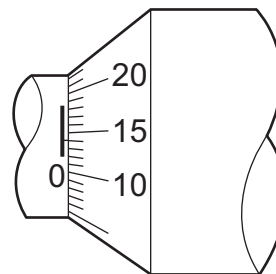


diagram 2

What is the diameter of the wire?

- A** 1.90 mm **B** 2.45 mm **C** 2.59 mm **D** 2.73 mm

126 9702/11/O/N/18/Q5

A digital meter has an accuracy of $\pm 1\%$.

The meter is used to measure the current in an electrical circuit.

The reading on the meter varies between 3.04 A and 3.08 A.

What is the value of the current, with its uncertainty?

- A** $(3.06 \pm 0.02) \text{ A}$ **C** $(3.06 \pm 0.05) \text{ A}$
B $(3.06 \pm 0.04) \text{ A}$ **D** $(3.06 \pm 0.07) \text{ A}$

127 9702/12/O/N/18/Q1

A car is travelling at a speed of 20 m s^{-1} . The table contains values for the kinetic energy and the momentum of the car.

Which values are reasonable estimates?

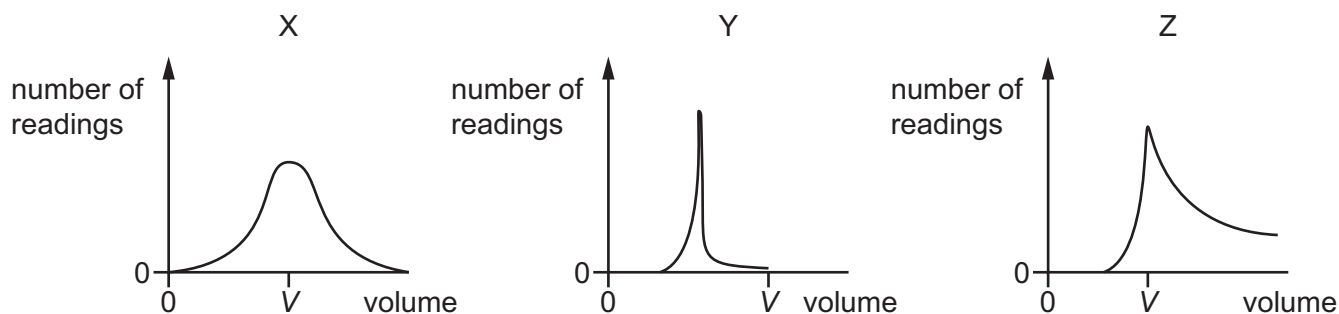
	kinetic energy / J	momentum / kg m s^{-1}
A	3×10^5	3×10^4
B	3×10^5	5×10^6
C	2×10^7	3×10^4
D	2×10^7	5×10^6

128 9702/12/O/N/18/Q5

Students take readings of the volume of a liquid using three different pieces of measuring equipment X, Y and Z.

The true value of the volume of the liquid is V .

The students' results are shown.



How many pieces of equipment are precise and how many are accurate?

	number of precise pieces of equipment	number of accurate pieces of equipment
A	1	1
B	1	2
C	2	1
D	2	2

129 9702/13/O/N/18/Q1

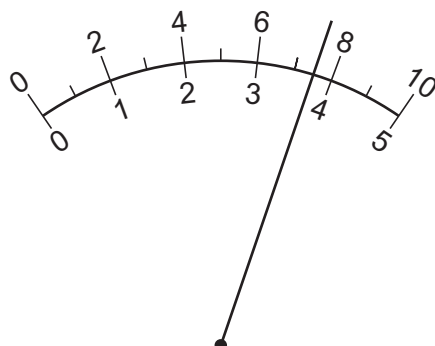
Which statement is **not** a reasonable estimate?

- A** Atmospheric pressure at sea level is about 1×10^5 Pa.
- B** Light takes 5×10^2 s to reach us from the Sun.
- C** The frequency of ultraviolet light is 3×10^{12} Hz.
- D** The lifespan of a man is about 2×10^9 s.

130 9702/13/O/N/18/Q4

An ammeter is calibrated so that it shows a full-scale deflection when it measures a current of 2.0 A.

The diagram shows the display of this ammeter when it is measuring a current.



Which current is the ammeter measuring?

- A** 0.75 A **B** 1.5 A **C** 3.8 A **D** 7.5 A

131 9702/13/O/N/18/Q5

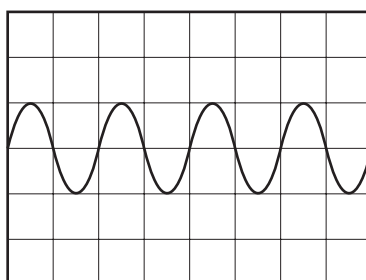
The width of a table is measured as (50.3 ± 0.1) cm. Its length is measured as (1.40 ± 0.01) m.

What is the area of the table and its absolute uncertainty?

- A** $(0.7 \pm 0.1) \text{ m}^2$ **C** $(0.704 \pm 0.011) \text{ m}^2$
B $(0.704 \pm 0.006) \text{ m}^2$ **D** $(70.4 \pm 0.6) \text{ m}^2$

132 9702/11/M/J/19/Q4

A whale produces sound waves of frequency 5 Hz. The waves are detected by a microphone and displayed on an oscilloscope.



What is the time-base setting on the oscilloscope?

- A** 0.1 ms div^{-1} **B** 1 ms div^{-1} **C** 10 ms div^{-1} **D** 100 ms div^{-1}

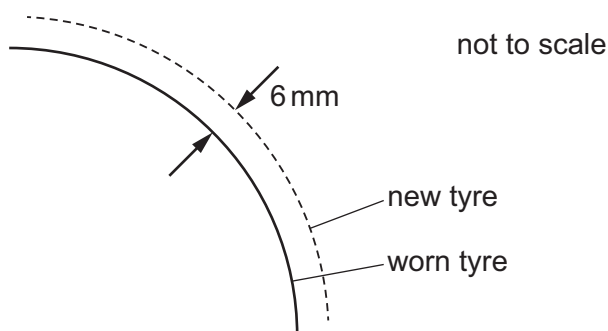
133 9702/11/M/J/19/Q5

The speed shown on a car's speedometer is proportional to the rate of rotation of the tyres.

The variation of the diameter of a tyre as it wears introduces an error in the speed shown on the speedometer.

A car has new tyres of diameter 600 mm. The speedometer is accurately calibrated for this diameter.

The tyres wear as shown, with 6 mm of material being removed from the outer surface.



What is the error in the speed shown on the speedometer after this wear has taken place?

- A** The speed shown is too high by 1%.
- B** The speed shown is too high by 2%.
- C** The speed shown is too low by 1%.
- D** The speed shown is too low by 2%.

134 9702/12/M/J/19/Q5

A student wishes to determine the density ρ of lead. She measures the mass and diameter of a small sphere of lead:

$$\text{mass} = (0.506 \pm 0.005) \text{ g}$$

$$\text{diameter} = (2.20 \pm 0.02) \text{ mm}.$$

What is the best estimate of the percentage uncertainty in her calculated value of ρ ?

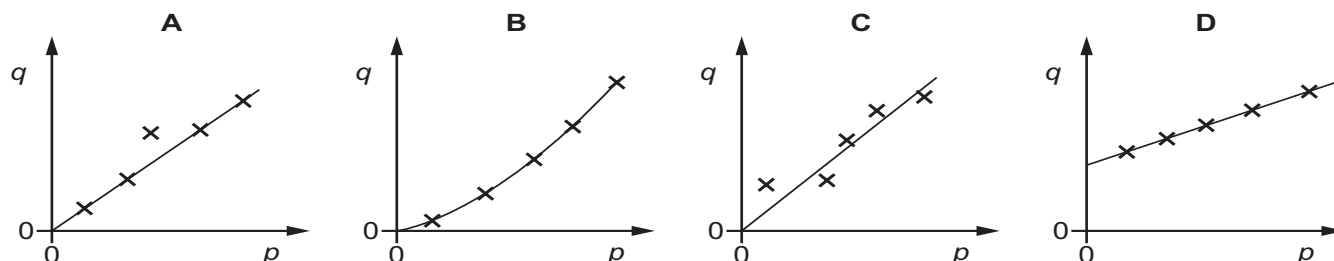
- A** 1.7%
- B** 1.9%
- C** 2.8%
- D** 3.7%

135 9702/12/M/J/19/Q6

Two quantities p and q are directly proportional to each other.

Experimental results are taken and plotted in a graph of q against p .

Which graph shows there were random errors in the measurements of p and q ?



136 9702/12/M/J/19/Q7

A man of mass 75.2 kg uses a set of weighing scales to measure his mass three times. He obtains the following readings.

	mass / kg
reading 1	80.2
reading 2	80.1
reading 3	80.2

Which statement best describes the precision and accuracy of the weighing scales?

- A** not precise to ± 0.1 kg and accurate to ± 0.1 kg
- B** not precise to ± 0.1 kg and not accurate to ± 0.1 kg
- C** precise to ± 0.1 kg and accurate to ± 0.1 kg
- D** precise to ± 0.1 kg and not accurate to ± 0.1 kg

137 9702/13/M/J/19/Q4

What is the approximate kinetic energy of an Olympic athlete when running at maximum speed during a 100 m race?

- A** 400 J
- B** 4000 J
- C** 40 000 J
- D** 400 000 J

138 9702/13/M/J/19/Q6

A micrometer screw gauge is used to measure the diameter of a small uniform steel sphere. The micrometer reading is $5.00 \text{ mm} \pm 0.01 \text{ mm}$.

What will be the percentage uncertainty in a calculation of the volume of the sphere, using these values?

- A** 0.2%
- B** 0.4%
- C** 0.6%
- D** 1.2%

139 9702/11/O/N/19/Q1

For which quantity is the magnitude a reasonable estimate?

- A** frequency of a radio wave 500 pHz
- B** mass of an atom 500 μg
- C** the Young modulus of a metal 500 kPa
- D** wavelength of green light 500 nm

140 9702/11/O/N/19/Q2

The speed of a wave in deep water depends on its wavelength L and the acceleration of free fall g .

What is a possible equation for the speed v of the wave?

- A** $v = \sqrt{\left(\frac{gL}{2\pi}\right)}$ **B** $v = \frac{gL}{4\pi^2}$ **C** $v = 2\pi\sqrt{\left(\frac{g}{L}\right)}$ **D** $v = \frac{2\pi g}{L}$

141 9702/11/O/N/19/Q4

A student intends to measure accurately the diameter of a wire (known to be approximately 1 mm) and the internal diameter of a pipe (known to be approximately 2 cm).

What are the most appropriate instruments for the student to use to make these measurements?

	wire	pipe
A	calipers	calipers
B	calipers	micrometer
C	micrometer	calipers
D	micrometer	micrometer

142 9702/11/O/N/19/Q5

The power P dissipated in a resistor of resistance R is calculated using the expression

$$P = \frac{V^2}{R}$$

where V is the potential difference (p.d.) across the resistor. The percentage uncertainty in V is 5% and in R is 2%.

What is the percentage uncertainty in P ?

- A** 3% **B** 7% **C** 8% **D** 12%

143 9702/12/O/N/19/Q1

A cyclist has a speed of 5 m s^{-1} and a small car has a speed of 12 m s^{-1} .

Which statement does **not** give a reasonable estimate?

- A** The kinetic energy of the cyclist is $1 \times 10^3 \text{ J}$.
- B** The kinetic energy of the car is $7 \times 10^4 \text{ J}$.
- C** The momentum of the cyclist is $4 \times 10^2 \text{ kg m s}^{-1}$.
- D** The momentum of the car is $2 \times 10^5 \text{ kg m s}^{-1}$.

144 9702/12/O/N/19/Q4

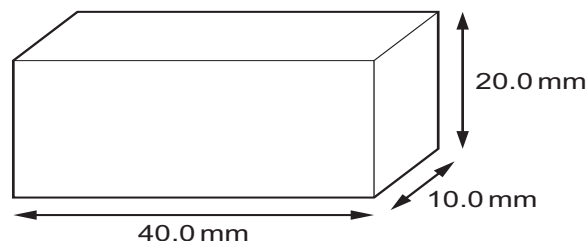
A micrometer is used to measure the 28.50 mm width of a plastic ruler. The micrometer reads to the nearest 0.01 mm.

What is the correct way to record this reading?

- | | |
|---------------------------------------|---|
| A $0.02850 \pm 0.01 \text{ m}$ | C $(2.850 \pm 0.001) \times 10^{-2} \text{ m}$ |
| B $0.0285 \pm 0.001 \text{ m}$ | D $(2.850 \pm 0.001) \times 10^{-3} \text{ m}$ |

145 9702/12/O/N/19/Q5

The sides of a wooden block are measured with calipers. The lengths of the sides are measured as 20.0 mm, 40.0 mm and 10.0 mm.



The calipers can measure with an absolute uncertainty of $\pm 0.1 \text{ mm}$.

What is the percentage uncertainty in the calculated volume of the block?

- | | | | |
|---------------|---------------|---------------|--------------|
| A 0.3% | B 1.8% | C 3.8% | D 30% |
|---------------|---------------|---------------|--------------|

146 9702/13/O/N/19/Q4

What could reduce systematic errors?

- A** averaging a large number of measurements
- B** careful calibration of measuring instruments
- C** reducing the sample size
- D** repeating measurements

147 9702/13/O/N/19/Q5

The power loss P in a resistor is calculated using the formula $P = \frac{V^2}{R}$.

The percentage uncertainty in the potential difference V is 3% and the percentage uncertainty in the resistance R is 2%.

What is the percentage uncertainty in P ?

- A** 4% **B** 7% **C** 8% **D** 11%

148. 9702/12/F/M/20/Q5

A micrometer is used to measure the diameters of two cylinders.

diameter of first cylinder = (12.78 ± 0.02) mm

diameter of second cylinder = (16.24 ± 0.03) mm

The difference in the diameters is calculated.

What is the uncertainty in this difference?

- A** 0.01 mm **B** 0.02 mm **C** 0.03 mm **D** 0.05 mm
-

149. 9702/11/M/J/20/Q5

A measurement is taken correctly but with a ruler at a significantly higher temperature than that at which the ruler was calibrated. The higher temperature causes the ruler to expand.

What describes the effect on the measurement caused by the higher temperature and how the measurement may be improved?

- A** The measurement will be subject to a random error. The measurement can be made more accurate by taking the average of several repeated measurements.
- B** The measurement will be subject to a random error. The measurement can be made more precise by taking the average of several repeated measurements.
- C** The measurement will be subject to a systematic error. The measurement can be made more accurate by taking the average of several repeated measurements.
- D** The measurement will be subject to a systematic error. The measurement can be made more precise by taking the average of several repeated measurements.

150. 9702/12/M/J/20/Q3

A galvanometer of resistance $5\ \Omega$ is to be used in a null method.

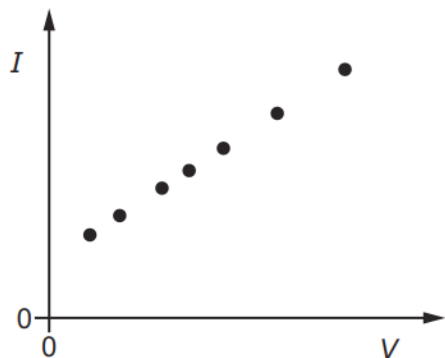
In order to protect the galvanometer from damage due to an excessive initial current, resistors of resistance $0.5\ \Omega$ and $1\ \text{k}\Omega$ are available.

Which arrangement would provide this protection?

- A** the $0.5\ \Omega$ resistor in series with the galvanometer
- B** the $0.5\ \Omega$ resistor in parallel with the galvanometer and this combination placed in series with the $1\ \text{k}\Omega$ resistor
- C** the $1\ \text{k}\Omega$ resistor in parallel with the galvanometer
- D** the $1\ \text{k}\Omega$ resistor in parallel with the galvanometer and this combination placed in series with the $0.5\ \Omega$ resistor

151. 9702/12/M/J/20/Q4

Readings are made of the current I for different voltages V across a fixed resistor. The results are plotted on a graph to show the variation of I with V .



What is the best description of the errors in the readings?

- A** both systematic and random
- B** neither systematic nor random
- C** random only
- D** systematic only

152. 9702/13/M/J/20/Q4

A circuit is set up in order to determine the resistance of a $12\ \text{V}$, $1.2\ \text{W}$ lamp when operating normally. An analogue ammeter and an analogue voltmeter are used.

Which ranges for the meters would be most suitable?

	ammeter range /A	voltmeter range /V
A	0–0.5	0–20
B	0–0.5	0–100
C	0–10	0–20
D	0–10	0–100

153. 9702/13/M/J/20/O5

Two liquid-in-glass thermometers in a well-mixed liquid are individually observed by 10 different students. All agree that one thermometer reads 21 °C and the other thermometer reads 23 °C.

What is a possible explanation for the difference?

- A The liquid is not all at the same temperature.
- B The readings are not precise.
- C There is a random error affecting the readings.
- D There is a systematic error affecting the readings.

154. 9702/11/O/N/20/Q5

The diameter of a spherical golf ball is measured with calipers and found to be (4.11 ± 0.01) cm.

The volume of a sphere is $V = \frac{1}{6}\pi d^3$, where d is the diameter of the sphere.

What is the volume of the golf ball?

- A $(36.35 \pm 0.01)\text{cm}^3$
- B $(36.35 \pm 0.03)\text{cm}^3$
- C $(36.35 \pm 0.09)\text{cm}^3$
- D $(36.4 \pm 0.3)\text{cm}^3$

155. 9702/12/O/N/20/Q1

A student uses the volume of a metal coin in order to determine the density of the metal.

What is **not** needed in order to determine an estimate of the volume of the coin?

- A estimate of the diameter
- B estimate of the mass
- C estimate of the thickness
- D use of the formula for the volume of a cylinder

156. 9702/12/O/N/20/Q4

A student wishes to measure a distance of about 10 cm to a precision of 0.01 cm.

Which measuring instrument should be used?

- A metre rule
- B micrometer
- C tape measure
- D vernier calipers

157. 9702/12/O/N/20/Q5

A steel ball is dropped and falls through a vertical height h . The time t taken to fall is measured using light gates.

The results are given in the table.

h	$(4.05 \pm 0.01)\text{m}$
t	$(0.91 \pm 0.02)\text{s}$

The acceleration of free fall g is calculated using the equation shown.

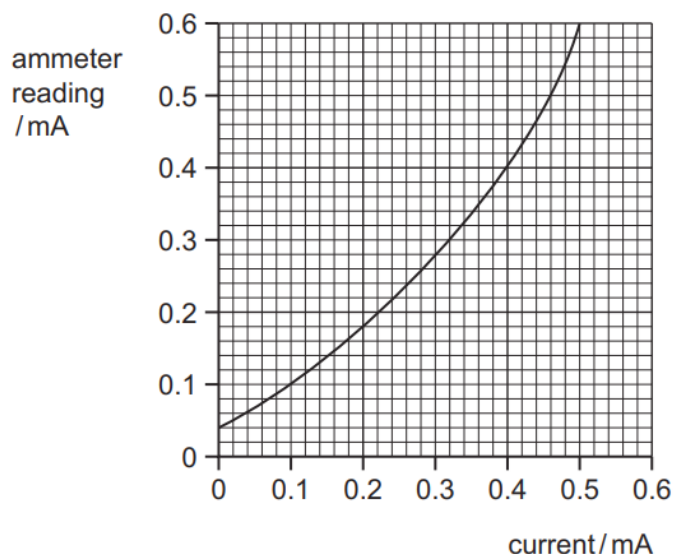
$$h = \frac{1}{2}gt^2$$

What is the percentage uncertainty in the value of g ?

- A** 2.4% **B** 4.6% **C** 5.1% **D** 9.3%

158. 9702/13/O/N/20/Q4

A calibration curve is shown for an ammeter whose scale is inaccurate.



Two readings taken on the meter at different times during an experiment are 0.13 mA and 0.47 mA.

By how much did the current really increase between taking the two readings?

- A** 0.30 mA **B** 0.34 mA **C** 0.40 mA **D** 0.44 mA

159. 9702/13/O/N/20/Q5

A student measures the length l and the period T of oscillation of a simple pendulum. He then uses the equation shown to calculate the acceleration of free fall g .

$$T = 2\pi \sqrt{\frac{l}{g}}$$

His measurements are shown.

l	$(87.3 \pm 0.2) \text{ cm}$
T	$(1.9 \pm 0.05) \text{ s}$

What is the percentage uncertainty in his calculated value of g ?

- A** 2.4% **B** 2.9% **C** 5.5% **D** 7.2%

160. 9702/12/F/M/21/Q4

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.

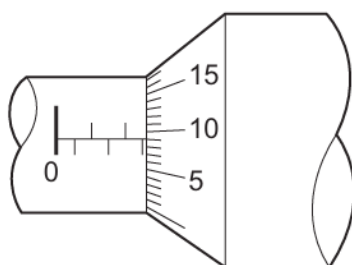


diagram 1

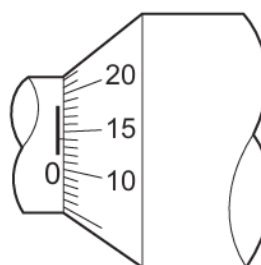


diagram 2

What is the diameter of the wire?

- A** 1.95 mm **B** 2.45 mm **C** 2.59 mm **D** 2.73 mm

161. 9702/12/F/M/21/Q5

A student measures the current and the potential difference for a resistor in a circuit.

$$\text{current} = (50.00 \pm 0.01) \text{ mA}$$

$$\text{potential difference} = (500.0 \pm 0.1) \text{ mV}$$

The measurements are used to calculate the resistance of the resistor.

What is the percentage uncertainty in the calculated resistance?

- A** 0.0002% **B** 0.0004% **C** 0.02% **D** 0.04%

162. 9702/11/M/J/21/Q5

A micrometer screw gauge is used to measure the diameter of a wire.

The reading on the micrometer with the jaws closed is (-0.05 ± 0.02) mm.

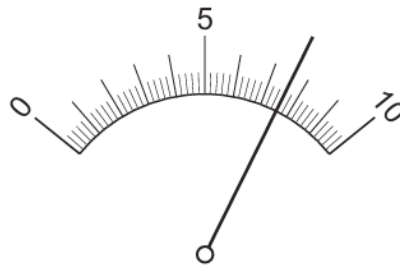
The reading with the wire in position between the two jaws is $(+1.03 \pm 0.02)$ mm.

What is the diameter of the wire?

- A** (0.98 ± 0.02) mm
- B** (1.08 ± 0.02) mm
- C** (0.98 ± 0.04) mm
- D** (1.08 ± 0.04) mm

163. 9702/12/M/J/21/Q4

An analogue ammeter with a range of 0–250 mA is connected into an electrical circuit. The diagram shows the ammeter's display.

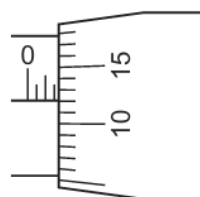


What is the reading on the ammeter?

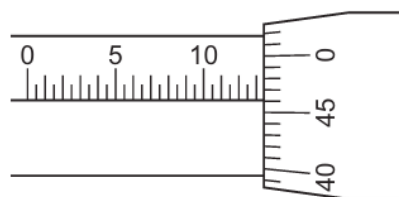
- A** 76 mA
- B** 165 mA
- C** 183 mA
- D** 190 mA

164. 9702/13/M/J/21/Q4

The diagram shows two readings on a micrometer.



reading 1



reading 2

What is the difference between the two readings?

- A** 10.34 mm
- B** 11.84 mm
- C** 12.34 mm
- D** 12.84 mm

165. 9702/13/M/J/21/Q5

The diameter of a circular disc is measured as (7.0 ± 0.1) mm.

What is the area of the disc and the absolute uncertainty in the area?

	area of disc / mm ²	absolute uncertainty / mm ²
A	38.5	± 0.5
B	38	± 1
C	154	± 2
D	154	± 4

166. 9702/11/O/N/21/Q5

Four possible sources of error in a series of measurements are listed.

- 1 an analogue meter whose scale is read from different angles
- 2 a meter which always measures 5% too high
- 3 a meter with a needle that is not frictionless, so the needle sometimes sticks slightly
- 4 a meter with a zero error

Which errors are random and which are systematic?

	random error	systematic error
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 4	1 and 3
D	3 and 4	1 and 2

167. 9702/12/O/N/21/Q5

A student measures the time T for one complete oscillation of a pendulum of length l .

Her results are shown in the table.

l/m	T/s
0.420 ± 0.001	1.3 ± 0.1

She uses the formula

$$T = 2\pi \sqrt{\frac{l}{g}}$$

to calculate the acceleration of free fall g .

What is the best estimate of the percentage uncertainty in the value of g ?

- A** 0.02% **B** 4% **C** 8% **D** 16%

168. 9702/13/O/N/21/Q5

After measuring the width of a shelf to be 305 mm, it is found that the graduations on the ruler used are 1.0% further apart than they should be.

Which type of measurement error is this and what is the true width of the shelf?

	type of error	true width / mm
A	random	302
B	random	308
C	systematic	302
D	systematic	308

169. 9702/12/F/M/22/Q3

A man of mass 75.2 kg uses a set of weighing scales to measure his mass three times. He obtains the following readings.

	mass / kg
reading 1	80.2
reading 2	80.1
reading 3	80.2

Which statement describes the precision and accuracy of the weighing scales?

- A** not precise to ± 0.1 kg and accurate to ± 0.1 kg
- B** not precise to ± 0.1 kg and not accurate to ± 0.1 kg
- C** precise to ± 0.1 kg and accurate to ± 0.1 kg
- D** precise to ± 0.1 kg and not accurate to ± 0.1 kg

170. 9702/11/M/J/22/Q3

A value for the acceleration of free fall on Earth is given as $(10 \pm 2) \text{ m s}^{-2}$.

Which statement is correct?

- A** The value is accurate but not precise.
- B** The value is both precise and accurate.
- C** The value is neither precise nor accurate.
- D** The value is precise but not accurate.

171. 9702/12/M/J/22/Q3

Which statement about systematic errors is **not** correct?

- A A systematic error can be caused by using an incorrectly calibrated instrument.
- B One particular type of systematic error can affect all the measurements by the same amount.
- C The effect of a systematic error can be reduced by repeating and averaging the measurements.
- D Zero error is a type of systematic error.

172. 9702/13/M/J/22/Q3

A micrometer screw gauge is used to measure the diameter of a small uniform steel sphere. The measurement of the diameter is $5.00 \text{ mm} \pm 0.01 \text{ mm}$.

What is the percentage uncertainty in the calculated volume of the sphere, using these values?

- A 0.2% B 0.4% C 0.6% D 1.2%

173. 9702/11/O/N/22/Q3

In an experiment to determine the acceleration of free fall g , the time t taken for a ball to fall through distance s is measured. The percentage uncertainty in the measurement of s is 2%. The percentage uncertainty in the measurement of t is 3%.

The value of g is determined using the equation shown.

$$g = \frac{2s}{t^2}$$

What is the percentage uncertainty in the calculated value of g ?

- A 1% B 5% C 8% D 11%

174. 9702/12/O/N/22/Q3

A spring is suspended from a fixed point and a force is applied. The position of a pointer attached to the bottom of the spring against a vertical ruler is recorded.

Before the force is applied, the position of the pointer is $(225 \pm 2) \text{ mm}$.

After the force is applied, the position of the pointer is $(250 \pm 2) \text{ mm}$.

The extension of the spring is determined.

What is the percentage uncertainty in the extension?

- A 1.6% B 1.8% C 8.0% D 16%

175. 9702/13/O/N/22/Q3

A digital meter is used to measure the current in an electric circuit.

The reading on the meter fluctuates (varies) between 3.04 A and 3.08 A. The readings on the meter have an accuracy of $\pm 1\%$.

What is the true value of the current, with its uncertainty?

- A $(3.06 \pm 0.02) \text{ A}$
- B $(3.06 \pm 0.04) \text{ A}$
- C $(3.06 \pm 0.05) \text{ A}$
- D $(3.06 \pm 0.07) \text{ A}$

176. 9702/12/F/M/23/Q3

A hollow cylinder, which is open at both ends, has a radius of $(3.0 \pm 0.1) \text{ cm}$ and a length of $(15.0 \pm 0.1) \text{ cm}$.

What is the value, with its absolute uncertainty, of the surface area of the cylinder?

- A $(280 \pm 10) \text{ cm}^2$
- B $(282.7 \pm 0.2) \text{ cm}^2$
- C $(420 \pm 30) \text{ cm}^2$
- D $(424.1 \pm 0.3) \text{ cm}^2$

177. 9702/11/M/J/23/Q3

A copper pipe has a true diameter of 42.03 mm.

A builder measures the diameter of the pipe five times using digital calipers. The measurements are shown.

diameter / mm
48.01
47.99
48.01
48.00
47.99

What describes the builder's measurements?

- A accurate and precise
- B accurate but not precise
- C not precise and not accurate
- D precise but not accurate

178. 9702/12/M/J/23/Q3

Two measurements for a solid sphere are shown.

$$\text{mass} = (32.5 \pm 0.1)\text{g}$$

$$\text{diameter} = (1.87 \pm 0.04)\text{cm}$$

These values are used to determine the density of the sphere.

What is the percentage uncertainty in the density?

- A** 2.4% **B** 4.6% **C** 6.1% **D** 6.7%

179. 9702/13/M/J/23/Q3

A desk has a true width of 50.0 cm.

Two students, X and Y, measure the width of the desk.

Student X uses a tape measure and records a width of $(49.5 \pm 0.5)\text{cm}$.

Student Y uses a metre rule and records a width of $(51.4 \pm 0.1)\text{cm}$.

Which statement about the measurement of student X is correct?

- A** It is less accurate and less precise than the measurement of student Y.
B It is less accurate but more precise than the measurement of student Y.
C It is more accurate and more precise than the measurement of student Y.
D It is more accurate but less precise than the measurement of student Y.

180. 9702/13/O/N/23/Q3

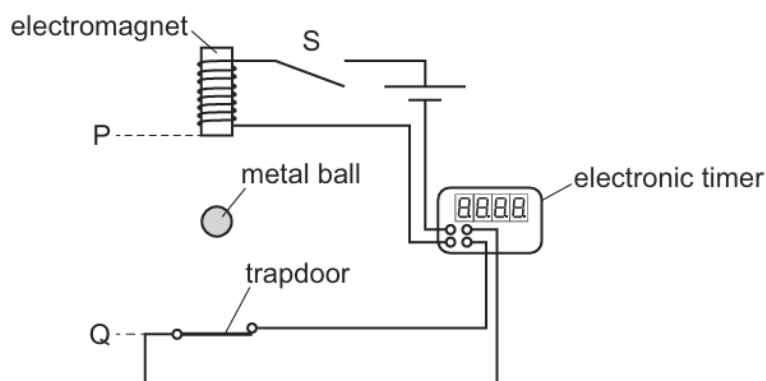
A set of repeated measurements is made of a fixed quantity. An average of these measurements is calculated.

What is the effect of averaging on the random error and the systematic error in the measurements?

- A** Random error and systematic error are both reduced.
B Random error and systematic error are both unaffected.
C Random error is reduced but systematic error is unaffected.
D Random error is unaffected but systematic error is reduced.

181. 9702/11/O/N/20/Q3

A student determines the acceleration of free fall by using a small metal ball, as shown.



When switch S is opened, the ball is released from an electromagnet and an electronic timer is started. The ball then falls vertically downwards. The timer stops when the ball hits a trapdoor. The student measures the distance PQ between the electromagnet and the trapdoor. This distance and the reading on the timer are then used to calculate the acceleration of free fall.

Which statement about errors in the experiment is correct?

- A The random error can be reduced by adding the diameter of the ball to the distance PQ.
- B The random error can be reduced by subtracting the diameter of the ball from the distance PQ.
- C The systematic error can be reduced by adding the diameter of the ball to the distance PQ.
- D The systematic error can be reduced by subtracting the diameter of the ball from the distance PQ.

182. 9702/12/O/N/23/Q3

A student takes measurements to determine the constant acceleration of a model car moving from rest in a straight line. The measured values with their absolute uncertainties are shown.

quantity	measured value	uncertainty
displacement	16.5 m	± 0.1 m
time	15.0 s	± 1.0 s

The student uses the equation $s = \frac{1}{2}at^2$ to calculate the acceleration of the car.

What is the acceleration and its absolute uncertainty?

- A $(0.11 \pm 0.01) \text{ m s}^{-2}$
- B $(0.11 \pm 0.02) \text{ m s}^{-2}$
- C $(0.15 \pm 0.01) \text{ m s}^{-2}$
- D $(0.15 \pm 0.02) \text{ m s}^{-2}$

ANSWERS

1. D	49. D	97. B	144. A
2. C	50. D	98. D	145. B
3. B	51. D	99. C	146. B
4. D	52. B	100. D	147. C
5. D	53. B	101. B	148. D
6. B	54. D	102. B	149. D
7. C	55. B	103. A	150. B
8. C	56. A	104. D	151. D
9. C	57. B	105. D	152. A
10. A	58. C	106. B	153. D
11. B	59. A	107. C	154. D
12. D	60. A	108. B	155. B
13. D	61. D	109. C	156. D
14. C	62. A	110. D	157. B
15. A	63. C	111. C	158. A
16. D	64. D	112. B	159. C
17. B	65. D	113. B	160. B
18. B	66. A	114. B	161. D
19. C	67. D	115. A	162. D
20. D	68. C	116. A	163. D
21. C	69. C	117. C	164. B
22. C	70. D	118. D	165. B
23. D	71. D	119. D	166. B
24. A	72. B	120. A	167. D
25. D	73. D	121. D	168. D
26. C	74. B	122. B	169. D
27. A	75. C	123. B	170. A
28. D	76. C	124. B	171. C
29. C	77. B	125. B	172. B
30. A	78. D	126. C	173. C
31. C	79. D	127. A	174. D
32. D	80. B	128. A	175. C
33. D	81. C	129. C	176. A
34. D	82. D	130. B	177. D
35. B	83. B	131. B	178. D
36. B	84. C	132. D	179. D
37. A	85. B	133. B	180. C
38. B	86. C	134. D	181. D
39. B	87. C	135. C	182. D
40. C	88. C	136. D	
41. A	89. D	137. B	
42. B	90. C	138. C	
43. C	91. D	139. D	
44. C	92. C	140. A	
45. D	93. D	141. C	
46. C	94. D	142. D	
47. B	95. D	143. D	
48. A	96. A		

- 1 (a) Two of the SI base quantities and their units are mass (kg) and length (m). Name three other SI base quantities and their units.

1. quantity unit

2. quantity unit

3. quantity unit

[3]

- (b) The pressure p due to a liquid of density ρ is related to the depth h by the expression

$$p = \rho gh,$$

where g is the acceleration of free fall.

Use this expression to determine the derived units of pressure. Explain your working.

[5]

- 2 A unit is often expressed with a prefix. For example, the gram may be written with the prefix 'kilo' as the kilogram. The prefix represents a power-of-ten. In this case, the power-of-ten is 10^3 .

Complete Fig. 1.1 to show each prefix with its symbol and power-of-ten.

prefix	symbol	power-of-ten
kilo	k	10^3
nano	n	
centi		10^{-2}
.....	M	10^6
.....	T	10^{12}

Fig. 1.1

[4]

- 3 (a) Two of the SI base quantities are mass and time. State three other SI base quantities.

1.

2.

3.

[3]

- (b) A sphere of radius r is moving at speed v through air of density ρ . The resistive force F acting on the sphere is given by the expression

$$F = Br^2\rho v^k$$

where B and k are constants without units.

- (i) State the SI base units of F , ρ and v .

F

ρ

v [3]

- (ii) Use base units to determine the value of k .

$k =$ [2]

9702/22/O/N/10/Q1

- 4 (a) (i) Distinguish between vector quantities and scalar quantities.

.....

 [2]

- (ii) State whether each of the following is a vector quantity or a scalar quantity.

1. temperature

..... [1]

2. acceleration of free fall

..... [1]

3. electrical resistance

..... [1]

- (b) A block of wood of weight 25 N is held stationary on a slope by means of a string, as shown in Fig. 1.1.

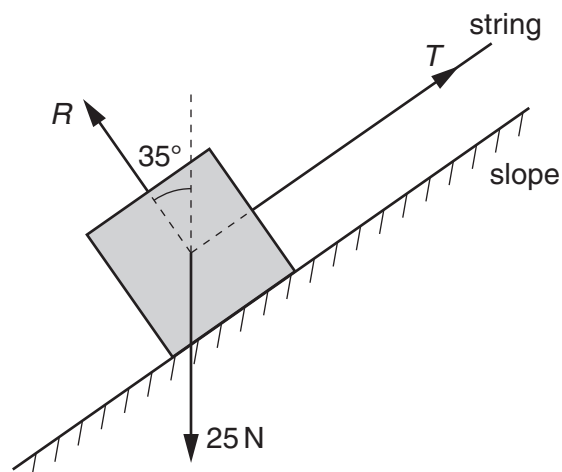


Fig. 1.1

The tension in the string is T and the slope pushes on the block with a force R that is normal to the slope.

Either by scale drawing on Fig. 1.1 or by calculation, determine the tension T in the string.

$T = \dots\dots\dots$ N [3]

- 5 (a) Distinguish between *scalar* quantities and *vector* quantities.

9702/22/M/J/11/Q1

.....

 [2]

- (b) In the following list, underline **all** the scalar quantities.

acceleration force kinetic energy mass power weight [1]

- 6 (a) (i) State the SI base units of volume.

base units of volume [1]

- (ii) Show that the SI base units of pressure are $\text{kg m}^{-1} \text{s}^{-2}$.

[1]

- (b) The volume V of liquid that flows through a pipe in time t is given by the equation

$$\frac{V}{t} = \frac{\pi P r^4}{8 C l}$$

where P is the pressure difference between the ends of the pipe of radius r and length l
 The constant C depends on the frictional effects of the liquid.
 Determine the base units of C .

base units of C [3]

- 7 (e) The velocity vector diagram for an aircraft heading due north is shown to scale in Fig. 1.1. There is a wind blowing from the north-west.

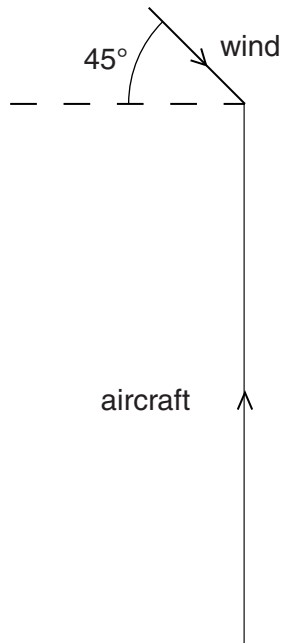


Fig. 1.1

The speed of the wind is 36 m s^{-1} and the speed of the aircraft is 250 m s^{-1} .

- (i) Draw an arrow on Fig. 1.1 to show the direction of the resultant velocity of the aircraft. [1]
- (ii) Determine the magnitude of the resultant velocity of the aircraft.

resultant velocity = m s^{-1} [2]

- 8 (a) State the SI base units of force.

9702/23/M/J/13/Q1

.....[1]

- (b) Two wires each of length l are placed parallel to each other a distance x apart, as shown in Fig. 1.1.

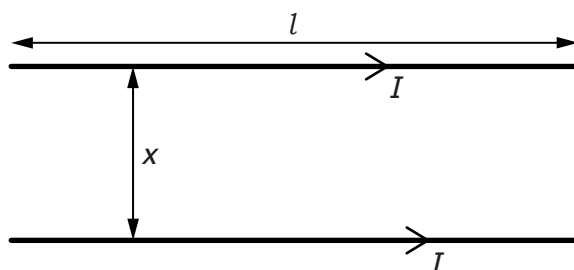


Fig. 1.1

Each wire carries a current I . The currents give rise to a force F on each wire given by

$$F = \frac{KI^2l}{x}$$

where K is a constant.

- (i) Determine the SI base units of K .

units of K [2]

- (ii) On Fig. 1.2, sketch the variation with x of F . The quantities I and l remain constant.



Fig. 1.2

[2]

- (iii) The current I in both of the wires is varied.

On Fig. 1.3, sketch the variation with I of F . The quantities x and l remain constant.



Fig. 1.3

[1]

- 9 (a) State two SI base units other than the kilogram, metre and second. 9702/22/O/N/13/Q1

1.

2. [2]

- (b) A metal wire has original length l_0 . It is then suspended and hangs vertically as shown in Fig. 1.1.

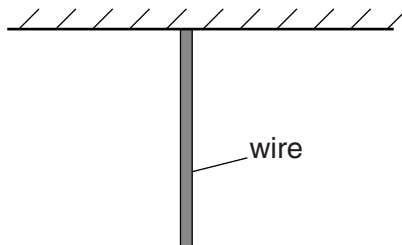


Fig. 1.1

The weight of the wire causes it to stretch. The elastic potential energy stored in the wire is E .

- (i) Show that the SI base units of E are $\text{kg m}^2 \text{s}^{-2}$.

[2]

- (ii) The elastic potential energy E is given by

$$E = C\rho^2 g^2 A l_0^3$$

where ρ is the density of the metal,
 g is the acceleration of free fall,
 A is the cross-sectional area of the wire
and C is a constant.

Determine the SI base units of C .

SI base units of C [3]

10 (a) Show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

9702/22/M/J/14/Q1

[3]

(b) The rate of flow of thermal energy $\frac{Q}{t}$ in a material is given by

$$\frac{Q}{t} = \frac{CAT}{x}$$

where A is the cross-sectional area of the material,
 T is the temperature difference across the thickness of the material,
 x is the thickness of the material,
 C is a constant.

Determine the SI base units of C .

base units [4]

11 (a) Underline **all** the base quantities in the following list.

9702/23/M/J/14/Q1

ampere charge current mass second temperature weight [2]

(b) The potential energy E_p stored in a stretched wire is given by

$$E_p = \frac{1}{2} C \sigma^2 V$$

where C is a constant,
 σ is the strain,
 V is the volume of the wire.

Determine the SI base units of C .

base units [3]

- 12 (a) Mass, length and time are SI base quantities.
State two other base quantities.

9702/21/O/N/14/Q1

1.

2.

[2]

- (b) A mass m is placed on the end of a spring that is hanging vertically, as shown in Fig. 1.1.

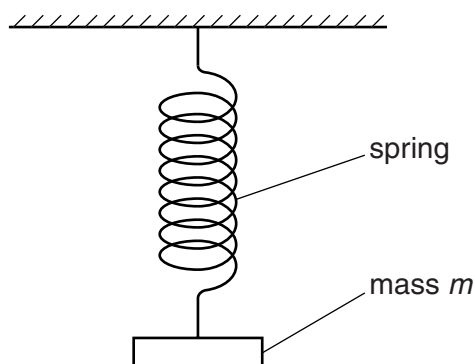


Fig. 1.1

The mass is made to oscillate vertically. The time period of the oscillations of the mass is T .
The period T is given by

$$T = C \sqrt{\frac{m}{k}}$$

where C is a constant and k is the spring constant.
Show that C has no units.

[3]

- 13 (c) An object B is on a horizontal surface. Two forces act on B in this horizontal plane. A vector diagram for these forces is shown to scale in Fig. 1.1.

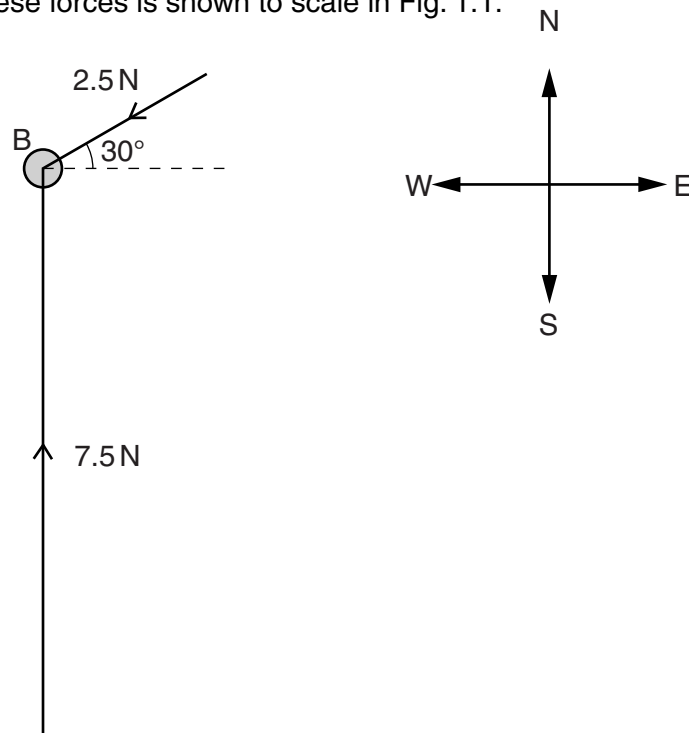


Fig. 1.1

A force of 7.5 N towards north and a force of 2.5 N from 30° north of east act on B. The mass of B is 750 g.

- (i) On Fig. 1.1, draw an arrow to show the approximate direction of the resultant of these two forces. [1]

- (ii) 1. Show that the magnitude of the resultant force on B is 6.6 N. [1]

2. Calculate the magnitude of the acceleration of B produced by this resultant force.

magnitude = ms^{-2} [2]

- (iii) Determine the angle between the direction of the acceleration and the direction of the 7.5 N force.

angle = ° [1]

14 (a) Force is a vector quantity. State three other vector quantities.

9702/23/O/N/14/Q3

1.
2.
3.

[2]

(b) Three coplanar forces X , Y and Z act on an object, as shown in Fig. 3.1.

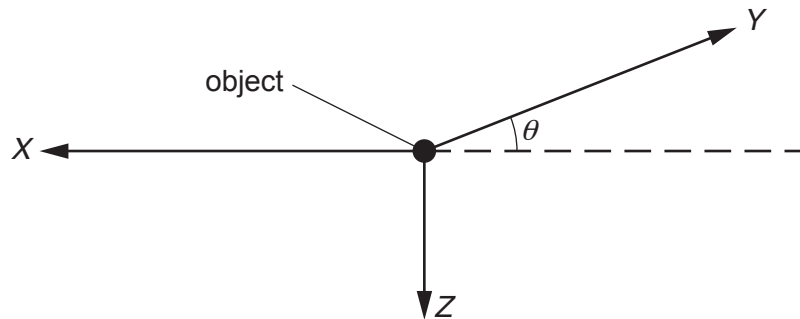


Fig. 3.1

The force Z is vertical and X is horizontal. The force Y is at an angle θ to the horizontal. The force Z is kept constant at 70 N.

In an experiment, the magnitude of force X is varied. The magnitude and direction of force Y are adjusted so that the object remains in equilibrium.

Fig. 3.2 shows the variation of the magnitude of force Y with the magnitude of force X .

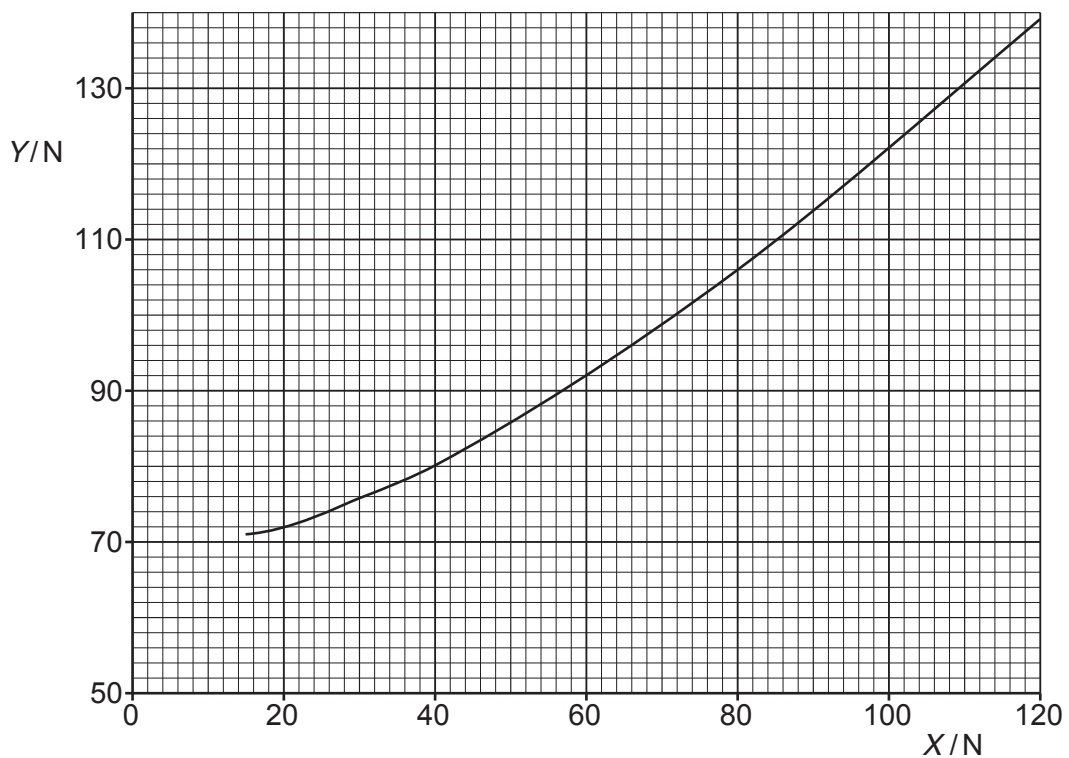


Fig. 3.2

- (i) Use Fig. 3.2 to estimate the magnitude of Y for $X = 0$.

$Y =$ N [1]

- (ii) State and explain the value of θ for $X = 0$.

.....
.....
..... [2]

- (iii) The magnitude of X is increased to 160 N. Use resolution of forces to calculate the value of

1. angle θ ,

$\theta =$ ° [2]

2. the magnitude of force Y .

$Y =$ N [2]

- (c) The angle θ decreases as X increases. Explain why the object cannot be in equilibrium for $\theta = 0$.

.....
.....
..... [1]

- 15 (a)** The kilogram, metre and second are SI base units.
State two other base units.

1.

2.

- (b)** Determine the SI base units of [2]

(i) stress,

(ii) the Young modulus. SI base units [2]

SI base units [1]

- 16 (a)** Use the definition of power to show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

[2]

- (b)** Use an expression for electrical power to determine the SI base units of potential difference.

units [2]

- 17 (a)** Use the definition of work done to show that the SI base units of energy are $\text{kg m}^2 \text{s}^{-2}$.

[2]

- (b)** Define potential difference.

.....

..... [1]

- (c)** Determine the SI base units of resistance. Show your working.

units [3]

- 18 (a) The distance between the Sun and the Earth is 1.5×10^{11} m. State this distance in Gm.

distance = Gm [1]

- (b) The distance from the centre of the Earth to a satellite above the equator is 42.3 Mm. The radius of the Earth is 6380 km.
A microwave signal is sent from a point on the Earth directly below the satellite.
Calculate the time taken for the microwave signal to travel to the satellite and back.

time = s [2]

- (c) The speed v of a sound wave through a gas of density ρ and pressure P is given by

$$v = \sqrt{\frac{CP}{\rho}}$$

where C is a constant.

Show that C has no unit.

[3]

- (d) Underline all the scalar quantities in the list below.

acceleration energy momentum power weight

[1]

- (e) A boat travels across a river in which the water is moving at a speed of 1.8 m s^{-1} . The velocity vectors for the boat and the river water are shown to scale in Fig. 1.1.

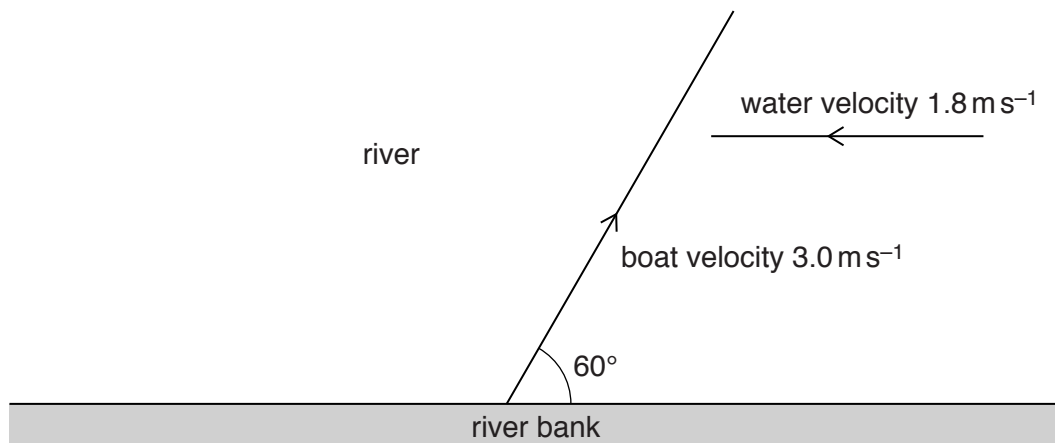


Fig. 1.1 (shown to scale)

In still water the speed of the boat is 3.0 m s^{-1} . The boat is directed at an angle of 60° to the river bank.

- (i) On Fig. 1.1, draw a vector triangle or a scale diagram to show the resultant velocity of the boat.

[2]

- (ii) Determine the magnitude of the resultant velocity of the boat.

resultant velocity = m s^{-1} [2]

19 (a) A list of quantities that are either scalars or vectors is shown in Fig. 1.1.

9702/23/M/J/16/Q1

quantity	scalar	vector
distance	✓	
energy		
momentum		
power		
time		
weight		

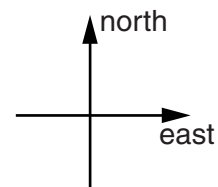
Fig. 1.1

Complete Fig. 1.1 to indicate whether each quantity is a scalar or a vector.
One line has been completed as an example.

[2]

(b) A girl runs 120 m due north in 15 s. She then runs 80 m due east in 12 s.

- (i)** Sketch a vector diagram to show the path taken by the girl. Draw and label her resultant displacement R.



[1]

(ii) Calculate, for the girl,

1. the average speed,

average speed = ms^{-1} [1]

2. the magnitude of the average velocity v and its angle with respect to the direction of the initial path.

magnitude of v = ms^{-1}

angle = $^{\circ}$
[3]

20 (a) (i) Define *pressure*.

9702/22/O/N/16/Q1

.....
.....[1]

(ii) Show that the SI base units of pressure are $\text{kg m}^{-1} \text{s}^{-2}$.

[1]

- (b) Gas flows through the narrow end (nozzle) of a pipe. Under certain conditions, the mass m of gas that flows through the nozzle in a short time t is given by

$$\frac{m}{t} = kC \sqrt{\rho P}$$

where k is a constant with no units,
 C is a quantity that depends on the nozzle size,
 ρ is the density of the gas arriving at the nozzle,
 P is the pressure of the gas arriving at the nozzle.
 Determine the base units of C .

base units[3]

-
- 21 (a) State two SI base units other than kilogram, metre and second. 9702/22/M/J/17/Q1

1.

2.

[1]

- (b) Determine the SI base units of resistivity.

base units[3]

- 22** The drag force F_D acting on a sphere moving through a fluid is given by the expression

$$F_D = K\rho v^2$$

where K is a constant,
 ρ is the density of the fluid
 and v is the speed of the sphere.
 Determine the SI base units of K .

base units[3]

- 23 (a) (i)** Define *power*.

.....
[1]

- (ii)** Show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

- (b) All bodies radiate energy. The power P radiated by a body is given by

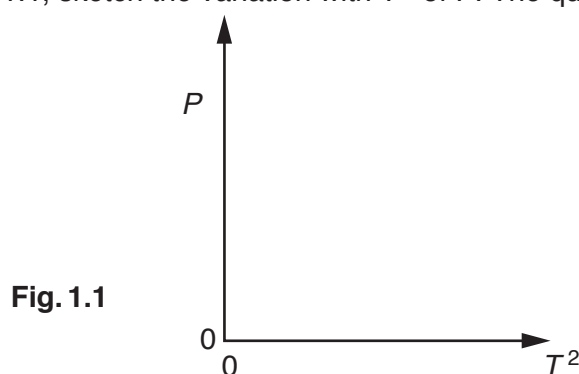
$$P = kAT^4$$

where T is the thermodynamic temperature of the body,
 A is the surface area of the body
and k is a constant.

- (i) Determine the SI base units of k .

base units[2]

- (ii) On Fig. 1.1, sketch the variation with T^2 of P . The quantity A remains constant.



[1]

- 24 (a)** State what is meant by a *scalar* quantity and by a *vector* quantity.

9702/21/M/J/18/Q1

scalar:.....

.....

vector:.....

.....

[2]

- (b) Complete Fig. 1.1 to indicate whether each of the quantities is a vector or a scalar.

quantity	vector or scalar
power	
temperature	
momentum	

Fig. 1.1

[2]

- (c) An aircraft is travelling in wind. Fig. 1.2 shows the velocities for the aircraft in still air and for the wind.

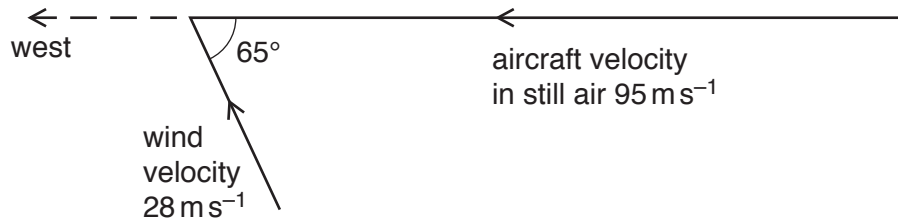


Fig. 1.2

The velocity of the aircraft in still air is 95 m s^{-1} to the west.
The velocity of the wind is 28 m s^{-1} from 65° south of east.

- (i) On Fig. 1.2, draw an arrow, labelled R, in the direction of the resultant velocity of the aircraft. [1]
- (ii) Determine the magnitude of the resultant velocity of the aircraft.

magnitude of velocity = m s^{-1} [2]

25 (a) Define *force*.

.....[1]

(b) State the SI base units of force.

.....[1]

(c) The force F between two point charges is given by $F = \frac{Q_1 Q_2}{4\pi r^2 \epsilon}$

where Q_1 and Q_2 are the charges,

r is the distance between the charges,

ϵ is a constant that depends on the medium between the charges.

Use the above expression to determine the base units of ϵ .

base units[2]

26 (a) (i) Define *resistance*.

.....
[1]

(ii) A potential difference of 0.60 V is applied across a resistor of resistance 4.0 GΩ.

Calculate the current, in pA, in the resistor.

current = pA [2]

(b) The energy E transferred when charge Q moves through an electrical component is given by the equation

$$E = QV$$

where V is the potential difference across the component.

Use the equation to determine the SI base units of potential difference.

SI base units [3]

- 27 (a) Distinguish between vector and scalar quantities.

.....

 [2]

- (b) The electric field strength E at a distance x from an isolated point charge Q is given by the equation

$$E = \frac{Q}{x^2 b}$$

where b is a constant.

- (i) Use the definition of electric field strength to show that E has SI base units of $\text{kg m A}^{-1} \text{s}^{-3}$.

[2]

- (ii) Use the units for E given in (b)(i) to determine the SI base units of b .

SI base units of b [2]

Hy The volume of fuel in the tank of a car is monitored using a meter as illustrated in Fig. 1.1.

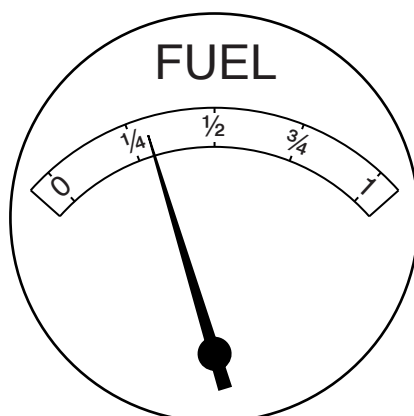


Fig. 1.1

The meter has an analogue scale. The meter reading for different volumes of fuel in the tank is shown in Fig. 1.2.

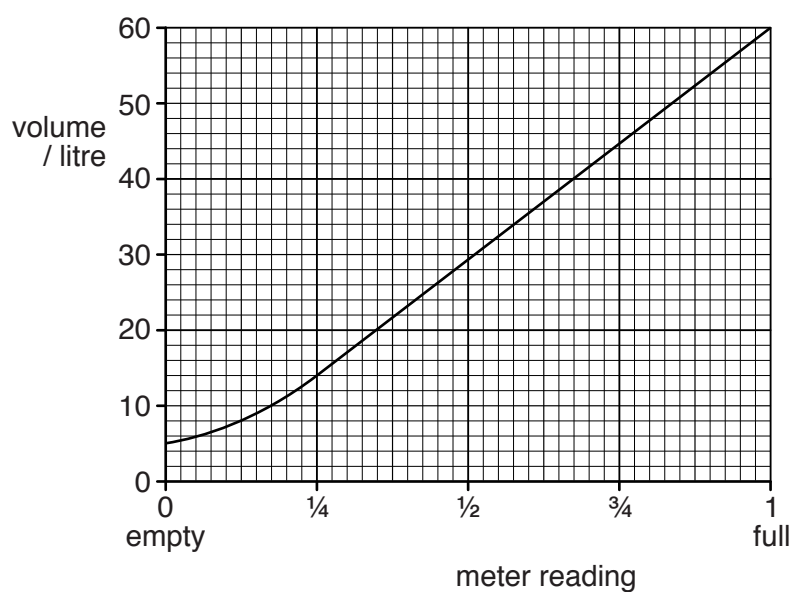


Fig. 1.2

The meter is calibrated in terms of the fraction of the tank that remains filled with fuel.

(a) The car uses 1.0 litre of fuel when travelling 14 km. The car starts a journey with a full tank of fuel.

(i) Calculate the volume of fuel remaining in the tank after a journey of 210 km.

volume = litres [2]

(ii) Use your answer to **(i)** and Fig. 1.2 to determine the change in the meter reading during the 210 km journey.

from *full* to [1]

(b) There is a systematic error in the meter.

(i) State the feature of Fig. 1.2 that indicates that there is a systematic error.

.....
..... [1]

(ii) Suggest why, for this meter, it is an advantage to have this systematic error.

.....
..... [1]

- Hφ** A simple pendulum may be used to determine a value for the acceleration of free fall g . Measurements are made of the length L of the pendulum and the period T of oscillation. The values obtained, with their uncertainties, are as shown.

$$T = (1.93 \pm 0.03) \text{ s}$$

$$L = (92 \pm 1) \text{ cm}$$

- (a)** Calculate the percentage uncertainty in the measurement of

- (i)** the period T ,

uncertainty = % [1]

- (ii)** the length L .

uncertainty = % [1]

- (b)** The relationship between T , L and g is given by

$$g = \frac{4\pi^2 L}{T^2}.$$

Using your answers in **(a)**, calculate the percentage uncertainty in the value of g .

uncertainty = % [1]

- (c)** The values of L and T are used to calculate a value of g as 9.751 ms^{-2} .

- (i)** By reference to the measurements of L and T , suggest why it would not be correct to quote the value of g as 9.751 ms^{-2} .

.....

..... [1]

- (ii)** Use your answer in **(b)** to determine the absolute uncertainty in g .

Hence state the value of g , with its uncertainty, to an appropriate number of significant figures.

$g = \dots \pm \dots \text{ ms}^{-2}$ [2]

30 A metal wire has a cross-section of diameter approximately 0.8 mm.

(a) State what instrument should be used to measure the diameter of the wire.

.....[1]

(b) State how the instrument in **(a)** is

(i) checked so as to avoid a systematic error in the measurements,

.....
[1]

(ii) used so as to reduce random errors.

.....

[2]

31 Measurements made for a sample of metal wire are shown in Fig. 1.1. **9702/21/M/J/11/Q1**

quantity	measurement	uncertainty
length	1750 mm	± 3 mm
diameter	0.38 mm	± 0.01 mm
resistance	7.5Ω	$\pm 0.2 \Omega$

Fig. 1.1

(a) State the appropriate instruments used to make each of these measurements.

(i) length [1]

(ii) diameter [1]

(iii) resistance [1]

(b) **(i)** Show that the resistivity of the metal is calculated to be $4.86 \times 10^{-7} \Omega \text{ m}$.

[2]

(ii) Calculate the uncertainty in the resistivity.

uncertainty = $\pm$ $\Omega \text{ m}$ [4]

(c) Use the answers in (b) to express the resistivity with its uncertainty to the appropriate number of significant figures.

resistivity = $\pm$ $\Omega \text{ m}$ [1]

9702/22/M/J/12/Q1

32 The volume V of liquid flowing in time t through a pipe of radius r is given by the equation

$$\frac{V}{t} = \frac{\pi P r^4}{8 C l}$$

where P is the pressure difference between the ends of the pipe of length l , and C depends on the frictional effects of the liquid.

An experiment is performed to determine C . The measurements made are shown in Fig. 1.1.

Fig. 1.1

$\frac{V}{t} / 10^{-6} \text{ m}^3 \text{ s}^{-1}$	$P / 10^3 \text{ N m}^{-2}$	$r / \text{ mm}$	$l / \text{ m}$
1.20 ± 0.01	2.50 ± 0.05	0.75 ± 0.01	0.250 ± 0.001

(a) Calculate the value of C .

$C =$ N s m^{-2} [2]

(b) Calculate the uncertainty in C .

uncertainty = Ns m^{-2} [3]

(c) State the value of C and its uncertainty to the appropriate number of significant figures.

$C = \dots\dots\dots \pm \dots\dots\dots \text{Ns m}^{-2}$ [1]

9702/23/O/N/12/Q1

33 (a) The spacing between two atoms in a crystal is $3.8 \times 10^{-10} \text{ m}$. State this distance in pm.

spacing = pm [1]

(b) Calculate the time of one day in Ms.

time = Ms [1]

(c) The distance from the Earth to the Sun is 0.15 Tm. Calculate the time in minutes for light to travel from the Sun to the Earth.

time = min [2]

(d) Underline all the vector quantities in the list below.

distance	energy	momentum	weight	work	[1]
----------	--------	----------	--------	------	-----

- 34 A source of radio waves sends a pulse towards a reflector. The pulse returns from the reflector and is detected at the same point as the source. The emitted and reflected pulses are recorded on a cathode-ray oscilloscope (c.r.o.) as shown in Fig. 2.1.

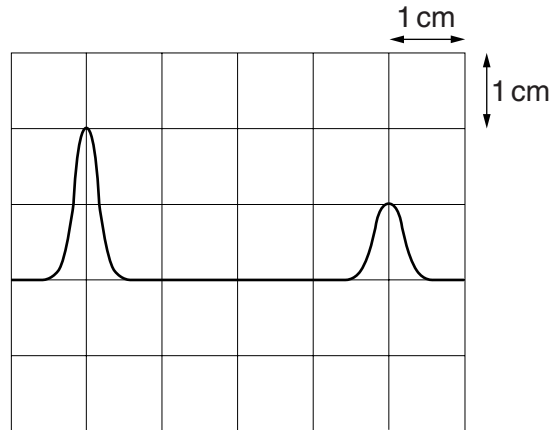


Fig. 2.1

The time-base setting is $0.20 \mu\text{s cm}^{-1}$.

- (a) Using Fig. 2.1, determine the distance between the source and the reflector.

distance = m [4]

- (b) Determine the time-base setting required to produce the same separation of pulses on the c.r.o. when sound waves are used instead of radio waves.
The speed of sound is 300 m s^{-1} .

.....

 [3]

- 35 A cylindrical disc is shown in Fig. 1.1.

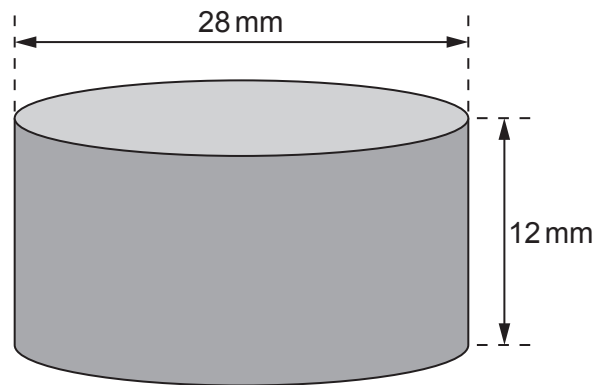


Fig. 1.1

The disc has diameter 28 mm and thickness 12 mm.
The material of the disc has density $6.8 \times 10^3 \text{ kg m}^{-3}$.

Calculate, to two significant figures, the weight of the disc.

weight = N [4]

- 36** The time T for a satellite to orbit the Earth is given by

$$T = \sqrt{\left(\frac{KR^3}{M}\right)}$$

where R is the distance of the satellite from the centre of the Earth,
 M is the mass of the Earth,
 and K is a constant.

- (a)** Determine the SI base units of K .

SI base units of K [2]

- (b)** Data for a particular satellite are given in Fig. 2.1.

quantity	measurement	uncertainty
T	$8.64 \times 10^4 \text{ s}$	$\pm 0.5\%$
R	$4.23 \times 10^7 \text{ m}$	$\pm 1\%$
M	$6.0 \times 10^{24} \text{ kg}$	$\pm 2\%$

Fig. 2.1

Calculate K and its actual uncertainty in SI units.

$K = \dots\dots\dots \pm \dots\dots\dots$ SI units [4]

37 A coin is made in the shape of a thin cylinder, as shown in Fig. 2.1.

9702/22/M/J/14/Q2

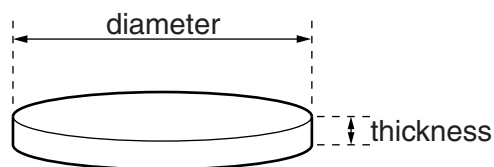


Fig. 2.1

Fig. 2.2 shows the measurements made in order to determine the density ρ of the material used to make the coin.

quantity	measurement	uncertainty
mass	9.6 g	± 0.5 g
thickness	2.00 mm	± 0.01 mm
diameter	22.1 mm	± 0.1 mm

Fig. 2.2

(a) Calculate the density ρ in kg m^{-3} .

$$\rho = \dots\dots\dots \text{kg m}^{-3} \quad [3]$$

(b) (i) Calculate the percentage uncertainty in ρ .

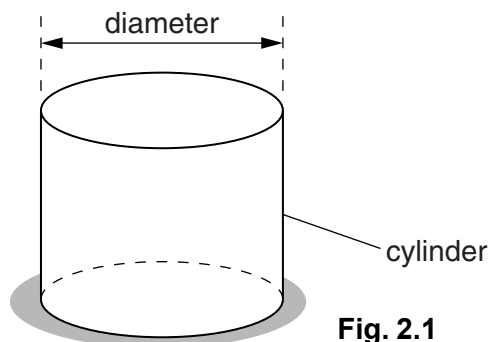
$$\text{percentage uncertainty} = \dots\dots\dots [3]$$

(ii) State the value of ρ with its actual uncertainty.

$$\rho = \dots\dots\dots \pm \dots\dots\dots \text{kg m}^{-3} \quad [1]$$

..... [1]

- (b) A cylinder is placed on a horizontal surface, as shown in Fig. 2.1.



The following measurements were made on the cylinder:

mass = 5.09 ± 0.01 kg

diameter = 9.4 ± 0.1 cm.

- (i) Calculate the pressure produced by the cylinder on the surface.

pressure = Pa [3]

- (ii) Calculate the actual uncertainty in the pressure.

actual uncertainty = Pa [3]

- (iii) State the pressure, with its actual uncertainty.

pressure = $\pm$ Pa [1]

- 39 The resistance R of a uniform metal wire is measured for different lengths l of the wire. The variation with l of R is shown in Fig. 3.1.

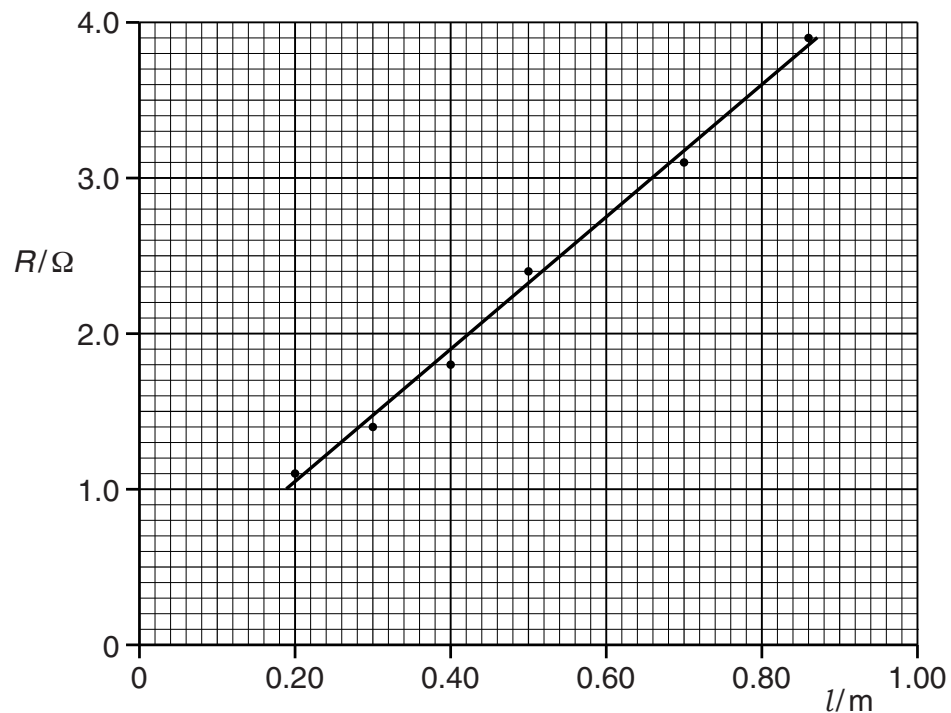


Fig. 3.1

- (a) The points shown in Fig. 3.1 do not lie on the best-fit line. Suggest a reason for this.

.....
[1]

- (b) Determine the gradient of the line shown in Fig. 3.1.

gradient = [2]

- (c) The cross-sectional area of the wire is 0.12 mm^2 .

Use your answer in (b) to determine the resistivity of the metal of the wire.

resistivity = $\Omega \text{ m}$ [3]

- 40 A microphone detects a musical note of frequency f . The microphone is connected to a cathode-ray oscilloscope (c.r.o.). The signal from the microphone is observed on the c.r.o. as illustrated in Fig. 2.1.

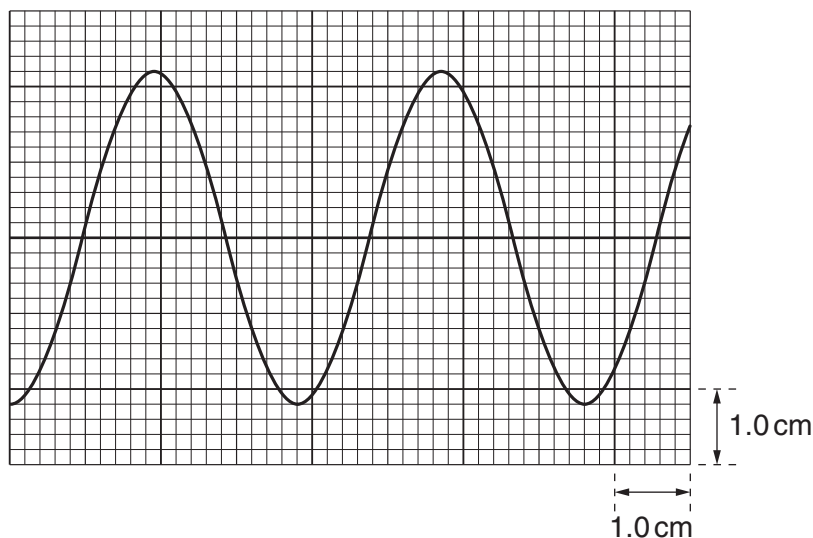


Fig. 2.1

The time-base setting of the c.r.o. is 0.50 ms cm^{-1} . The Y-plate setting is 2.5 mV cm^{-1} .

- (a) Use Fig. 2.1 to determine

- (i) the amplitude of the signal,

amplitude = mV [2]

- (ii) the frequency f ,

$f = \dots\dots\dots$ Hz [3]

- (iii) the actual uncertainty in f caused by reading the scale on the c.r.o.

actual uncertainty = Hz [2]

- (b) State f with its actual uncertainty.

$f = \dots\dots\dots \pm \dots\dots\dots$ Hz [1]

- 41 Fig. 4.1 shows the values obtained in an experiment to determine the Young modulus E of a metal in the form of a wire.

quantity	value	instrument
diameter d	0.48 mm	
length l	1.768 m	
load F	5.0 N to 30.0 N in 5.0 N steps	
extension e	0.25 mm to 1.50 mm	

Fig. 4.1

- (a) (i) Complete Fig. 4.1 with the name of an instrument that could be used to measure each of the quantities. [3]

- (ii) Explain why a series of values of F , each with corresponding extension e , are measured.

.....

.....[1]

- 42 The speed v of a transverse wave on a uniform string is given by the expression

$$v = \sqrt{\frac{Tl}{m}}$$

where T is the tension in the string, l is its length and m is its mass.

An experiment is performed to determine the speed v of the wave. The measurements are shown in Fig. 1.1.

quantity	measurement	uncertainty
T	1.8N	$\pm 5\%$
l	126cm	$\pm 1\%$
m	5.1g	$\pm 2\%$

Fig. 1.1

- (a) State an appropriate instrument to measure the length l .

..... [1]

- (b) (i) Use the data in Fig. 1.1 to calculate the speed v .

$v =$ ms^{-1} [2]

- (ii) Use your answer in (b)(i) and the data in Fig. 1.1 to determine the value of v , with its absolute uncertainty, to an appropriate number of significant figures.

$v =$ $\pm$ ms^{-1} [3]

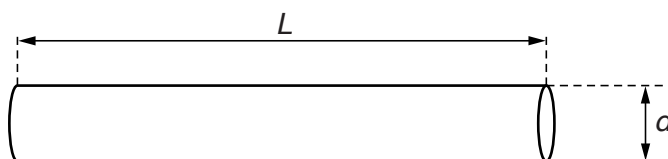
- (i) the mass, in kg, of a wooden metre rule,

mass = kg [1]

- (ii) the volume, in
- cm^3
- , of a cricket ball or a tennis ball.

volume = cm^3 [1]

- (b) A metal wire of length
- L
- has a circular cross-section of diameter
- d
- , as shown in Fig. 1.1.

**Fig. 1.1**

The volume V of the wire is given by the expression

$$V = \frac{\pi d^2 L}{4}.$$

The diameter, length and mass M are measured to determine the density of the metal of the wire. The measured values are:

$$d = 0.38 \pm 0.01 \text{ mm},$$

$$L = 25.0 \pm 0.1 \text{ cm},$$

$$M = 0.225 \pm 0.001 \text{ g}.$$

Calculate the density of the metal, with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

density = kg m^{-3} [5]

- 44 (a) Describe the effects, one in each case, of systematic errors and random errors when using a micrometer screw gauge to take readings for the diameter of a wire.

systematic errors:

.....

random errors:

.....[2]

- (b) Distinguish between precision and accuracy when measuring the diameter of a wire.

precision:

.....

accuracy:

.....[2]

- 45 (a) Define *density*.

.....

.....[1]

- (b) The mass m of a metal sphere is given by the expression $m = \frac{\pi d^3 \rho}{6}$

where ρ is the density of the metal and d is the diameter of the sphere.

Data for the density and the mass are given in Fig. 1.1.

quantity	value	uncertainty
ρ	8100 kg m^{-3}	$\pm 5\%$
m	7.5 kg	$\pm 4\%$

Fig. 1.1

- (i) Calculate the diameter d .

$d = \dots\dots\dots \text{ m}$ [1]

- (ii) Use your answer in (i) and the data in Fig. 1.1 to determine the value of d , with its absolute uncertainty, to an appropriate number of significant figures.

$d = \dots\dots\dots \pm \dots\dots\dots \text{ m}$ [3]

- 46 (a) Determine the SI base units of stress.
Show your working.

9702/21/M/J/17/Q1

base units[2]

- (b) A beam PQ is clamped so that the beam is horizontal. A mass M of 500 g is hung from end Q and the beam bends slightly, as illustrated in Fig. 1.1.

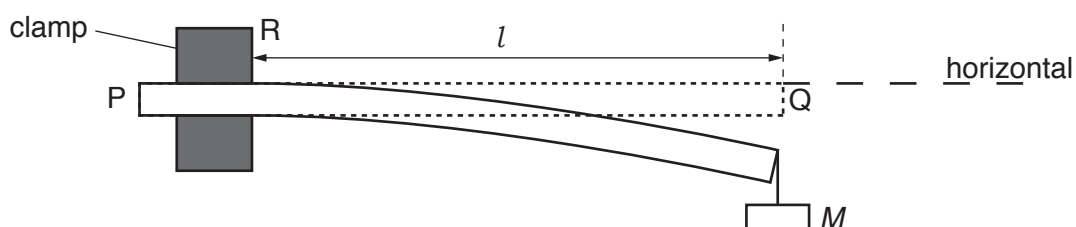


Fig. 1.1

The length l of the beam from the edge of the clamp R to end Q is 60.0 cm. The width b of the beam is 30.0 mm and the thickness d of the beam is 5.00 mm. The material of the beam has Young modulus E .

The mass M is made to oscillate vertically. The time period T of the oscillations is 0.58 s.

The period T is given by the expression

$$T = 2\pi \sqrt{\frac{4Ml^3}{Ebd^3}}.$$

- (i) Determine E in GPa.

$E =$ GPa [3]

(ii) The quantities used to determine E should be measured with accuracy and with precision.

1. Explain the difference between accuracy and precision.

accuracy:

.....

precision:

.....

[2]

2. In a particular experiment, the quantities l and T are measured with the same percentage uncertainty. State and explain which of these two quantities contributes more to the uncertainty in the value of E .

.....

.....[1]

47 1. State what is meant by *precision*.

9702/22/M/J/17/Q1(c)

.....

.....

2. Explain why the precision in the value of the resistivity is improved by using a micrometer screw gauge rather than a metre rule to measure the diameter of the wire.

.....

.....

.....

[2]

- 48** One end of a wire is connected to a fixed point. A load is attached to the other end so that the wire hangs vertically.

The diameter d of the wire and the load F are measured as $d = 0.40 \pm 0.02 \text{ mm}$,
 $F = 25.0 \pm 0.5 \text{ N}$.

- (a)** For the measurement of the diameter of the wire, state

- (i)** the name of a suitable measuring instrument,

.....[1]

- (ii)** how random errors may be reduced when using the instrument in **(i)**.

.....
[2]

- (b)** The stress σ in the wire is calculated by using the expression $\sigma = \frac{4F}{\pi d^2}$.

- (i)** Show that the value of σ is $1.99 \times 10^8 \text{ N m}^{-2}$.

[1]

- (ii)** Determine the percentage uncertainty in σ .

percentage uncertainty =% [2]

- (iii)** Use the information in **(b)(i)** and your answer in **(b)(ii)** to determine the value of σ , with its absolute uncertainty, to an appropriate number of significant figures.

$\sigma = \dots \pm \dots \text{ N m}^{-2}$ [2]

- 49 (a) An analogue voltmeter is used to take measurements of a constant potential difference across a resistor. For these measurements, describe **one** example of

(i) a systematic error,
[1]

(ii) a random error.
[1]

- (b) The potential difference across a resistor is measured as $5.0\text{ V} \pm 0.1\text{ V}$. The resistor is labelled as having a resistance of $125\ \Omega \pm 3\%$.

- (i) Calculate the power dissipated by the resistor.

power = W [2]

- (ii) Calculate the percentage uncertainty in the calculated power.

percentage uncertainty = % [2]

- (iii) Determine the value of the power, with its absolute uncertainty, to an appropriate number of significant figures.

power = $\pm$ W [2]

50 (a) Define *velocity*.

.....
[1]

(b) The speed v of a sound wave through a gas of pressure P and density ρ is given by the equation

$$v = \sqrt{\frac{kP}{\rho}}$$

where k is a constant that has no units.

An experiment is performed to determine the value of k . The data from the experiment are shown in Fig. 1.1.

quantity	value	uncertainty
v	$3.3 \times 10^2 \text{ m s}^{-1}$	$\pm 3\%$
P	$9.9 \times 10^4 \text{ Pa}$	$\pm 2\%$
ρ	1.29 kg m^{-3}	$\pm 4\%$

Fig. 1.1

(i) Use data from Fig. 1.1 to calculate k .

$k = \dots\dots\dots$ [2]

(ii) Use your answer in (b)(i) and data from Fig. 1.1 to determine the value of k , with its absolute uncertainty, to an appropriate number of significant figures.

$k = \dots\dots\dots \pm \dots\dots\dots$ [3]

- 51 (a) The diameter d of a cylinder is measured as $0.0125 \text{ m} \pm 1.6\%$.

Calculate the absolute uncertainty in this measurement.

absolute uncertainty = m [1]

- (b) The cylinder in (a) stands on a horizontal surface. The pressure p exerted on the surface by the cylinder is given by

$$p = \frac{4W}{\pi d^2}.$$

The measured weight W of the cylinder is $0.38 \text{ N} \pm 2.8\%$.

- (i) Calculate the pressure p .

$p = \dots\dots\dots \text{Nm}^{-2}$ [1]

- (ii) Determine the absolute uncertainty in the value of p .

absolute uncertainty = Nm^{-2} [2]

52 (a) Make estimates of:

(i) the mass, in g, of a new pencil

mass = g [1]

(ii) the wavelength of ultraviolet radiation.

wavelength = m [1]

(b) The period T of the oscillations of a mass m suspended from a spring is given by

$$T = 2\pi \sqrt{\frac{m}{k}}$$

where k is the spring constant of the spring.

The manufacturer of a spring states that it has a spring constant of $25 \text{ N m}^{-1} \pm 8\%$. A mass of $200 \times 10^{-3} \text{ kg} \pm 4 \times 10^{-3} \text{ kg}$ is suspended from the end of the spring and then made to oscillate.

(i) Calculate the period T of the oscillations.

$T =$ s [1]

(ii) Determine the value of T , with its absolute uncertainty, to an appropriate number of significant figures.

$T =$ $\pm$ s [3]

- 53 (a) Use an expression for work done, in terms of force, to show that the SI base units of energy are $\text{kg m}^2 \text{s}^{-2}$.

[2]

- (b) (i) The energy E stored in an electrical component is given by

$$E = \frac{Q^2}{2C}$$

where Q is charge and C is a constant.

Use this equation and the information in (a) to determine the SI base units of C .

SI base units [2]

- (ii) Measurements of a constant current in a wire are taken using an analogue ammeter.

For these measurements, describe one possible cause of:

1. a random error

.....

2. a systematic error.

.....

[2]

54. 9702/23/O/N/20/Q1

(a) An electromagnetic wave has a wavelength of $85\text{ }\mu\text{m}$.

(i) State the wavelength, in m, of the wave.

wavelength = m [1]

(ii) Calculate the frequency, in THz, of the wave.

frequency = THz [2]

(iii) State the name of the region of the electromagnetic spectrum that contains this wave.

..... [1]

(b) The current I in a coil of wire produces a magnetic field. The energy E stored in the magnetic field is given by

$$E = \frac{I^2 L}{2}$$

where L is a constant.

The manufacturer of the coil states that the value of L , in SI base units, is $7.5 \times 10^{-6} \pm 5\%$.
The current I in the coil is measured as $(0.50 \pm 0.02)\text{ A}$.

The values of L and I are used to calculate E .

Determine the percentage uncertainty in the value of E .

percentage uncertainty = % [2]

[Total: 6]

55. 9702/21/M/J/21/Q1

(a) Define *density*.

.....
 [1]

(b) Fig. 1.1 shows a solid pyramid with a square base.

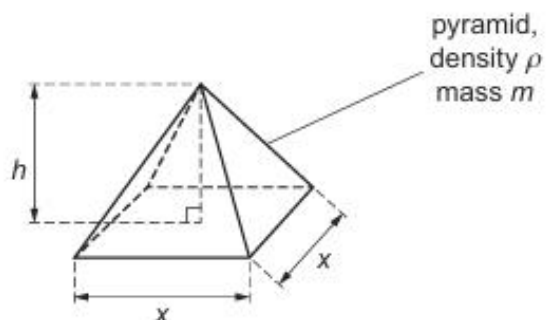


Fig. 1.1

The mass m of the pyramid is given by

$$m = \frac{1}{3} \rho h x^2$$

where ρ is the density of the material of the pyramid,
 h is the height, and
 x is the length of each side of the base.

Measurements are taken as shown in Table 1.1.

Table 1.1

quantity	measurement	percentage uncertainty
m	19.5 g	$\pm 2\%$
x	4.0 cm	$\pm 5\%$
h	4.8 cm	$\pm 4\%$

(i) Calculate the absolute uncertainty in length x .

absolute uncertainty = cm [1]

- (ii) The density ρ is calculated from the measurements in Table 1.1.

Determine the percentage uncertainty in the calculated value of ρ .

percentage uncertainty = % [2]

- (c) The square base of the pyramid in (b) rests on the horizontal surface of a bench.

Use data from Table 1.1 to calculate the average pressure of the pyramid on the surface of the bench. The uncertainty in your answer is not required.

pressure = Pa [3]

[Total: 7]

56. 9702/23/M/J/21/Q1

- (a) A property of a vector quantity, that is not a property of a scalar quantity, is direction.
For example, velocity has direction but speed does not.

- (i) State **two** other scalar quantities and **two** other vector quantities.

scalar quantities: and

vector quantities: and

[2]

- (ii) State **two** properties that are possessed by both scalar and vector physical quantities.

1.

2.

[2]

- (b) A ship at sea is travelling with a velocity of 13 m s^{-1} in a direction 35° east of north in still water, as shown in Fig. 1.1.

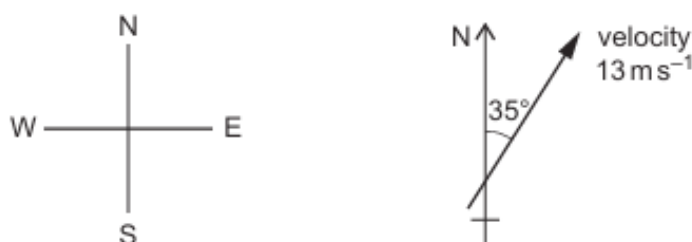


Fig. 1.1

- (i) Determine the magnitudes of the components of the velocity of the ship in the north and the east directions.

north component of velocity = m s^{-1}

east component of velocity = m s^{-1}

[2]

- (ii) The ship now experiences a tidal current. The water in the sea moves with a velocity of 2.7 m s^{-1} to the west.

Calculate the resultant velocity component of the ship in the east direction.

resultant east component of velocity = m s^{-1} [1]

- (iii) Use your answers in (b)(i) and (b)(ii) to determine the magnitude of the resultant velocity of the ship.

magnitude of resultant velocity = m s^{-1} [2]

- (iv) Use your answers in (b)(i) and (b)(ii) to determine the angle between north and the resultant velocity of the ship.

angle = $^{\circ}$ [2]

[Total: 11]

57. 9702/21/O/N/21/Q1

(a) Define *density*.

.....
 [1]

(b) A smooth pebble, made from uniform rock, has the shape of an elongated sphere as shown in Fig. 1.1.

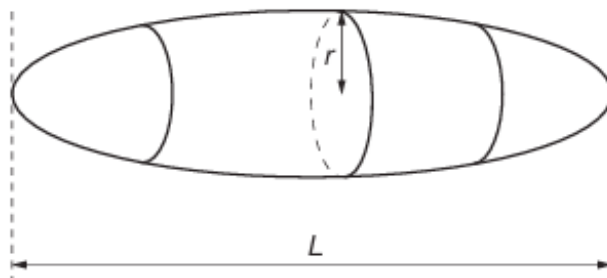


Fig. 1.1

The length of the pebble is L . The cross-section of the pebble, in the plane perpendicular to L , is circular with a maximum radius r .

A student investigating the density of the rock makes measurements to determine the values of L , r and the mass M of the pebble as follows:

$$\begin{aligned} L &= (0.1242 \pm 0.0001)\text{m} \\ r &= (0.0420 \pm 0.0004)\text{m} \\ M &= (1.072 \pm 0.001)\text{kg}. \end{aligned}$$

(i) State the name of a measuring instrument suitable for making this measurement of L .

..... [1]

(ii) Determine the percentage uncertainty in the measurement of r .

percentage uncertainty = % [1]

- (c) The density ρ of the rock from which the pebble in (b) is composed is given by

$$\rho = \frac{Mr^n}{kL}$$

where n is an integer and k is a constant, with no units, that is equal to 2.094.

- (i) Use SI base units to show that n is equal to -2 .

[2]

- (ii) Calculate the percentage uncertainty in ρ .

percentage uncertainty = % [3]

- (iii) Determine ρ with its absolute uncertainty. Give your values to the appropriate number of significant figures.

$\rho = (\text{.....} \pm \text{.....}) \text{ kg m}^{-3}$ [3]

[Total: 11]

58. 9702/22/O/N/21/Q1

- (a) A unit may be stated with a prefix that represents a power-of-ten multiple or submultiple.

Complete Table 1.1 to show the name and symbol of each prefix and the corresponding power-of-ten multiple or submultiple.

Table 1.1

prefix	power-of-ten multiple or submultiple
kilo (k)	10^3
tera (T)	
()	10^{-12}

[2]

- (b) In the following list, underline all the units that are SI base units.

ampere

coulomb

metre

newton

[1]

- (c) The potential difference V between the two ends of a uniform metal wire is given by

$$V = \frac{4\rho LI}{\pi d^2}$$

where d is the diameter of the wire,

I is the current in the wire,

L is the length of the wire,

and ρ is the resistivity of the metal.

For a particular wire, the percentage uncertainties in the values of some of the above quantities are listed in Table 1.2.

Table 1.2

quantity	percentage uncertainty
d	$\pm 3.0\%$
I	$\pm 2.0\%$
L	$\pm 2.5\%$
V	$\pm 3.5\%$

The quantities listed in Table 1.2 have values that are used to calculate ρ as $4.1 \times 10^{-7} \Omega \text{m}$.

For this value of ρ , calculate:

- (i) the percentage uncertainty

percentage uncertainty =% [2]

- (ii) the absolute uncertainty.

absolute uncertainty = Ωm [1]

[Total: 6]

59. 9702/22/M/J/22/Q1

(a) In the following list, underline all units that are SI base units.

ampere

degree Celsius

kilogram

newton

[1]

(b) Fig. 1.1 shows a horizontal beam clamped at one end with a block attached to the other end.

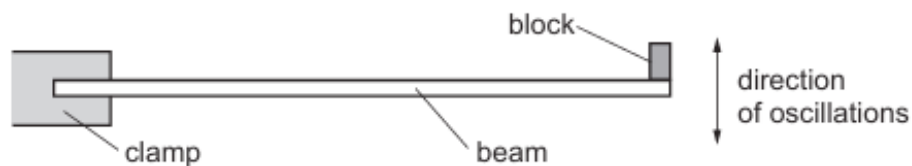


Fig. 1.1

The block is made to oscillate vertically.

The Young modulus E of the material of the beam is given by

$$E = \frac{kM}{T^2}$$

where M is the mass of the block,

T is the period of the oscillations

and k is a constant.

A student determines the values and percentage uncertainties of k , M and T .

Table 1.1 lists the percentage uncertainties.

Table 1.1

quantity	percentage uncertainty
k	$\pm 2.1\%$
M	$\pm 0.6\%$
T	$\pm 1.5\%$

The student uses the values of k , M and T to calculate the value of E as $8.245 \times 10^9 \text{ Pa}$.

(i) Calculate the percentage uncertainty in the value of E .

percentage uncertainty = % [2]

- (ii) Use your answer in **(b)(i)** to determine the value of E , with its absolute uncertainty, to an appropriate number of significant figures.

$$E = (\dots\dots\dots \pm \dots\dots\dots) \times 10^9 \text{ Pa} \quad [2]$$

[Total: 5]

60. 9702/23/M/J/22/Q1

A solid metal sphere has a diameter of $(3.42 \pm 0.02)\text{cm}$ and a mass of $(67 \pm 2)\text{g}$.

(a) Calculate the density, in g cm^{-3} , of the metal.

density = g cm^{-3} [3]

(b) Determine the percentage uncertainty in the density.

percentage uncertainty =% [2]

[Total: 5]

61. 9702/21/O/N/22/Q1

- (a) The boxes in Fig. 1.1 contain terms on the left-hand side and examples of these terms on the right-hand side.

Draw a line between each term on the left and the correct example on the right.

base quantity	coulomb
base unit	electric current
derived quantity	force
derived unit	kilogram

Fig. 1.1

[2]

- (b) A set of experimental measurements is described as precise and not accurate.

State what is meant by:

- (i) precise

.....
..... [1]

- (ii) not accurate.

.....
..... [1]

- (c) An object of mass m travels with speed v in a circle of radius r . The force F acting on the object is given by

$$F = \frac{mv^2}{r}.$$

The percentage uncertainties of three of the quantities are given in Table 1.1.

Table 1.1

quantity	percentage uncertainty
F	$\pm 3\%$
m	$\pm 4\%$
r	$\pm 5\%$

The value of v is determined from F , m and r .

- (i) Calculate the percentage uncertainty in v .

percentage uncertainty = % [2]

- (ii) The value of v is 15.0 ms^{-1} .

Calculate the absolute uncertainty in v .

absolute uncertainty = ms^{-1} [1]

[Total: 7]

62. 9702/22/O/N/22/Q1

- (a)** State what is meant by work done.

.....
..... [1]

- (b)** Use the answer to **(a)** to determine the SI base units of power.

SI base units [2]

- (c)** The maximum useful output power P of a car travelling on a horizontal road is given by

$$P = v^3 b$$

where v is the maximum speed of the car and b is a constant.

For the car,

$$P = 84 \text{ kW} \pm 5\%$$

and $b = 0.56 \pm 7\%$ in SI units.

- (i)** Calculate the value of v .

$v = \dots\dots\dots \text{ m s}^{-1}$ [2]

- (ii)** Determine the absolute uncertainty in the value of v .

absolute uncertainty = $\dots\dots\dots \text{ m s}^{-1}$ [2]

[Total: 7]

63. 9702/23/O/N/22/Q1

The rate of flow Q of a liquid along a narrow pipe of length L and radius r is given by

$$Q = \frac{\alpha r^4}{L}$$

where α is a constant.

An experiment is carried out to determine the value of α . The data from the experiment are shown in Table 1.1.

Table 1.1

quantity	value	percentage uncertainty
Q	$2.72 \times 10^{-8} \text{ m}^3 \text{ s}^{-1}$	$\pm 3\%$
r	$7.1 \times 10^{-5} \text{ m}$	$\pm 2\%$
L	$2.5 \times 10^{-2} \text{ m}$	$\pm 4\%$

- (a) Use information in Table 1.1 to show that the SI base unit of α is s^{-1} .

[1]

- (b) Show that the percentage uncertainty in α is 15%.

[1]

- (c) Calculate α with its absolute uncertainty. Give your answer to an appropriate number of significant figures.

$$\alpha = (\dots\dots\dots \pm \dots\dots\dots) \times 10^7 \text{ s}^{-1} \quad [3]$$

[Total: 5]

64. 9702/22/F/M/23/Q1

- (a) Underline **all** the SI base units in the following list.

ampere coulomb current kelvin newton [1]

- (b) A toy car moves in a horizontal straight line. The displacement s of the car is given by the equation

$$s = \frac{v^2}{2a}$$

where a is the acceleration of the car and v is its final velocity.

State **two** conditions that apply to the motion of the car in order for the above equation to be valid.

1

2

[2]

- (c) An experiment is performed to determine the acceleration of the car in (b). The following measurements are obtained:

$$s = 3.89 \text{ m} \pm 0.5\%$$

$$v = 2.75 \text{ m s}^{-1} \pm 0.8\%$$

- (i) Calculate the acceleration a of the car.

$$a = \dots \text{ m s}^{-2} \quad [1]$$

- (ii) Determine the percentage uncertainty, to two significant figures, in a .

$$\text{percentage uncertainty} = \dots \% \quad [2]$$

- (iii) Use your answers in (c)(i) and (c)(ii) to determine the absolute uncertainty in the calculated value of a .

absolute uncertainty = ms^{-2} [1]

[Total: 7]

65. 9702/21/M/J/23/Q1

(a) (i) Define power.

.....
..... [1]

(ii) Use the definition of power to show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

[1]

(b) The intensity I of a sound wave moving through a gas is given by

$$I = f^2 A^2 v k$$

where f is the frequency of the wave,
 A is the amplitude of the wave,
 v is the speed of the wave
and k is a constant that depends on the gas.

Determine the SI base units of k .

SI base units [3]

[Total: 5]

66. 9702/22/M/J/23/Q1

- (a) (i)** Define pressure.

.....
 [1]

- (ii)** Use the answer to **(a)(i)** to show that the SI base units of pressure are $\text{kg m}^{-1} \text{s}^{-2}$.

[1]

- (b)** A horizontal pipe has length L and a circular cross-section of radius R . A liquid of density ρ flows through the pipe. The mass m of liquid flowing through the pipe in time t is given by

$$m = \frac{\pi(p_2 - p_1)R^4 \rho t}{8kL}$$

where p_1 and p_2 are the pressures at the ends of the pipe and k is a constant.

Determine the SI base units of k .

SI base units [3]

- (c)** An experiment is performed to determine the value of k by measuring the values of the other quantities in the equation in **(b)**.

The values of L and R each have a percentage uncertainty of 2%.

State and explain, quantitatively, which of these two quantities contributes more to the percentage uncertainty in the calculated value of k .

.....

 [1]

67. 9702/23/M/J/23/Q1

A well has a depth of 36 m from ground level to the surface of the water in the well, as shown in Fig. 1.1.

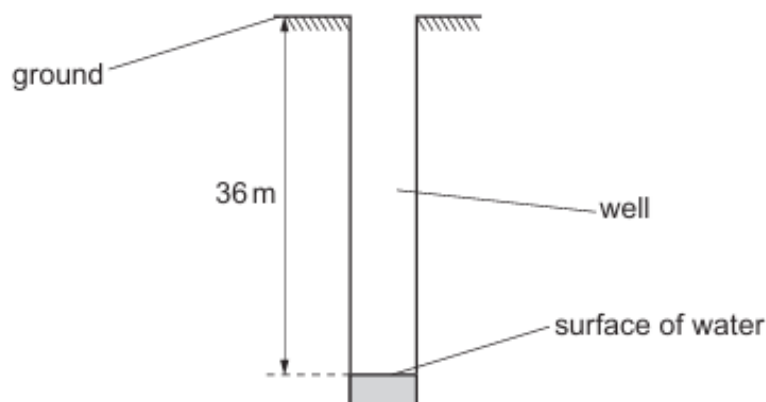


Fig. 1.1 (not to scale)

A student wishes to find the depth of the well. The student plans to drop a stone down the well and record the time taken from releasing the stone to hearing the splash made by the stone as it enters the water.

- (a) Assume that air resistance is negligible and that the stone is released from rest.

Calculate the time taken for the stone to fall from ground level to the surface of the water.

time = s [2]

- (b) The time recorded by the student using a stop-watch is not equal to the time in (a).

Suggest **three** possible reasons, other than the effect of air resistance, for this difference.

1

.....

2

.....

3

.....

[3]

- (c) The student repeats the experiment three times and uses the results to calculate the depth of the well. The values are shown in Table 1.1.

Table 1.1

	1st experiment	2nd experiment	3rd experiment
depth/m	54.4	53.9	54.1

The true depth of the well is 36.0 m. Explain why these results may be described as precise but not accurate.

.....

.....

.....

..... [2]

[Total: 7]

68. 9702/21/O/N/23/Q1

- (a) Compare scalar and vector quantities.

.....

.....

..... [2]

- (b) The radius of a small sphere is determined from a measurement of the volume of the sphere. The sphere is submerged in water, displacing some of the water into a measuring cylinder as shown in Fig. 1.1.

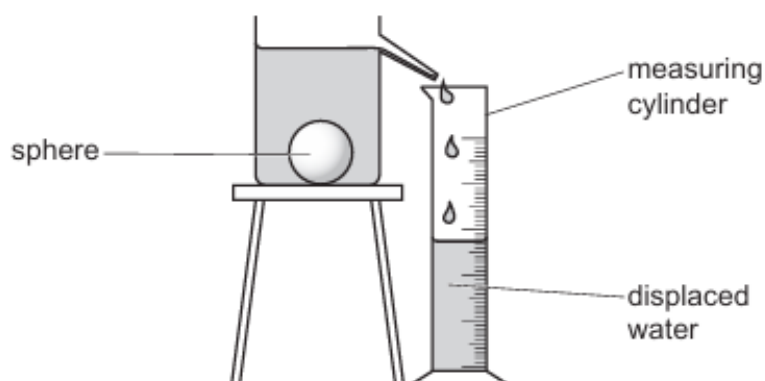


Fig. 1.1 (not to scale)

The measured volume of displaced water is $(28.0 \pm 0.5) \text{ cm}^3$.

Calculate:

- (i) the radius, in cm, of the sphere

radius = cm [1]

- (ii) the percentage uncertainty in the radius of the sphere.

percentage uncertainty = % [2]

[Total: 5]

69. 9702/23/O/N/23/Q1

- (a) Table 1.1 lists some physical quantities. Complete the table by placing a tick (✓) next to the scalar quantities.

Table 1.1

acceleration	
charge	
momentum	
power	
upthrust	

[1]

- (b) A uniform cylinder has diameter D , length L and mass M .
The density ρ of the cylinder is given by

$$\rho = \frac{4M}{\pi D^2 L}.$$

Table 1.2 shows the data obtained from an experiment to determine the density of the cylinder.

Table 1.2

quantity	measurement	percentage uncertainty
D	$(26.2 \pm 0.1)\text{mm}$	%
L	$(162 \pm 1)\text{mm}$	%
M	$(247 \pm 1)\text{g}$	0.4%

- (i) Calculate the percentage uncertainties in D and L . Write your answers in Table 1.2.

[1]

(ii) Calculate the density of the cylinder. Give your answer to three significant figures.

density = kg m^{-3} [2]

(iii) Calculate the percentage uncertainty in the density.

percentage uncertainty =% [2]

[Total: 6]

ANSWERS

- 1 (a) e.g. time (s), current (A), temperature (K), amount of substance (mol),
luminous intensity (cdl)
1 each, max 3 B3 [3]
- (b) density = mass / volume C1
unit of density: kg m^{-3} C1
unit of acceleration: m s^{-2} C1
unit of pressure: $\text{kg m}^{-3} \text{ m s}^{-2} \text{ m}$ B1
 $\text{kg m}^{-1} \text{ s}^{-2}$ B1 [5]
(allow 4/5 for solution in terms of only dimensions)
-
- 2 10^{-9} B1
c B1
mega B1
tera B1 [4]
-
- 3 (a) length, current, temperature, amount of substance, (luminous intensity)
any three, 1 each B3 [3]
- (b) (i) $F: \text{kg m s}^{-2}$ B1
 $\rho: \text{kg m}^{-3}$ B1
 $v: \text{m s}^{-1}$ B1 [3]
- (ii) some working e.g. $\text{kg m s}^{-2} = \text{m}^2 \text{ kg m}^{-3} (\text{m s}^{-1})^k$ M1
hence $k = 2$ A1 [2]
-
- 4 (a) (i) scalar quantity has magnitude (allow size) B1
vector quantity has magnitude and direction B1 [2]
- (ii) 1. temperature: scalar B1 [1]
2. acceleration: vector B1 [1]
3. resistance: scalar B1 [1]
- (b) either triangle / parallelogram with correct shape C1
tension = 14.3 N (allow $\pm 0.5 \text{ N}$) A2 [3]
- (if $> \pm 0.5 \text{ N}$ but $\leq \pm 1 \text{ N}$, allow 1 mark)
- or $R = 25 \cos 35^\circ$ (C1)
 $T = R \tan 35^\circ$ (C1)
 $T = 14.3 \text{ N}$ (A1)
- or $T = 25 \sin 35^\circ$ (C2)
 $T = 14.3 \text{ N}$ (A1)
- or R and T resolved vertically and horizontally (C2)
leading to $T = 14.3 \text{ N}$ (A1)
-

5 (a) scalar has only magnitude vector has magnitude and direction	B1 B1 [2]
(b) kinetic energy, mass, power all three underlined	B1 [1]
6 (a) (i) V units: m^3 (allow metres cubed or cubic metres) (ii) Pressure units: $\text{kg m s}^{-2} / \text{m}^2$ (allow use of $P = \rho gh$) Units: $\text{kg m}^{-1} \text{s}^{-2}$	A1 [1] M1 A0 [1]
(b) V / t units: $\text{m}^3 \text{s}^{-1}$ Clear substitution of units for P , r^4 and l	B1 M1
$C = \frac{\pi P r^4}{8 V t^{-1} l} = \frac{\text{kg m}^{-1} \text{s}^{-2} \text{m}^4}{\text{m}^3 \text{s}^{-1} \text{m}}$	
Units: $\text{kg m}^{-1} \text{s}^{-1}$ (δ or π in final answer –1. Use of dimensions max 2/3)	A1 [3]
7 (e) (i) arrow to the right of plane direction (about 4° to 24°) (ii) scale diagram drawn or use of cosine formula $v^2 = 250^2 + 36^2 - 2 \times 250 \times 36 \times \cos 45^\circ$ or resolving $v = [(36 \cos 45^\circ)^2 + (250 - 36 \sin 45^\circ)^2]^{1/2}$ resultant velocity = 226 (220 – 240 for scale diagram) m s^{-1} allow one mark for values 210 to 219 or 241 to 250 m s^{-1} or use of formula ($v^2 = 51068$) $v = 230$ (226) m s^{-1}	B1 [1] C1 A1 [2]
8 (a) force: kg m s^{-2} (b) (i) I^2 : A^2 l : m x : m K : $\text{kg m s}^{-2} \text{A}^{-2}$	A1 [1] C1 A1 [2]
(ii) curve of the correct shape (for inverse proportionality) clearly approaching each axis but never touching the axis	M1 A1 [2]
(iii) curving upwards and through origin	A1 [1]
9 (a) kelvin / K ampere / amp / A [allow mole / mol and candela / Cd]	B1 B1 [2]
(b) (i) energy OR work = force $\times$ distance [allow any energy expression] units: $\text{kg m s}^{-2} \times \text{m}$ OR $\text{kg (m s}^{-1})^2$ for $\frac{1}{2} mv^2$ or mc^2 (ignore any numerical factor) $= \text{kg m}^2 \text{s}^{-2}$	C1 M1 A0 [2]
(ii) units: ρ : kg m^{-3} g : m s^{-2} A : m^2 l_0 : m C : $\text{kg m}^2 \text{s}^{-2} / \text{kg}^2 \text{m}^{-6} \text{m}^2 \text{s}^{-4} \text{m}^2 \text{m}^3$ [any subject] $= \text{kg}^{-1} \text{m s}^2$ (allow $\text{m s}^2 / \text{kg}$)	C1 C1 A1 [3]

10 (a) power = energy/time or work done/time	B1
force: kg m s^{-2} (including from mg in mgh or Fv)	
or kinetic energy ($\frac{1}{2}mv^2$): $\text{kg (m s}^{-1})^2$	B1
(distance: m and (time) $^{-1}$: s^{-1}) and hence power: $\text{kg m s}^{-2} \text{ m s}^{-1} = \text{kg m}^2 \text{ s}^{-3}$	B1 [3]
(b) Q/t : $\text{kg m}^2 \text{ s}^{-3}$	C1
A: m^2 and x: m and T: K	C1
correct substitution into $C = (Qx)/tAT$ or equivalent, or with cancellation	C1
units of C: $\text{kg m s}^{-3} \text{ K}^{-1}$	A1 [4]
11 (a) current, mass and temperature two correct 2/2, one omission or error 1/2	A2 [2]
(b) σ : no units, V: m^3	C1
E_p : $\text{kg m}^2 \text{ s}^{-2}$	C1
C: $\text{kg m}^2 \text{ s}^{-2} \times \text{m}^{-3} = \text{kg m}^{-1} \text{ s}^{-2}$	A1 [3]
12 (a) temperature current (allow amount of substance and luminous intensity)	B1
	B1 [2]
(b) base units of force constant: $\text{kg m s}^{-2} \text{ m}^{-1}$ or kg s^{-2}	B1
base units of time and mass: s and kg	C1
base units of C: $\text{s (kg s}^{-2} / \text{kg)}^{1/2}$ cancelling to show no units	B1 [3]
13 (c) (i) arrow drawn up to the left of 7.5 N force approximately 5° to 40° to west of north	A1 [1]
(ii) 1. correct vector triangle or working to show magnitude of resultant force = 6.6 N allow 6.5 to 6.7 N if scale diagram	M1 [1]
2. magnitude of acceleration = $6.6 / 0.75$ [scale diagram: $(6.5 \text{ to } 6.7) / 0.75$]	C1
= 8.8 m s^{-2} [scale diagram: $8.7 - 8.9 \text{ m s}^{-2}$]	A1 [2]
(iii) 19° [<u>use of scale diagram</u> allow 17° to 21° (a diagram must be seen)]	B1 [1]

14 (a)	displacement/velocity/acceleration/momentum/etc. three correct (none wrong) 2, two correct (none or one wrong) 1	A2	[2]
(b) (i)	$Y = 70\text{ N}$ [allow 71 N as $+\frac{1}{2}$ small square on graph]	A1	[1]
(ii)	$\theta = 90^\circ$ (for equilibrium) the direction of Y must be <u>opposite</u> to Z or using $Y \sin \theta = Z$, hence $\sin \theta = 70 / 70 = 1$, $\theta = 90^\circ$	M1	
(iii) 1.	$Y \cos \theta = 160$ and $Y \sin \theta = 70$ $\tan \theta = 70/160$ hence $\theta = 23.6^\circ$ (24°)	A1	[2]
2.	$Y = 160 / \cos 23.6^\circ$ or $70 / \sin 23.6^\circ$ $= 174.6$ or 175 or 170 N or: $160^2 + 70^2 = Y^2$ $Y = 174.6$ or 175 or 170 N	C1 A1	[2]
(c)	(equilibrium not possible as) there is no vertical component from Y to balance Z	(C1) (A1) B1	[1]
15 (a)	ampere kelvin (allow mole and candela)	B1 B1	[2]
(b) (i)	stress: N m^{-2} $\text{kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$	C1 A1	[2]
(ii)	Young modulus = stress/strain and <u>strain has no units</u> hence units: $\text{kg m}^{-1} \text{s}^{-2}$	B1	[1]
16 (a)	power = work/time or energy/time or (force $\times$ distance)/time $= \text{kg m s}^{-2} \times \text{m s}^{-1} = \text{kg m}^2 \text{s}^{-3}$	B1 A1	[2]
(b)	power = VI [or V^2/R and $V = IR$ or I^2R and $V = IR$] (units of V): $\text{kg m}^2 \text{s}^{-3} \text{A}^{-1}$	B1 B1	[2]
17 (a)	(work =) force $\times$ distance or force $\times$ displacement or ($W =$) $F \times d$ units of work: $\text{kg m s}^{-2} \times \text{m} = \text{kg m}^2 \text{s}^{-2}$	M1 A1	[2]
(b)	(p.d. =) $\frac{\text{work (done) or energy (transformed) (from electrical to other forms)}}{\text{charge}}$	B1	[1]
(c)	$R = V/I$ units of V : $\text{kg m}^2 \text{s}^{-2} / \text{A s}$ and units of I : A or $R = P/I^2$ [or $P = VI$ and $V = IR$] units of P : $\text{kg m}^2 \text{s}^{-3}$ and units of I : A or $R = V^2/P$ units of V : $\text{kg m}^2 \text{s}^{-2} / \text{A s}$ and units of P : $\text{kg m}^2 \text{s}^{-3}$ units of R : $(\text{kg m}^2 \text{s}^{-2} / \text{A}^2 \text{s}) = \text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$	B1 C1 (B1) (C1) (B1) (C1) A1	[3]

18	(a)	150 or 1.5×10^2 Gm	A1	[1]
	(b)	distance = $2 \times (42.3 - 6.38) \times 10^6$ (= 7.184×10^7 m) (time =) $7.184 \times 10^7 / (3.0 \times 10^8) = 0.24$ (0.239) s	C1 A1	[2]
	(c)	units of pressure P : $\text{kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$	M1	
		units of density ρ : kg m^{-3} and speed v : m s^{-1}	M1	
		simplification for units of C : $C = v^2 \rho / P$ units: $(\text{m}^2 \text{s}^{-2} \text{kg m}^{-3}) / \text{kg m}^{-1} \text{s}^{-2}$ and cancelling to give no units for C	A1	[3]
	(d)	energy and power (<i>both underlined and no others</i>)	A1	[1]
	(e) (i)	vector triangle of correct orientation	M1	
		three arrows for the velocities in the correct directions	A1	[2]
	(ii)	length measured from scale diagram 5.2 ± 0.2 cm or components of boat speed determined parallel and perpendicular to river flow	C1	
		velocity = 2.6 m s^{-1} (allow $\pm 0.1 \text{ m s}^{-1}$)	A1	[2]
19	(a)	scalars: energy, power and time	A1	
		vectors: momentum and weight	A1	[2]
	(b) (i)	triangle with right angles between 120 m and 80 m, <u>arrows</u> in correct direction and result displacement from start to finish <u>arrow</u> in correct direction and labelled R	B1	[1]
	(ii) 1.	average speed (= $200 / 27$) = 7.4 m s^{-1}	A1	[1]
	2.	resultant displacement (= $[120^2 + 80^2]^{1/2}$) = 144 (m) average velocity (= $144 / 27$) = $5.3(3) \text{ m s}^{-1}$	C1 A1	
		direction (= $\tan^{-1} 80 / 120$) = 34° (33.7)	A1	[3]
20	(a) (i)	force / area (normal to the force)	B1	[1]
	(ii)	($p = F / A$ so) units: $\text{kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$ allow use of other correct equations: e.g. ($\Delta p = \rho g \Delta h$ so) $\text{kg m}^{-3} \text{m s}^{-2} \text{m} = \text{kg m}^{-1} \text{s}^{-2}$ e.g. ($p = W / \Delta V$ so) $\text{kg m s}^{-2} \text{m} / \text{m}^3 = \text{kg m}^{-1} \text{s}^{-2}$	A1	[1]
	(b)	units for m : kg, t : s and ρ : kg m^{-3}	C1	
		units of C : kg / s ($\text{kg m}^{-3} \text{kg m}^{-1} \text{s}^{-2}$) ^{1/2} or units of C^2 : $\text{kg}^2 / \text{s}^2 \text{kg m}^{-3} \text{kg m}^{-1} \text{s}^{-2}$	C1	
		units of C : m^2	A1	[3]

21 (a) kelvin, mole, ampere, candela **B1**
any two

(b) use of resistivity = RA / l and $V = IR$ (to give $\rho = VA / Il$) **C1**

units of V : (work done / charge) $\text{kg m}^2 \text{s}^{-2} (\text{A s})^{-1}$ **C1**

units of resistivity: $(\text{kg m}^2 \text{s}^{-3} \text{A}^{-1} \text{A}^{-1} \text{m})$ **A1**
 $= \text{kg m}^3 \text{s}^{-3} \text{A}^{-2}$

or

use of $R = \rho L / A$ and $P = I^2 R$ (gives $\rho = PA / I^2 L$) **(C1)**

units of P : $\text{kg m}^2 \text{s}^{-3}$ **(C1)**

units of resistivity: $(\text{kg m}^2 \text{s}^{-3} \times \text{m}^2) / (\text{A}^2 \times \text{m})$ **(A1)**
 $= \text{kg m}^3 \text{s}^{-3} \text{A}^{-2}$

22 units of F : kg ms^{-2} **C1**

units of ρ : kg m^{-3} **and** units of v : ms^{-1} **C1**

units of K : $\text{kg ms}^{-2} / [\text{kg m}^{-3} (\text{ms}^{-1})^2]$ **A1**
 $= \text{m}^2$

23 (a) (i) work (done) / time (taken) **or** energy (transferred) / time (taken) **B1**

(ii) Correct substitution of base units of all quantities into any correct equation for power. **A1**

Examples:

$$(P = E / t \text{ or } W / t \text{ gives}) \text{ kg m}^2 \text{s}^{-2} / \text{s} = \text{kg m}^2 \text{s}^{-3}$$

$$(P = Fs / t \text{ or } mgh / t \text{ gives}) \text{ kg m s}^{-2} \text{m} / \text{s} = \text{kg m}^2 \text{s}^{-3}$$

$$(P = \frac{1}{2}mv^2 / t \text{ gives}) \text{ kg (m s}^{-1})^2 / \text{s} = \text{kg m}^2 \text{s}^{-3}$$

$$(P = Fv \text{ gives}) \text{ kg m s}^{-2} \text{m s}^{-1} = \text{kg m}^2 \text{s}^{-3}$$

$$(P = VI \text{ gives}) \text{ kg m}^2 \text{s}^{-2} \text{A}^{-1} \text{s}^{-1} \text{A} = \text{kg m}^2 \text{s}^{-3}$$

(b) (i) units of A : m^2 **and** units of T : K **C1**

units of k : $\text{kg m}^2 \text{s}^{-3} / \text{m}^2 \text{K}^4$ **A1**
 $= \text{kg s}^{-3} \text{K}^{-4}$

(ii) curve from the origin with increasing gradient **B1**

24 (a) a scalar has magnitude (only) **B1**
a vector has magnitude and direction **B1**

(b) power: scalar **B2**
temperature: scalar
momentum: vector

(two correct 1 mark, all three correct 2 marks)

(c) (i) arrow labelled R in a direction from 5° to 20° north of west **B1**

(c)(ii) $v^2 = 28^2 + 95^2 - (2 \times 28 \times 95 \times \cos 115^\circ)$ **C1**

or

$$v^2 = [(95 + 28 \cos 65^\circ)^2 + (28 \sin 65^\circ)^2]$$

$v = 110 \text{ ms}^{-1}$ (109.8 ms⁻¹) **A1**

or (scale diagram method)

triangle of velocities drawn **(C1)**

$v = 110 \text{ ms}^{-1}$ (allow 108–112 ms⁻¹) **(A1)**

25 (a) rate of change of momentum **B1**

(b) kg ms⁻² **A1**

(c) units for Q: As **and** for r: m **C1**

units for $\varepsilon = (\text{As} \times \text{As}) / (\text{kg ms}^{-2} \times \text{m}^2)$ **A1**

$$= \text{A}^2 \text{kg}^{-1} \text{m}^{-3} \text{s}^4$$

26 (a) (i) potential difference / current **B1**

(ii) $R = 4.0 \times 10^9 (\Omega)$ **C1**

$$I = 0.60 / 4.0 \times 10^9 = 1.5 \times 10^{-10} (\text{A})$$
 A1

$$I = 150 \text{ pA}$$

(b) units of energy: kg m² s⁻² **C1**

units of charge: As **C1**

units of potential difference: (kg m² s⁻² / As =) kg m² A⁻¹ s⁻³ **A1**

27 (a) scalar quantity has (only) magnitude **B1**

vector quantity has magnitude and direction **B1**

(b)(i) $E = F / Q$ **C1**

$$= \text{kg m s}^{-2} / \text{As} = \text{kg m A}^{-1} \text{s}^{-3}$$
 A1

(ii) $b = Q / x^2 E$ **C1**

$$= \text{As} / \text{m}^2 \text{kg m A}^{-1} \text{s}^{-3}$$

$$= \text{A}^2 \text{s}^4 \text{kg}^{-1} \text{m}^{-3}$$
 A1

28 (a)	(i)	car uses $210 / 14 = 15$ litres of fuel	C1	
		volume reading = 45 litres	A1	[2]
	(ii)	from 'full' to '3/4' mark	B1	[1]
(b)	(i)	line/graph does not pass through ('empty, 0) / there is an intercept (do not allow 'non-linear')	B1	[1]
	(ii)	(meter shows zero fuel when there is some left in the tank so) acts as a 'reserve'	B1	[1]
29 (a)	(i)	either 1.55% or 1.6% ... (not 1.5 or 2)	A1	[1]
	(ii)	either 1.09% or 1.1% ... (not 1.0 or 1)	A1	[1]
	(b)	answer of {(ii) + 2 × (i)} to any number of sig. fig. either 4.2% or 4.3%	A1	[1]
(c)	(i)	either the value has more significant figures than the data or uncertainty of ± 0.4 renders more than 2 s.f. meaningless)	B1	[1]
	(ii)	uncertainty in $g = \pm 0.41 / \pm 0.42$ to any number of s.f.	C1	
		$g = (9.8 \pm 0.4) \text{ m s}^{-2}$	A1	[2]
30 (a)		micrometer/screw gauge/digital callipers	B1	[1]
(b)	(i)	look/check for zero error	B1	[1]
	(ii)	take several readings	M1	
		around the circumference/along the wire	A1	[2]
31 (a)	(i)	metre rule / tape (not 'rule')	B1	[1]
	(ii)	micrometer (screw gauge) / digital caliper	B1	[1]
	(iii)	ammeter and voltmeter / ohmmeter / multimeter on 'ohm' setting	B1	[1]
(b)	(i)	resistivity = RA / L	C1	
		$= [7.5 \times \pi \times (0.38 \times 10^{-3})^2 / 4] / 1.75$	M1	
		$= 4.86 \times 10^{-7} \Omega \text{ m}$	A0	[2]
	(ii)	(uncertainty in R =) $[0.2 / 7.5] \times 100 = 2.7\%$		
		and (uncertainty in L =) $[3 / 1750] \times 100 = 0.17\%$	C1	
		(uncertainty in A =) $2 \times (0.01 / 0.38) \times 100 = 5.3 \%$	C1	
		total = 8.13%	C1	
(c)		uncertainty = $0.395 \times 10^{-7} (\Omega \text{ m})$	A1	[4]
		(missing 2 factor in uncertainty in A , then allow max 3/4)		
		resistivity = $(4.9 \times 10^{-7} \pm 0.4 \times 10^{-7}) \Omega \text{ m}$	A1	[1]

- 32 (a)** $\frac{V}{t} = \frac{\pi P r^4}{8 C l}$
 $C = [\pi \times 2.5 \times 10^3 \times (0.75 \times 10^{-3})^4] / (8 \times 1.2 \times 10^{-6} \times 0.25)$ C1
 $= 1.04 \times 10^{-3} \text{ N s m}^{-2}$ A1 [2]
- (b)** $4 \times \%r$ C1
 $\%C = \%P + 4 \times \%r + \%V/t + \%l$
 $= 2\% + 5.3\% + 0.83\% + 0.4\% (= 8.6\%)$ A1
 $\Delta C = \pm 0.089 \times 10^{-3} \text{ N s m}^{-2}$ A1 [3]
- (c)** $C = (1.04 \pm 0.09) \times 10^{-3} \text{ N s m}^{-2}$ A1 [1]

- 33 (a)** spacing = 380 or $3.8 \times 10^2 \text{ pm}$ B1 [1]
- (b)** time = 24×3600
time = 0.086 (0.0864) Ms B1 [1]
- (c)** time = distance / speed = $\frac{1.5 \times 10^{11}}{3 \times 10^8}$ C1
 $= 500 \text{ (s)} = 8.3 \text{ min}$ A1 [2]
- (d)** momentum and weight B1 [1]

- 34 (a)** $d = v \times t$ C1
 $t = 0.2 \times 4$ (allow $t = 0.2 \times 2$) C1
 $d = 3 \times 10^8 \times 0.8 \times 10^{-6}$ OR $3 \times 10^8 \times 0.4 \times 10^{-6}$ C1
 $d = 240 \text{ m}$ hence distance from source to reflector = 120 m A1 [4]
- (b)** speed of sound 300 cf speed of light 3×10^8 OR time = $240 / 300 (= 0.8)$
OR time = $120 / 300 (= 0.4)$ C1
sound slower by factor of 10^6 OR time for one division $0.8 / 4$
OR time for one division $0.4 / 2$ C1
time base setting 0.2 s cm^{-1} [unit required] A1 [3]

- 35** volume = $\pi (14 \times 10^{-3})^2 \times 12 \times 10^{-3} (= 7.389 \times 10^{-6} \text{ m}^3)$
density = mass / volume [any subject] C1
mass = $6.8 \times 10^3 \times 7.389 \times 10^{-6} = 0.0502$
weight = mg C1
 $= 0.0502 \times 9.81 = 0.49 \text{ N}$ (mark not awarded if not to **two** s.f.) A1 [4]

36 (a)	SI units for T : s, R : m and M : kg (or seen clearly in formula)	C1
	$K = T^2 M / R^3$ units: $\text{s}^2 \text{kg m}^{-3}$ (allow $\text{s}^2 \text{kg} / \text{m}^3$ or $\frac{\text{s}^2 \text{kg}}{\text{m}^3}$)	A1 [2]
(b)	% uncertainty in K : 1% (for T) + 3% (for R) + 2% (for M) OR = 6%	C1
	$K = [(86400)^2 \times 6 \times 10^{24}] / (4.23 \times 10^7)^3 = 5.918 \times 10^{11}$	C1
	6% of $K = 0.355 \times 10^{11}$	C1
	$K = (5.9 \pm 0.4) \times 10^{11}$ (SI units) correct power of ten required for both [incorrect % value then max. 1]	A1 [4]
37 (a)	$\rho = m / V$ $V = (\pi d^2 / 4) \times t = 7.67 \times 10^{-7} \text{m}^3$ $\rho = (9.6 \times 10^{-3}) / [\pi(22.1/2 \times 10^{-3})^2 \times 2.00 \times 10^{-3}]$ $\rho = 12513 \text{kg m}^{-3}$ (allow 2 or more s.f.)	C1 A1 [3]
(b) (i)	$\Delta \rho / \rho = \Delta m / m + \Delta t / t + 2\Delta d / d$ $= 5.21\% + 0.50\% + 0.905\%$ [or correct fractional uncertainties] $= 6.6\%$ (6.61%)	C1 C1 A1 [3]
(ii)	$\rho = 12500 \pm 800 \text{kg m}^{-3}$	A1 [1]
38 (a)	pressure = force / area (normal to the force) [clear ratio essential]	B1 [1]
(b) (i)	$P = mg / A = (5.09 \times 9.81) / A$ $A = (\pi d^2 / 4) = \pi \times (9.4 \times 10^{-2})^2 / 4 (= 0.00694 \text{m}^2)$ $P = 49.93 / 0.00694$ $= 7200$ (7195) Pa (minimum of 2 s.f. required)	C1 C1 A1 [3]
(ii)	$\Delta P / P = \Delta m / m + 2\Delta d / d$ $= 0.01 / 5.09 + (2 \times 0.1) / 9.4 (= 0.0020 + 0.021 \text{ or } 2.3\%)$ $\Delta P = 170$ (165 to 167) Pa	C1 C1 A1 [3]
(iii)	$P = 7200 \pm 200 \text{Pa}$	A1 [1]
39 (a)	<u>random</u> error (in the measurements) of the length OR resistance	B1 [1]
(b)	gradient = $(3.6 - 1.9) / (0.8 - 0.4)$ $= 4.25$	C1 A1 [2]
(c)	$R = \rho l / A$ $\rho = \text{gradient} \times \text{area} = 4.25 \times 0.12 \times 10^{-6}$ $= 5.1(0) \times 10^{-7} \Omega \text{m}$	C1 C1 A1 [3]

40	(a) (i)	amplitude scale reading 2.2 (cm) amplitude = $2.2 \times 2.5 = 5.5 \text{ mV}$	C1 A1 [2]
	(ii)	time period scale reading = 3.8 (cm) time period = $3.8 \times 0.5 \times 10^{-3} = 0.0019 \text{ (s)}$ frequency $f = 1 / 0.0019 = 530 \text{ (526) Hz}$	C1 C1 A1 [3]
	(iii)	uncertainty in reading = ± 0.2 in 3.8 (cm) or 5.3% or 0.2 in 7.6 (cm) or 2.6% [allow other variations of the distance on the x-axis] actual uncertainty = 5.3% of 526 = 27.7 or 28 Hz or 2.6% of 526 = 13 or 14	M1 A1 [2]
	(b)	frequency = $530 \pm 30 \text{ Hz}$ or $530 \pm 10 \text{ Hz}$	A1 [1]
41	(a) (i)	diameter and extension: micrometer (screw gauge) or digital calipers length: tape measure or metre rule load: spring balance or Newton meter	B1 B1 B1 [3]
	(ii)	to reduce the effect of random errors or to plot a graph to check for zero error in measurement of extension or to see if limit of proportionality is exceeded	B1 [1]
42	(a)	metre rule/tape measure	B1
	(b) (i)	$v = [(1.8 \times 126 \times 10^{-2}) / 5.1 \times 10^{-3}]^{1/2}$ $= 21.1 \text{ (ms}^{-1}\text{)}$	C1 A1
	(ii)	percentage uncertainty = 4% or fractional uncertainty = 0.04 $\Delta v = 0.04 \times 21.1$ $= 0.84$ $v = 21.1 \pm 0.8 \text{ (ms}^{-1}\text{)}$	C1 C1 A1
	(a) (i)	$(50 \text{ to } 200) \times 10^{-3} \text{ kg}$ or $(0.05 \text{ to } 0.2) \text{ kg}$	B1 [1]
	(ii)	$(50 \text{ to } 300) \text{ cm}^3$	B1 [1]
43	(b)	density = mass/volume or $\rho = M/V$	C1
		$V = [\pi(0.38 \times 10^{-3})^2 \times 25.0 \times 10^{-2}] / 4 (= 2.835 \times 10^{-8} \text{ m}^3)$	C1
		$\rho = (0.225 \times 10^{-3}) / 2.835 \times 10^{-8}$ $= 7940 \text{ (kg m}^{-3}\text{)}$	A1
		$\Delta\rho/\rho = 2(0.01/0.38) + (0.1/25.0) + (0.001/0.225) [= 0.061]$ or $\%\rho = 5.3\% + 0.40\% + 0.44\% (= 6.1\%)$	C1
		$\Delta\rho = 0.061 \times 7940 = 480 \text{ (kg m}^{-3}\text{)}$	
		density = $(7.9 \pm 0.5) \times 10^3 \text{ kg m}^{-3}$ or $(7900 \pm 500) \text{ kg m}^{-3}$	A1 [5]

44	(a) systematic: the reading is larger or smaller than (or varying from) the true reading by a constant amount random: scatter in readings about the true reading	B1	
		B1	[2]
	(b) precision: the size of the smallest division (on the measuring instrument) or 0.01 mm for the micrometer	B1	
	accuracy: how close (diameter) value is to the true (diameter) value	B1	[2]
45	(a) (density =) mass / volume		
	(b) (i) $d = [(6 \times 7.5) / (\pi \times 8100)]^{1/3}$ $= 0.12(1) \text{ m}$	A1	[1]
	(ii) percentage uncertainty = $(4 + 5) / 3$ (= 3%) or fractional uncertainty = $(0.04 + 0.05) / 3$ (= 0.03)	C1	
	absolute uncertainty (= 0.03×0.121) = 0.0036	C1	
	$d = 0.121 \pm 0.004 \text{ m}$	A1	[3]
46	(a) (stress =) force / area or $\text{kg m s}^{-2} / \text{m}^2$ $= \text{kg m}^{-1} \text{ s}^{-2}$	B1	
		A1	
	(b) (i) $0.58 = 2\pi \times [(4 \times 0.500 \times 0.600^3) / (E \times 0.0300 \times 0.00500^3)]^{0.5}$	C1	
	$E = [4\pi^2 \times 4 \times 0.500 \times (0.600)^3] / [(0.58)^2 \times 0.0300 \times (0.00500)^3]$	C1	
	$= 1.35 \times 10^{10} \text{ (Pa)}$		
	$= 14 \text{ (13.5) GPa}$	A1	
	(ii) 1 (accuracy determined by) the closeness of the value(s)/measurement(s) to the true value	B1	
	(precision determined by) the range of the values/measurements	B1	
	2 l is (cubed so) $3 \times$ (percentage/fractional) uncertainty	B1	
	and T is (squared so) $2 \times$ (percentage / fractional) uncertainty		
	and (so) l contributes more		
47	1. precision is determined by the range in the measurements/values/readings/data/results	B1	
	2. metre rule measures to $\pm 1 \text{ mm}$ and micrometer to $\pm 0.01 \text{ mm}$ (so there is less (percentage) uncertainty/random error)	B1	

48	(a) (i) micrometer (screw gauge)/digital calipers	B1
	(a) (ii) take several readings (and average) along the wire or around the circumference	M1 A1
	(b) (i) $\sigma = 4 \times 25 / [\pi \times (0.40 \times 10^{-3})^2] = 1.99 \times 10^8 \text{ Nm}^{-2}$ or $\sigma = 25 / [\pi \times (0.20 \times 10^{-3})^2] = 1.99 \times 10^8 \text{ Nm}^{-2}$	A1
	(b) (ii) %F = 2% and %d = 5%	C1
	or $\Delta F / F = \frac{0.5}{25}$ and $\Delta d / d = \frac{0.02}{0.4}$ % σ = 2% + (2 $\times$ 5%)	A1
	or % σ = [0.02 + (2 $\times$ 0.05)] $\times$ 100 % σ = 12%	
	(b) (iii) absolute uncertainty = (12/100) $\times$ 1.99×10^8 = 2.4×10^7	C1
	$\sigma = 2.0 \times 10^8 \pm 0.2 \times 10^8 \text{ Nm}^{-2}$ or $2.0 \pm 0.2 \times 10^8 \text{ Nm}^{-2}$	A1
49	(a) (i) zero error or wrongly calibrated scale	B1
	(ii) reading scale from different angles or wrongly interpolating between scale readings/divisions	B1
	(b) (i) $P = V^2 / R$ or $P = VI$ and $V = IR$	C1
	$P = 5.0^2 / 125$ or 5.0×0.04 or $(0.04)^2 \times 125$ = 0.20 W	A1
	(ii) %V = 2% or $\Delta V / V = 0.02$	C1
	%P = (2 $\times$ 2%) + 3% or %P = (2 $\times$ 0.02 + 0.03) $\times$ 100 = 7%	A1
	(iii) absolute uncertainty in P = (7 / 100) $\times$ 0.20 = 0.014	C1
	power = $0.20 \pm 0.01 \text{ W}$ or $(2.0 \pm 0.1) \times 10^{-1} \text{ W}$	A1
50 (a)	(velocity =) change in displacement / time (taken)	B1
	(b)(i) $k = [1.29 \times (3.3 \times 10^2)^2] / 9.9 \times 10^4$ = 1.4	C1 A1
	(ii) percentage uncertainty = (3 $\times$ 2) + 4 + 2 (= 12%) or	C1
	fractional uncertainty = (0.03 $\times$ 2) + 0.04 + 0.02 (= 0.12)	
	$\Delta k = 0.12 \times 1.42$	C1
	= 0.17 (allow to 1 significant figure)	
	$k = 1.4 \pm 0.2$	A1

51 (a)	absolute uncertainty = $(1.6 / 100) \times 0.0125$ = $2 \times 10^{-4} \text{ m}$	A1
(b)(i)	$\rho = (4 \times 0.38) / (\pi \times 0.0125^2)$ = 3100 N m^{-2}	A1
(ii)	percentage uncertainty = $2.8 + (2 \times 1.6)$ (= 6%) or fractional uncertainty = $0.028 + (2 \times 0.016)$ (= 0.06) absolute uncertainty = 0.06×3100 = 190 N m^{-2} (<i>allow to 1 significant figure</i>)	C1 A1
52 (a)(i)	mass in range 1–20 g	A1
(ii)	wavelength in range $1 \times 10^{-8} \text{ m}$ to $4 \times 10^{-7} \text{ m}$	A1
(b)(i)	$T = 2\pi \times (200 \times 10^{-3} / 25)^{0.5}$ = 0.56 s	A1
(ii)	percentage uncertainty = $(2\% + 8\%) / 2$ (= 5%) or fractional uncertainty = $(0.02+0.08) / 2$ (= 0.05) $\Delta T = 0.56 \times 0.05$ = 0.028 (s) $T = (0.56 \pm 0.03) \text{ s}$	C1 C1 A1
53 (a)	(work =) force $\times$ displacement units: $\text{kg m s}^{-2} \times \text{m} = \text{kg m}^2 \text{ s}^{-2}$	C1 A1
(b)(i)	units of Q: As units of C: $\text{kg}^{-1} \text{ m}^{-2} \text{ A}^2 \text{ s}^4$	C1 A1
(ii)	1. e.g. reading scale from different angles (wrongly) interpolating between scale readings/divisions 2. e.g. zero error wrongly calibrated scale	B1 B1

54.

(a)(i)	wavelength = $8.5 \times 10^{-5} \text{ m}$
(a)(ii)	$f = v / \lambda \text{ or } c / \lambda$
	$= 3.0 \times 10^8 / 8.5 \times 10^{-5} (= 3.5 \times 10^{12})$
	$= 3.5 \text{ THz}$
(a)(iii)	infrared
(b)	(implied) percentage uncertainty in $I = 4\%$ or (implied) fractional uncertainty in $I = 0.04$
	percentage uncertainty in $E = 5\% + (4\% \times 2)$
	$= 13\%$

55.

(a)	mass / volume
(b)(i)	absolute uncertainty = $4.0 \times (5 / 100)$
	$= (\pm) 0.2 \text{ cm}$
(b)(ii)	percentage uncertainty = $2 + 4 + (5 \times 2)$
	$= (\pm) 16\%$
(c)	$p = F / A \text{ or } p = W / A$
	$p = (19.5 \times 10^{-3} \times 9.81) / (4.0 \times 10^{-2})^2$
	$= 120 \text{ Pa}$

56.

(a)(i)	two correct scalar quantities e.g. time, mass, distance, temperature
	two correct vector quantities e.g. force, acceleration, velocity, displacement
(a)(ii)	magnitude
	unit
(b)(i)	north component of velocity = 11 m s^{-1}
	east component of velocity = 7.5 m s^{-1}
(b)(ii)	velocity = $7.5 - 2.7$
	$= 4.8 \text{ m s}^{-1}$
(b)(iii)	velocity = $\sqrt{(11^2 + 4.8^2)}$
	$= 12 \text{ m s}^{-1}$
(b)(iv)	angle = $\tan^{-1} (4.8 / 11)$
	$= 24^\circ$

57.

(a)	mass / volume
(b)(i)	(vernier/digital) calipers
(b)(ii)	percentage uncertainty = $(0.0004 / 0.0420) \times 100$
	$= 1\%$
(c)(i)	$\text{kg m}^{-3} = \text{kg} \times \text{m}^n / \text{m}$ or $\text{kg m}^{-3} = \text{kg} \times \text{m}^n \times \text{m}^{-1}$
	$-3 = n - 1$ and (so) $n = -2$
(c)(ii)	$(\Delta\rho / \rho) = (\Delta M / M) + 2(\Delta r / r) + (\Delta L / L)$
	percentage uncertainty = $[(0.001 / 1.072) + 2 \times (0.0004 / 0.0420) + (0.0001 / 0.1242)] (\times 100)$
	$= 0.09\% + 2 \times 0.95\% + 0.08\%$
	$= 2\%$
(c)(iii)	$\rho = (1.072 \times 0.0420^{-2}) / (2.094 \times 0.1242)$
	$= 2337 \text{ (kg m}^{-3}\text{)}$
	$\Delta\rho = 0.021 \times 2337$
	$= 49 \text{ (kg m}^{-3}\text{)}$
	$\rho = (2340 \pm 50) \text{ kg m}^{-3}$

58.

(a)	10^{12}
	pico (p)
(b)	ampere and metre both underlined (and no other units underlined)
(c)(i)	percentage uncertainty = $3.5 + (3.0 \times 2) + 2.5 + 2.0$
	= 14%
(c)(ii)	absolute uncertainty = $4.1 \times 10^{-7} \times 14 / 100$ = $6 \times 10^{-8} \Omega \text{ m}$

59.

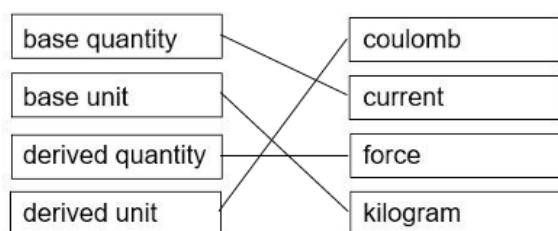
(a)	only ampere and kilogram underlined
(b)(i)	percentage uncertainty = $2.1 + 0.6 + (1.5 \times 2)$
	= 5.7%
(b)(ii)	absolute uncertainty = $(5.7 / 100) \times 8.245 \times 10^9$
	(= $4.7 \times 10^8 \text{ Pa}$ or $0.47 \times 10^9 \text{ Pa}$)
	$E = (8.2 \pm 0.5) \times 10^9 \text{ Pa}$

60.

(a)	$\rho = m / V$
	$V = (4 / 3) \times \pi \times r^3$
	= $(4 / 3) \times \pi \times (3.42 / 2)^3$
	(= 20.9 cm^3)
	$\rho = 67 / 20.9$ = 3.2 g cm^{-3}
(b)	$\% \rho = \%m + 3 \times \%d$
	= $[(2 / 67) \times 100] + [3 \times (0.02 / 3.42) \times 100]$
	= 3.0% + $3 \times 0.58\%$
	= 4.7% or 5%

61.

(a)



any two joined correctly

all four joined correctly

(b)(i)	the measurements have a small range
(b)(ii)	(average of the) measurements not close to the true value
(c)(i)	percentage uncertainty = $(3 + 5 + 4) / 2$ = 6%
(c)(ii)	absolute uncertainty = $(6 / 100) \times 15.0$ = 0.9 m s^{-1}

62.

(a)

force $\times$ displacement in the direction of the force

(b)

$$P = Fs / t$$

$$= (\text{kg m s}^{-2} \times \text{m}) / \text{s}$$

$$= \text{kg m}^2 \text{s}^{-3}$$

(c)(i)

$$84 \times 10^3 = v^3 \times 0.56$$

$$v = 53 \text{ m s}^{-1}$$

(c)(ii)

$$\text{percentage uncertainty} = (5\% + 7\%) / 3 (= 4\%)$$

or

$$\text{fractional uncertainty} = (0.05 + 0.07) / 3 (= 0.04)$$

$$\text{absolute uncertainty} = 0.04 \times 53$$

$$= (\pm) 2 \text{ m s}^{-1}$$

63.

(a)	(SI base unit of α =) $\text{m}^3 \text{s}^{-1} \times \text{m} / \text{m}^4 = \text{s}^{-1}$
(b)	(percentage uncertainty =) $3 + 4 + 2 \times 4 = 15$ (%)
(c)	$\alpha = QL / r^4$ $= 2.72 \times 10^{-8} \times 2.5 \times 10^{-2} / (7.1 \times 10^{-5})^4$ $= 2.7 \times 10^7$
	absolute uncertainty = $0.15 \times [2.7 \times 10^7]$ $= 0.4 \times 10^7$
	$\alpha = (2.7 \pm 0.4) \times 10^7 \text{s}^{-1}$

64.

(a)	only ampere and kelvin underlined
(b)	initial speed / velocity is zero
	(non-zero magnitude of) acceleration is constant / uniform (and in a straight line)
(c)(i)	$a = 2.75^2 / (2 \times 3.89)$ $= 0.97 \text{ m s}^{-2}$
(c)(ii)	percentage uncertainty = $(2 \times 0.8) + 0.5$ $= 2.1\%$
(c)(iii)	absolute uncertainty = $(2.1 / 100) \times 0.97$ $= 0.02 \text{ m s}^{-2}$

65.

(a)(i)	work done per unit time
(a)(ii)	($P = W / t$ gives) units: $\text{kg m}^2 \text{s}^{-2} / \text{s} = \text{kg m}^2 \text{s}^{-3}$
(b)	$(I = P / A \text{ so})$ units of I : $\text{kg m}^2 \text{s}^{-3} / \text{m}^2$ or kg s^{-3} units of f : s^{-1} and units of A : m and units of v : m s^{-1}
	units of k : $\text{kg s}^{-3} / [(\text{s}^{-1})^2 \text{m}^2 \text{m s}^{-1}]$ $= \text{kg m}^{-3}$

66.

(a)(i)	force / area (normal to the force)
(a)(ii)	$(p = F / A \text{ so units are}) \text{ kg m s}^{-2} / \text{m}^2 = \text{kg m}^{-1} \text{s}^{-2}$
(b)	unit of R : m and unit of t : s and unit of L : m unit of ρ : kg m^{-3} or $\rho = m / V$ base units of k : $(\text{kg m}^{-1} \text{s}^{-2} \times \text{m}^4 \times \text{kg m}^{-3} \times \text{s}) / (\text{kg} \times \text{m}) = \text{kg m}^{-1} \text{s}^{-1}$
(c)	R contributes $4 \times 2\%$ or 8% (and L contributes 2%) so R contributes more (to the percentage uncertainty in k)

67.

(a)	$t = \sqrt{(2s / g)}$ $= \sqrt{[(2 \times 36) / 9.81]}$ $= 2.7 \text{ s}$
(b)	- reaction time between hearing the splash and stopping the stop-watch - the sound (of the splash) takes time to reach the student or the stone hits the water at a different time to the sound being heard or the sound (of the splash) has to travel to the student - the student might not let go of the stone from ground level - the student might not let go of the stone and start the stop-watch at the same time - stop-watch may not be properly calibrated / has a zero error (local value of) g is not (exactly) $9.81 \text{ (m s}^{-2}\text{)}$ - stone given initial velocity / initial velocity not zero - stone does not fall (exactly) vertically / in a straight line <i>Any three points, 1 mark each</i>
(c)	precise: results are close together / have little scatter not accurate: the values are not close to / 50% different / (very) different from the true value

68.

(a)	scalar and vector have magnitude vector has direction (and scalar does not have direction)
(b)(i)	$r = [(3 \times 28) / 4\pi]^{1/3}$ $= 1.9 \text{ cm}$
(b)(ii)	percentage uncertainty in $V = (0.5 / 28) \times 100$ ($= 1.79\%$) percentage uncertainty in $r = 1.79 / 3$ $= 0.6\%$

69.

(a)	charge and power only ticked
(b)(i)	$\%D = 0.4\%$ and $\%L = 0.6\%$
(b)(ii)	$\rho = (4 \times 0.247) / [\pi \times (26.2 \times 10^{-3})^2 \times 0.162]$
	$\rho = 2.83 \times 10^3 \text{ kg m}^{-3}$
(b)(iii)	percentage uncertainty = $0.4 + (2 \times 0.4) + 0.6$
	= 1.8%